

Author profiling in social media

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MIARFID

Official Master's Degree
in Artificial Intelligence,
Pattern Recognition
and Digital Imaging

DSIC



UNIVERSITAT
POLITÈCNICA
DE VALÈNCIA

Author Profiling

Language and style varies among classes of authors

Forensics: who is behind an harassment

Security: who is behind a threat

Marketing: who is behind an opinion

Socio-political analysis: who is behind a stance

- **Gender & age**
- **Personality**
- **Native language and language variety**
- **Ideological/organizational affiliation**

Security

ARABIC AUTHOR PROFILING FOR CYBER SECURITY

FUNDED BY THE QATAR NATIONAL RESEARCH FUND



Profiling for security

- Profiling a deceiver
- Profiling an ironic person

In case of a potential threat:

- Profiling gender
- Profiling age
- Profiling native language
- Profiling language variety

Outline

- Profiling gender & age
- Author profiling at PAN-2013 and PAN-2014
- EmoGraph: graph-based discourse analysis
- Profiling bots at PAN-2019
- Profiling native language

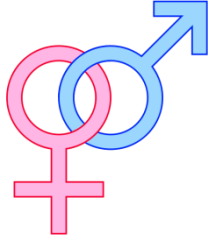


<https://pan.webis.de/>



Profiling gender & age





Which is female/male?

My aim in this article is to show that given a relevance theoretic approach to utterance interpretation, it is possible to develop a better understanding of what some of these so-called apposition markers indicate. It will be argued that the decision to put something in other words is essentially a decision about style, a point which is, perhaps, anticipated by Burton-Roberts when he describes loose apposition as a rhetorical device. However, he does not justify this suggestion by giving the criteria for classifying a mode of expression as a rhetorical device. Nor does he specify what kind of effects might be achieved by a reformulation or explain how it achieves those effects. In this paper I follow Sperber and Wilson's (1986) suggestion that rhetorical devices like metaphor, irony and repetition are particular means of achieving relevance. As I have suggested, the corrections that are made in unplanned discourse are also made in the pursuit of optimal relevance. However, these are made because the speaker recognises that the original formulation did not achieve optimal relevance .



The main aim of this article is to propose an exercise in stylistic analysis which can be employed in the teaching of English language. It details the design and results of a workshop activity on narrative carried out with undergraduates in a university department of English. The methods proposed are intended to enable students to obtain insights into aspects of cohesion and narrative structure: insights, it is suggested, which are not as readily obtainable through more traditional techniques of stylistic analysis. The text chosen for analysis is a short story by Ernest Hemingway comprising only 11 sentences. A jumbled version of this story is presented to students who are asked to assemble a cohesive and well formed version of the story. Their re-constructions are then compared with the original Hemingway version.

[examples: Moshe Koppel]

British National Corpus

- 920 documents labelled for
 - author gender
 - document genre
- Used 566 controlled for genre

	Male	Fem
Fiction (prose)	132	132
Non-fiction	151	151
Arts (general)	8	8
Arts (acad.)	12	12
Belief/Thought	12	12
Biography	27	27
Commerce	5	5
Leisure	8	8
Science (gen.)	13	13
Soc. Sci. (gen.)	26	26
Soc. Sci. (acad.)	19	19
World Affairs	21	21

M. Koppel, S. Argamon, and A. R. Shimoni. Automatically categorizing written texts by author gender. Literary and linguistic computing 17(4), 2002.

Distinguishing features: male vs. female style

Males use more

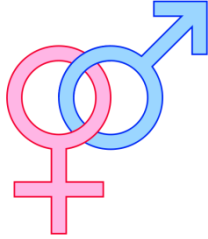
- Determiners
- Adjectives
- *of* modifiers (e.g. *pot of gold*)

Informational
features

Females use more

- Pronouns *
- *for* and *with*
- Negation
- Present tense

Involvedness
features



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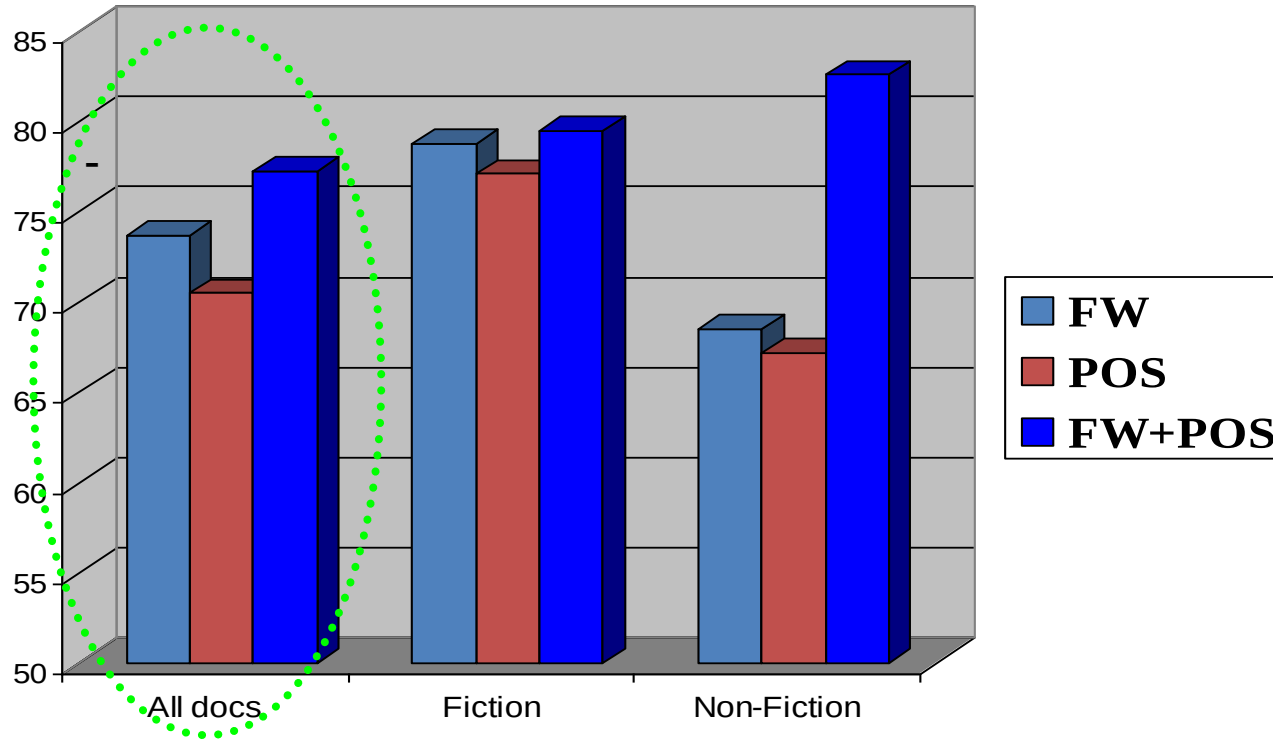
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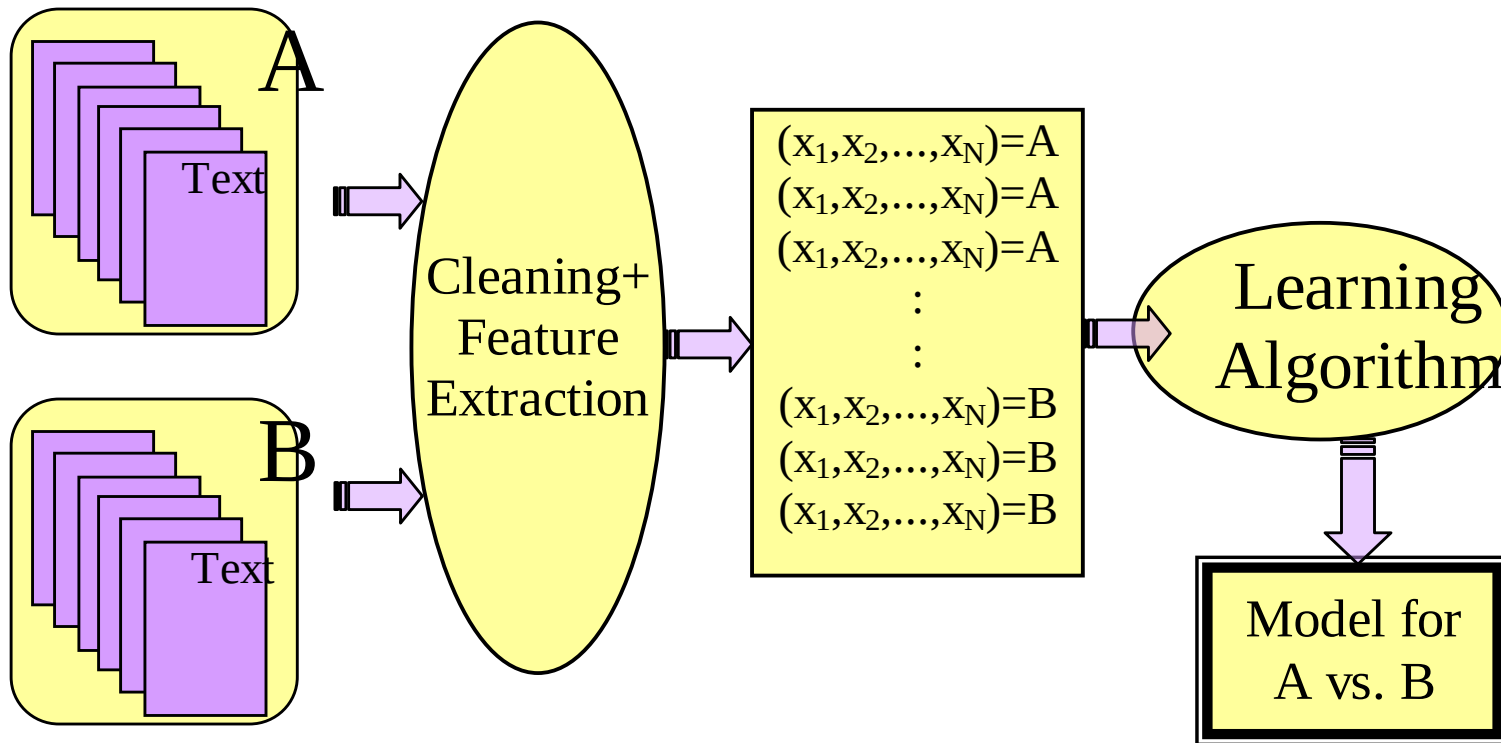
Results per feature set



- Handle fiction and non-fiction separately
- Use full feature set

POS: Part Of Speech FW: Function words (*and, of, the,..*)

Text categorization



10-fold cross-validation

- Break example set into 10 parts
- Train on nine parts, test on tenth
- Repeat ten times, each holding out different part
- Average results

Teen

Twenties

Thirties

Male

Female

Social media: example

Yesterday we had our second jazz competition. Thank God we weren't competing. We were sooo bad. Like, I was so ashamed, I didn't even want to talk to anyone after. I felt so rotten, and I wanted to cry, but...it's ok.

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Blog corpus

- Less-formal text
 - 85,000 blogs
 - blogger-provided profiles (gender, age, occupation, astrological sign)
 - harvested August 2004
 - all non-text ignored (formatting, quoting)

Blog corpus

	Gender		
Age	Female	Male	Total
unknown	12287	12259	24546
13-17	6949	4120	8240
18-22	7393	7690	15083
23-27	4043	6062	8086
28-32	1686	3057	4743
33-37	860	1827	1720
38-42	374	819	748
43-48	263	584	526
>48	314	906	1220
Total	9660	9660	19320

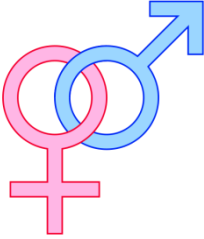
Final balanced corpus:

- 19,320 total blogs
 - 8240 in “10s”
 - 8086 in “20s”
 - 2994 in “30s”
- 681,288 total posts
- 141,106,859 total words

Gender and age classification

Features	Gender & age (accuracy)
Style & Content	80.0% - 77.4%
Style Words	77.0% - 69.4%
Content Words	73.0% - 76.2%

J. Schler, M. Koppel, S. Argamon, and J. W. Pennebaker. Effects of age and gender on blogging. In AAAI Spring Symposium: Computational Approaches to Analyzing Weblogs, pages 199–205. AAAI, 2006.



Men vs. women

LIWC category	male	female
job	<u>68.1±0.6</u>	56.5±0.5
money	<u>43.6±0.4</u>	37.1±0.4
sports	<u>31.2±0.4</u>	20.4±0.2
tv	<u>21.1±0.3</u>	15.9±0.2
sex	32.4±0.4	<u>43.2±0.5</u>
family	27.5±0.3	<u>40.6±0.4</u>
eating	23.9±0.3	<u>30.4±0.3</u>
friends	20.5±0.2	<u>25.9±0.3</u>
sleep	18.4±0.2	<u>23.5±0.2</u>
<i>pos-emotions</i>	248.2±1.9	<u>265.1±2</u>
<i>neg-emotions</i>	159.5±1.3	<u>178±1.4</u>

The lifecycle of the common blogger...

Word	10s	20s	30s
maths	105	3	2
homework	137	18	15
bored	384	111	47
sis	74	26	10
boring	369	102	63
awesome	292	128	57
mum	125	41	23
crappy	46	28	11
mad	216	80	53
dumb	89	45	22

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Word	10s	20s	30s
semester	22	44	18
apartment	18	123	55
drunk	77	88	41
beer	32	115	70
student	65	98	61
album	64	84	56
college	151	192	131
someday	35	40	28
dating	31	52	37
bar	45	153	111

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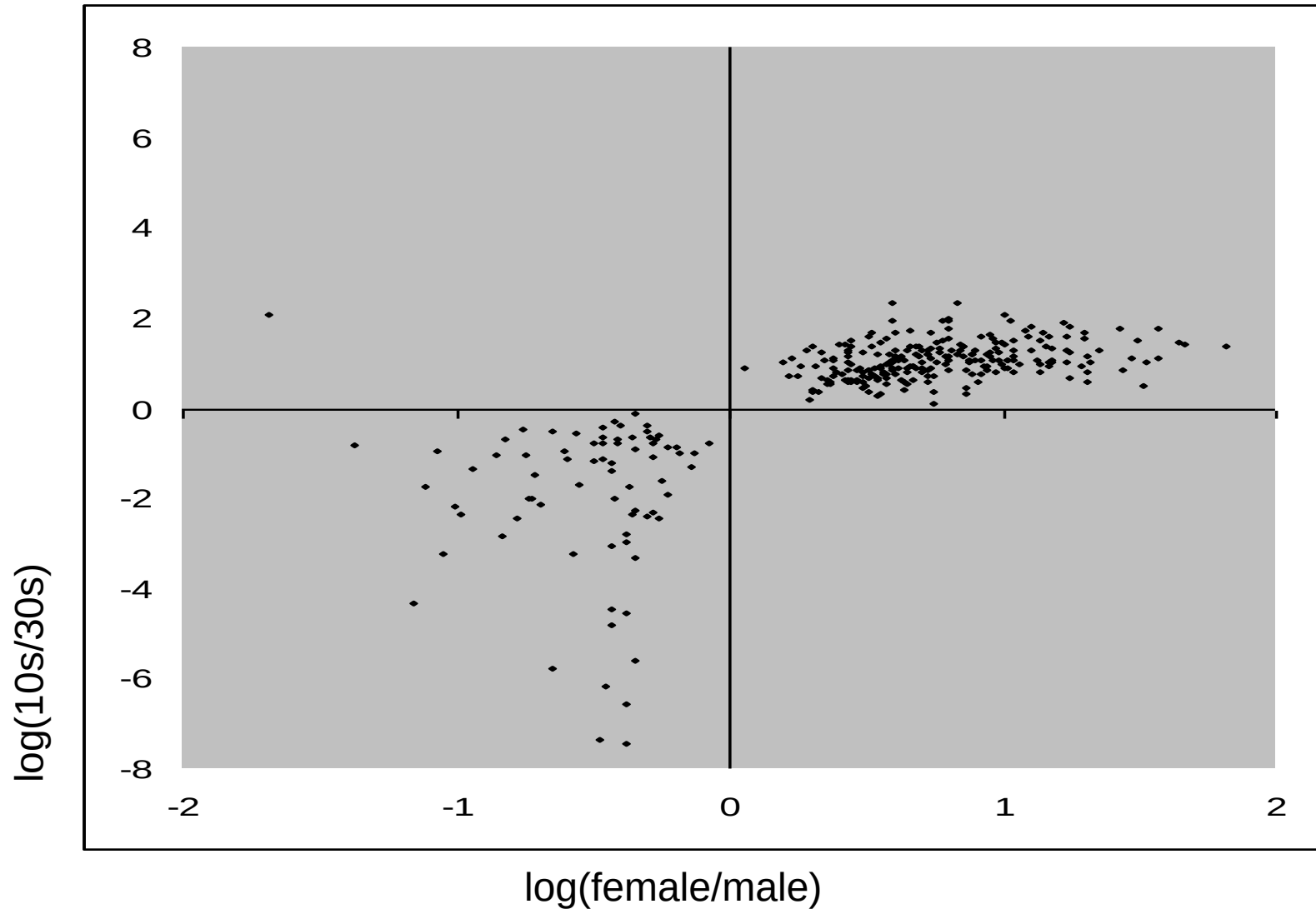
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Word	10s	20s	30s
marriage	27	83	141
development	16	50	82
campaign	14	38	70
tax	14	38	72
local	38	118	185
democratic	13	29	59
son	51	92	237
systems	12	36	55
provide	15	54	69
workers	10	35	46

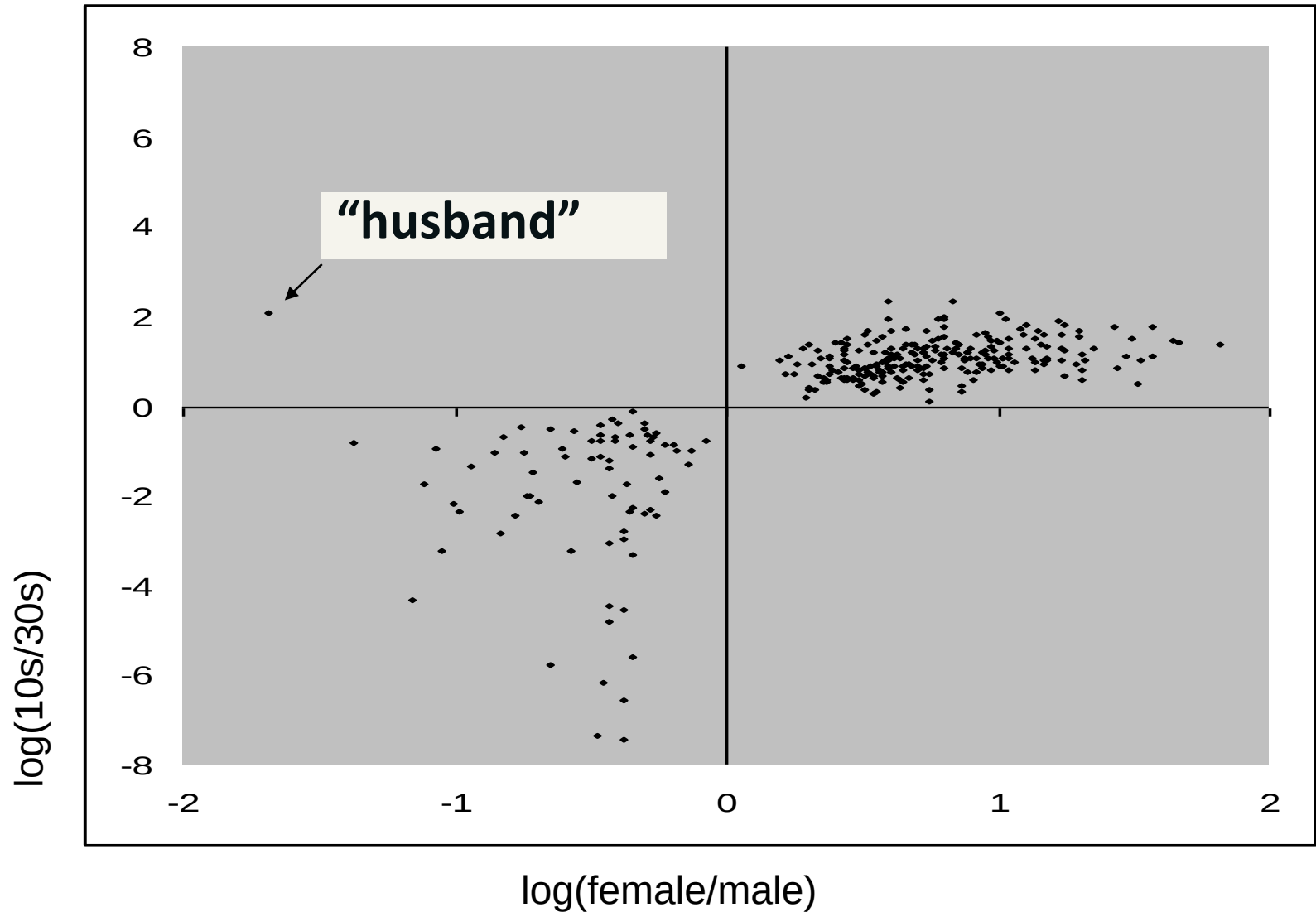
Relating age & gender

- Now...is there a linguistic connection between age and gender?
- Consider the most distinctive words for both Age and Gender:
 - Intersect the 1000 words with **highest Age information gain** and the 1000 words with **highest Gender information gain**
 - Total of 316 words
 - Plot $\log(30s/10s)$ vs. $\log(\text{male/female})$

Relating age & gender



Relating age & gender



Gender & age: pre PAN state of the art

AUTHOR	COLLECTION	FEATURES	RESULTS	OTHER CHARACTERISTICS
Argamon et al., 2002	British National Corpus	Part-of-speech	Gender: 80% accuracy	
Koppel et al., 2003	Blogs	Lexical and syntactic features	Gender: 80% accuracy	Self-labeling
Schler et al., 2006	Blogs	Stylistic features + content words with the highest information gain	Gender: 80% accuracy Age: 75% accuracy	
Goswami et al., 2009	Blogs	Slang + sentence length	Gender: 89.18 accuracy Age: 80.32 accuracy	
Zhang & Zhang, 2010	Segments of blog	Words, punctuation, average words/sentence length, POS, word factor analysis	Gender: 72.10 accuracy	
Nguyen et al., 2011, 2013	Blogs & Twitter	Unigrams, POS, LIWC	Correlation: 0.74 Mean absolute error: 4.1 - 6.8 years	Manual labeling Age as continuous variable
Peersman et al., 2011	Netlog	Unigrams, bigrams, trigrams and tetagrams	Gender+Age: 88.8 accuracy	Self-labeling, min 16 plus 16,18,25

Author profiling @

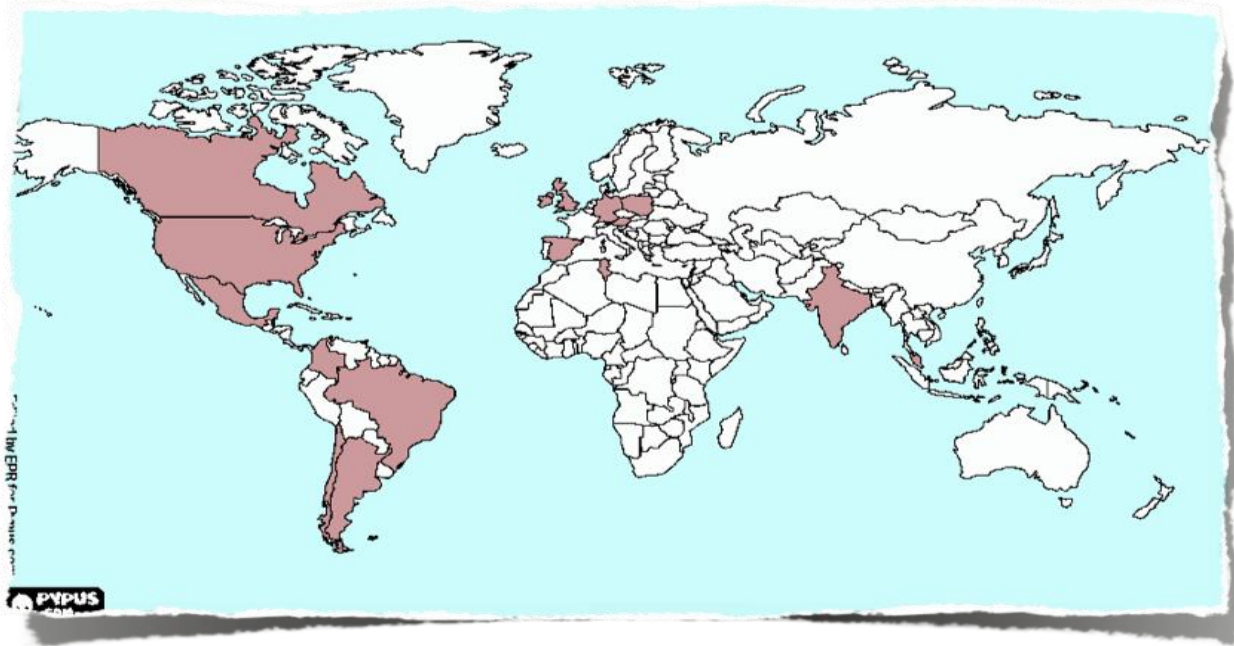
- **CLEF 2013: Age and gender in social media**
- **CLEF 2014: Age and gender in social media, Twitter, blogs, reviews**
- CLEF 2015: Age, gender, personality in Twitter
- CLEF 2016: Cross-genre age and gender
- FIRE 2016: Personality in source code
- CLEF 2017: Gender and language variety identification in Twitter
- FIRE 2017: Native Indian language identification
- FIRE 2017: Cross-genre gender identification in Russian
- CLEF 2018: Multimodal (text + image) gender in Twitter

Author profiling @

- **CLEF 2019: Bots and gender profiling on Twitter**
- CLEF 2020: Profiling fake news spreaders on Twitter
- CLEF 2021: Profiling hate speech spreaders on Twitter
- CLEF 2022: Profiling irony and stereotype spreaders on Twitter
- CLEF 2023: Profiling cryptocurrency influencers with few-shot learning

Author profiling: **PAN** @CLEF 2013

- Teams submitting results: 21 (Registered teams: 64)
- (Towards) **big data**: 400,000 social media texts including **chat lines of potential pedophiles** (task in 2012)



- **Age classes**: 10s (13-17), 20s (23-27), 30s (33-48)
- **Languages**: English and Spanish

Results: EN vs. ES

English			
Team	Total	Gender	Age
Meina	0.3894	0.5921	0.6491
Pastor L.	0.3813	0.5690	0.6572
Seifeddine	0.3677	0.5816	0.5897
Santosh	0.3508	0.5652	0.6408
Yong Lim	0.3488	0.5671	0.6098
Ladra	0.3420	0.5608	0.6118
Aleman	0.3292	0.5522	0.5923
Gillam	0.3268	0.5410	0.6031
Kern	0.3115	0.5267	0.5690
Cruz	0.3114	0.5456	0.5966
Pavan	0.2843	0.5000	0.6055
Caurcel Diaz	0.2840	0.5000	0.5679
H. Farias	0.2816	0.5671	0.5061
Jankowska	0.2814	0.5381	0.4738
Flekova	0.2785	0.5343	0.5287
Weren	0.2564	0.5044	0.5099
Sapkota	0.2471	0.4781	0.5415
De-Arteaga	0.2450	0.4998	0.4885
Moreau	0.2395	0.4941	0.4824
baseline	0.1650	0.5000	0.3333
Gopal Patra	0.1574	0.5683	0.2895
Cagnina	0.0741	0.5040	0.1234

Spanish			
Team	Total	Gender	Age
Santosh	0.4208	0.6473	0.6430
Pastor L.	0.4158	0.6299	0.6558
Cruz	0.3897	0.6165	0.6219
Flekova	0.3683	0.6103	0.5966
Ladra	0.3523	0.6138	0.5727
De-Arteaga	0.3145	0.5627	0.5429
Kern	0.3134	0.5706	0.5375
Yong Lim	0.3120	0.5468	0.5705
Sapkota	0.2934	0.5116	0.5651
Pavan	0.2824	0.5000	0.5643
Jankowska	0.2592	0.5846	0.4276
Meina	0.2549	0.5287	0.4930
Gillam	0.2543	0.4784	0.5377
Moreau	0.2539	0.4967	0.5049
Weren	0.2463	0.5362	0.4615
Cagnina	0.2339	0.5516	0.4148
Caurcel Diaz	0.2000	0.5000	0.4000
H. Farias	0.1757	0.4982	0.3554
baseline	0.1650	0.5000	0.3333
Aleman	0.1638	0.5526	0.2915
Seifeddine	0.0287	0.5455	0.0512
Gopal Patra	–	–	–

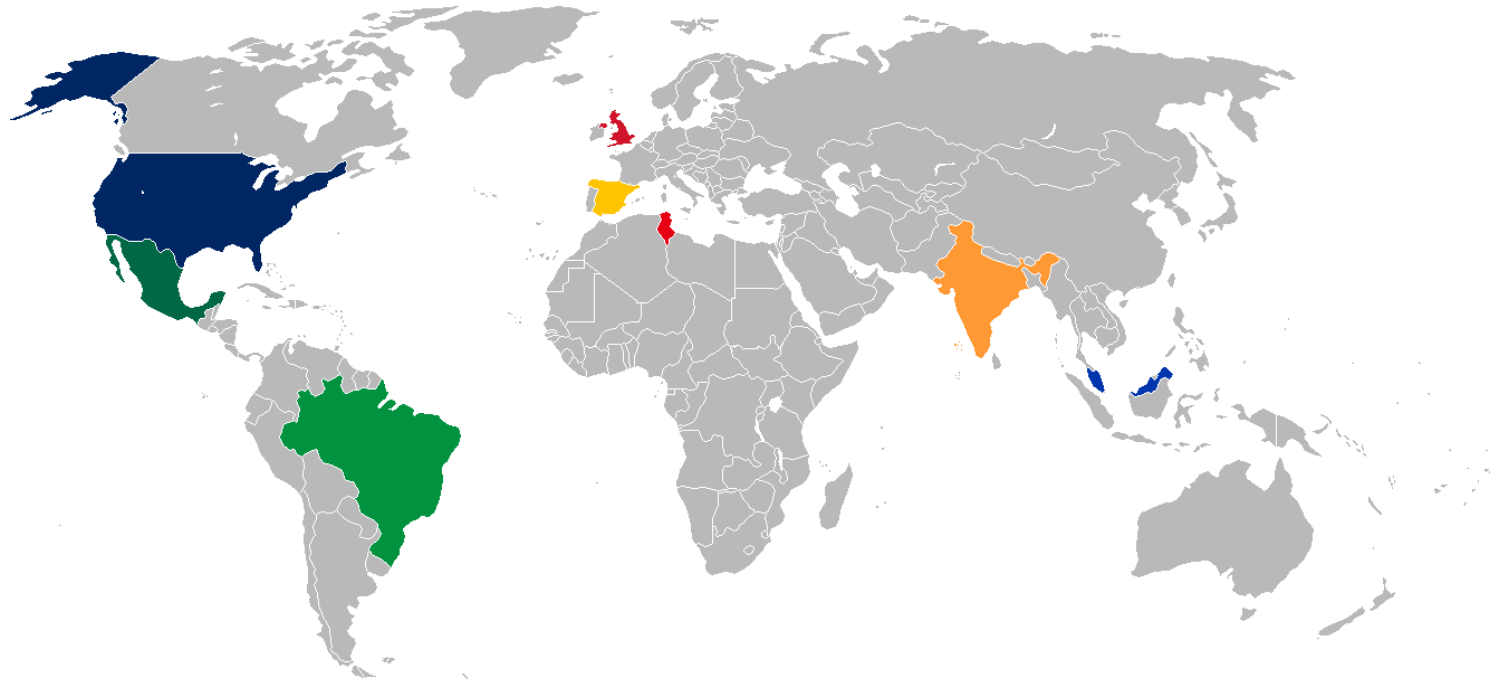
Features

- Stylistic: frequency of punctuation marks, capital letters,...
- Part of Speech
- Readability measures
- Dictionary-based words, topic-based words
- Collocations
- Character or word n-grams
- Slang words, character flooding
- Emoticons
- Emotion words

F. Rangel, P. Rosso, M. Koppel, E. Stamatatos, and G. Inches. Overview of the Author Profiling Task at PAN 2013 - Notebook for PAN at CLEF 2013. CEUR Workshop Proceedings Vol. 1179. 2013.

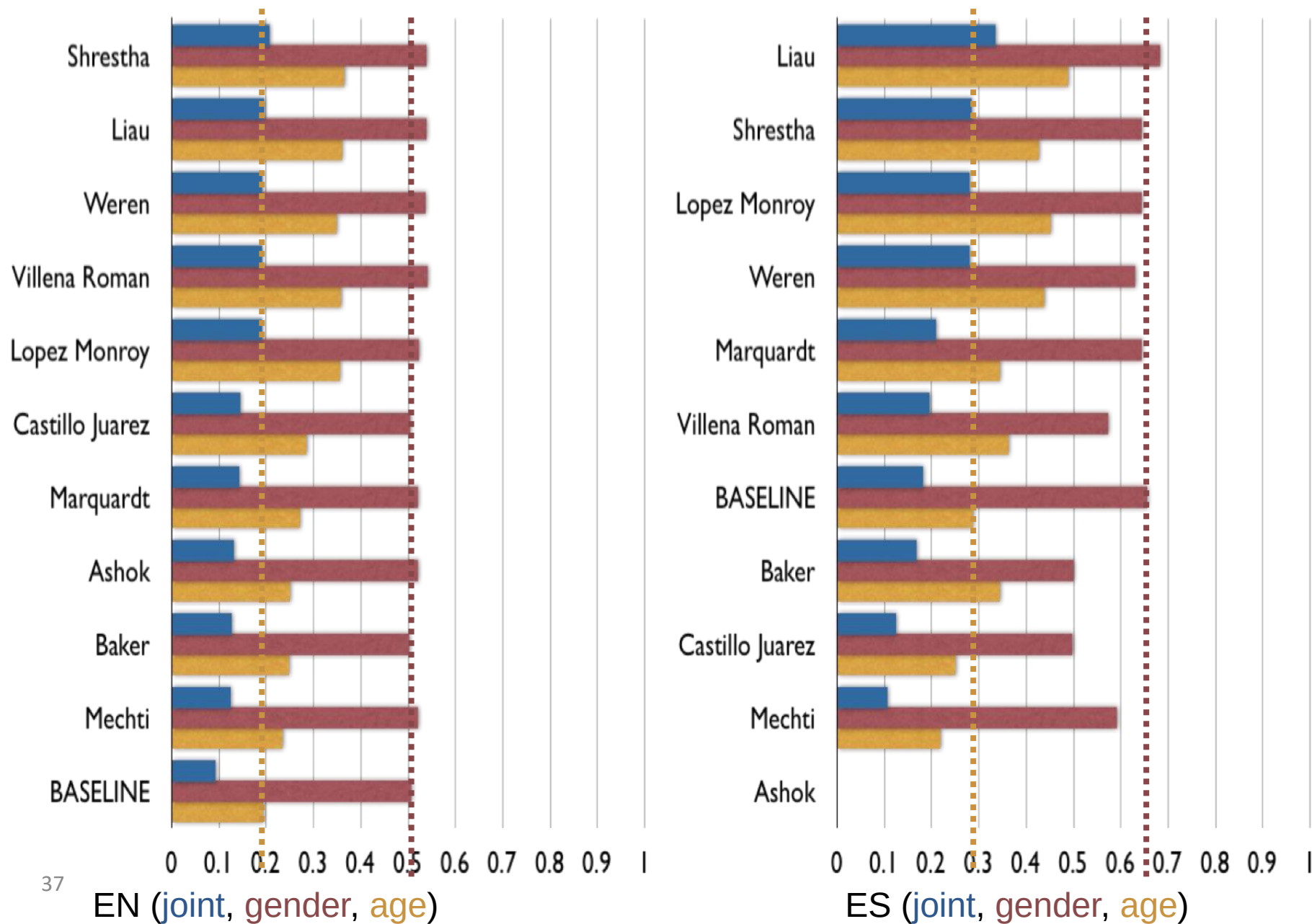
Author profiling **PAN** @CLEF 2014

- Teams submitting results: 10

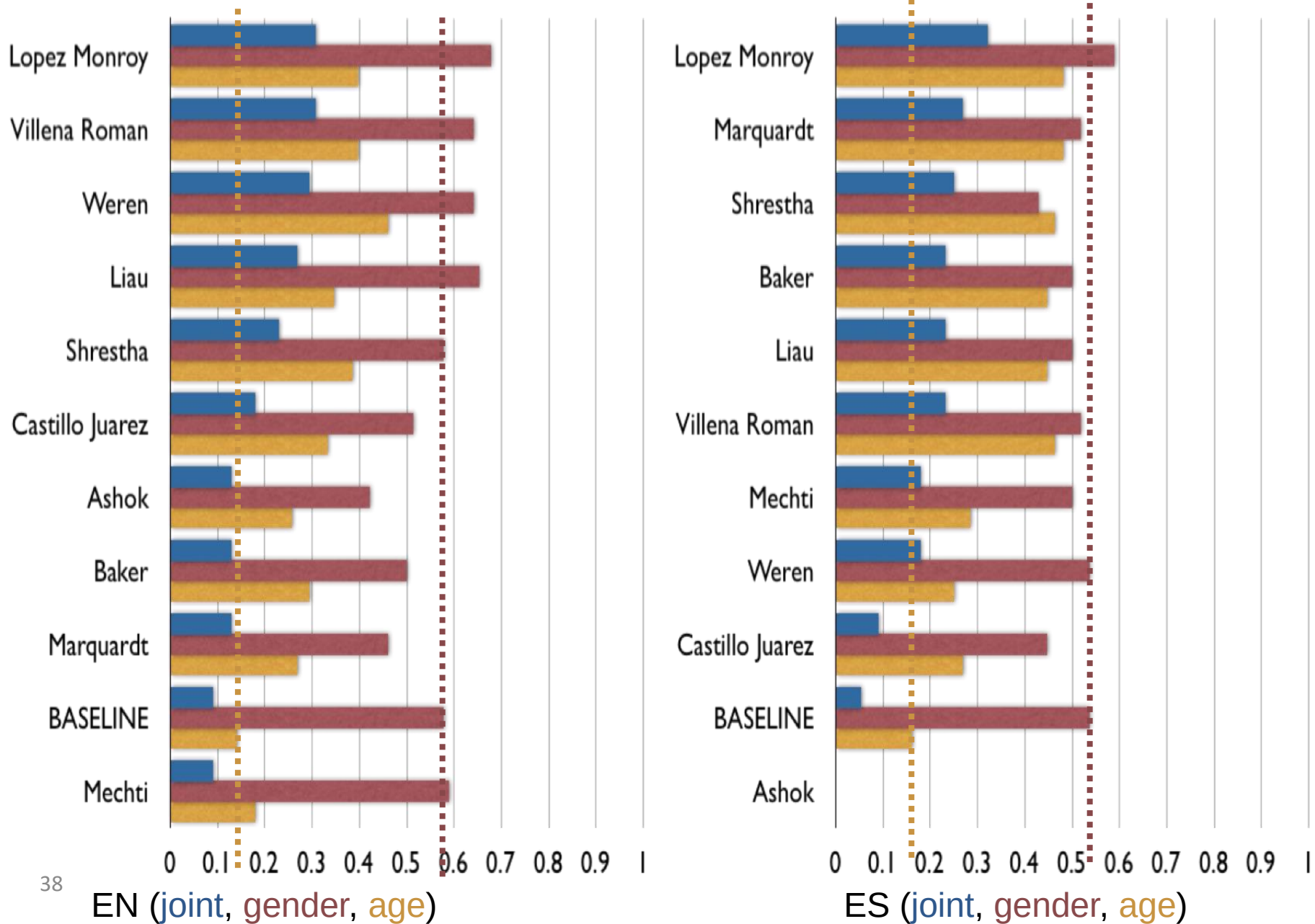


- **Social media + blogs + Twitter + reviews**
- **Age classes: 18-24, 25-34, 35-49, 50-64, 65+**
- **Languages: English and Spanish**

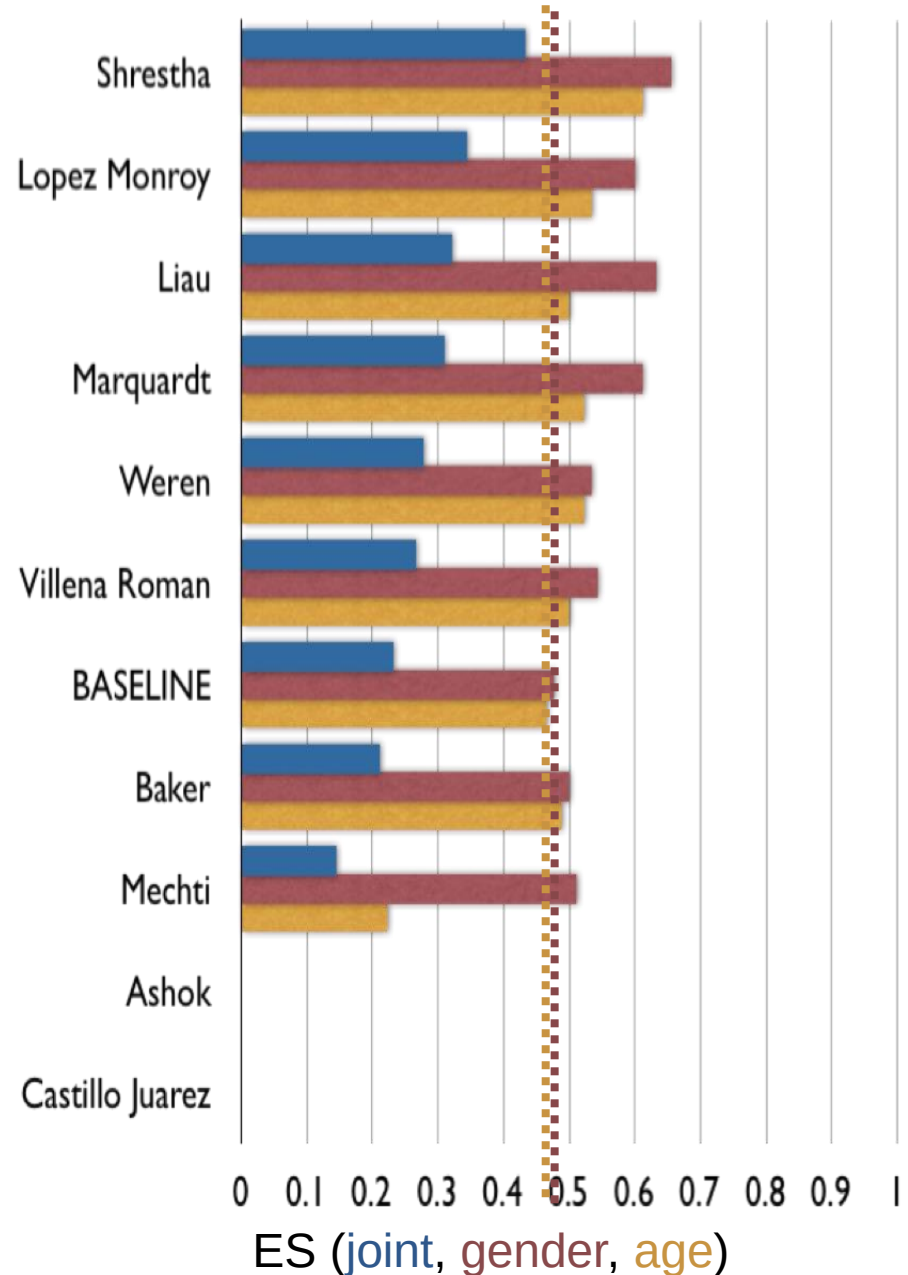
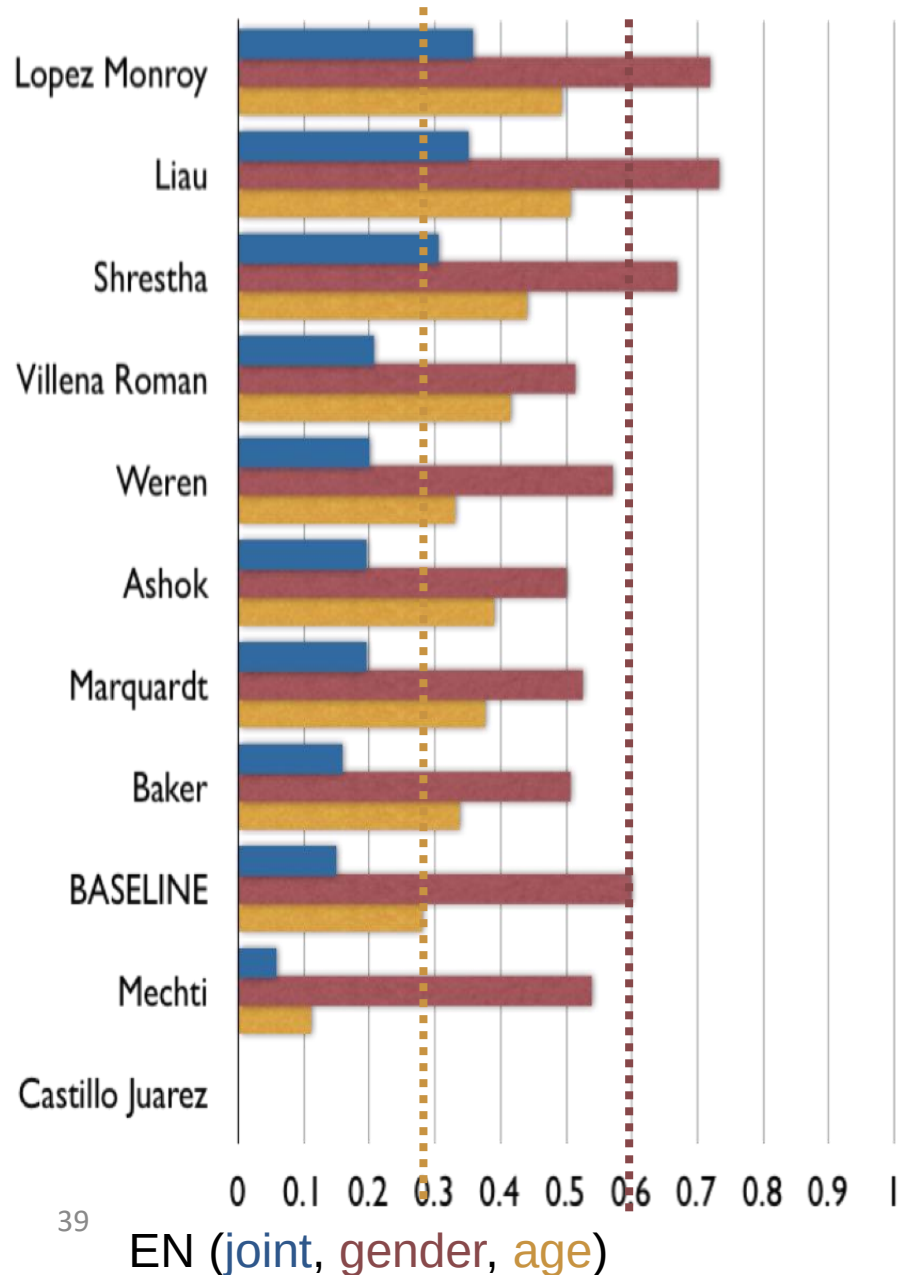
Results in social media



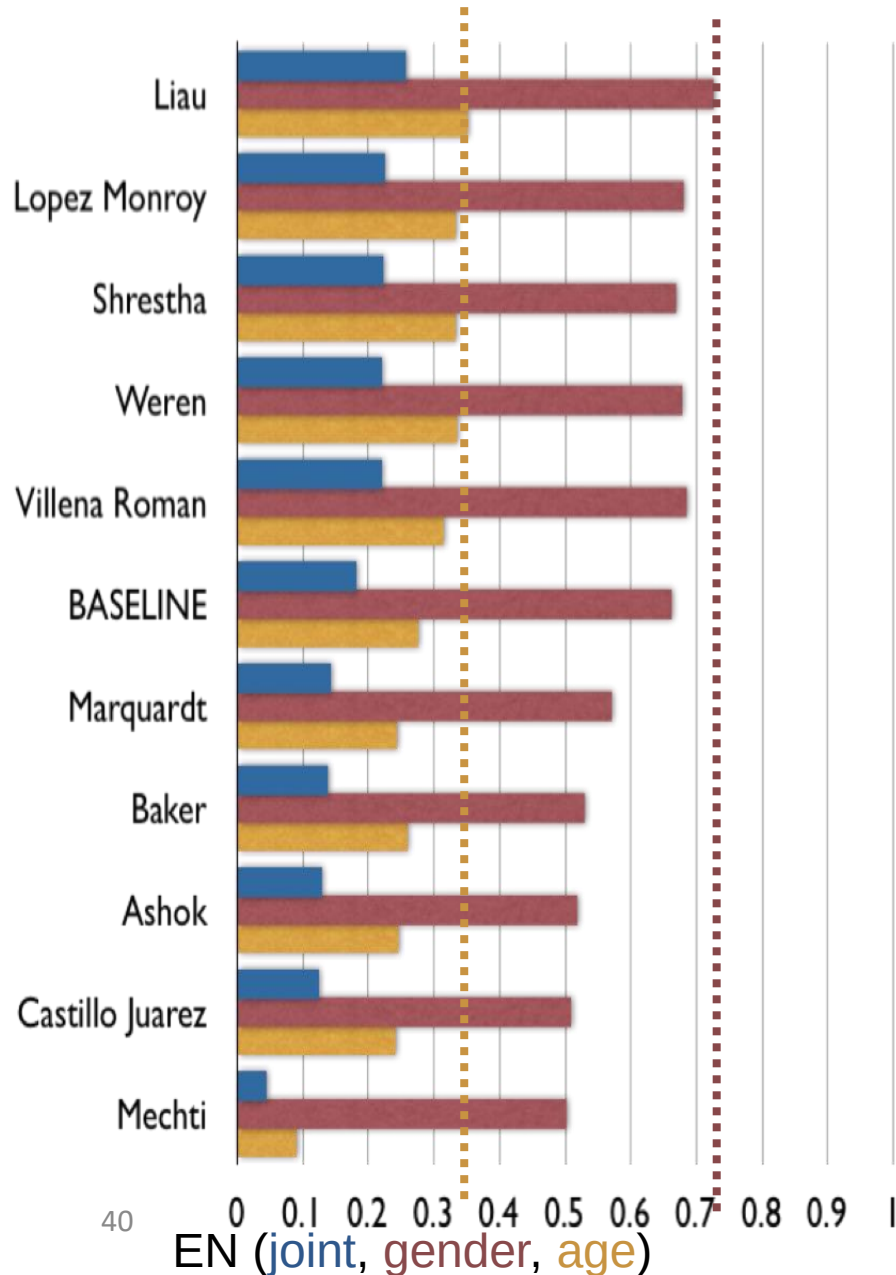
Results in blogs



Results in Twitter

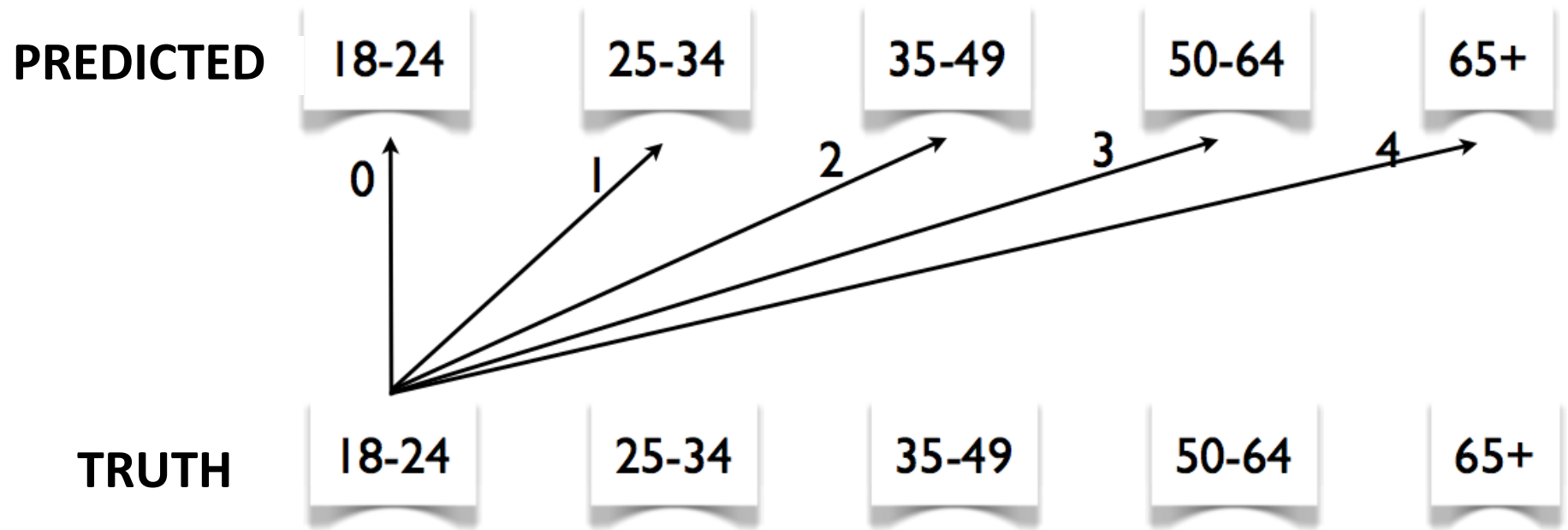


Results in reviews

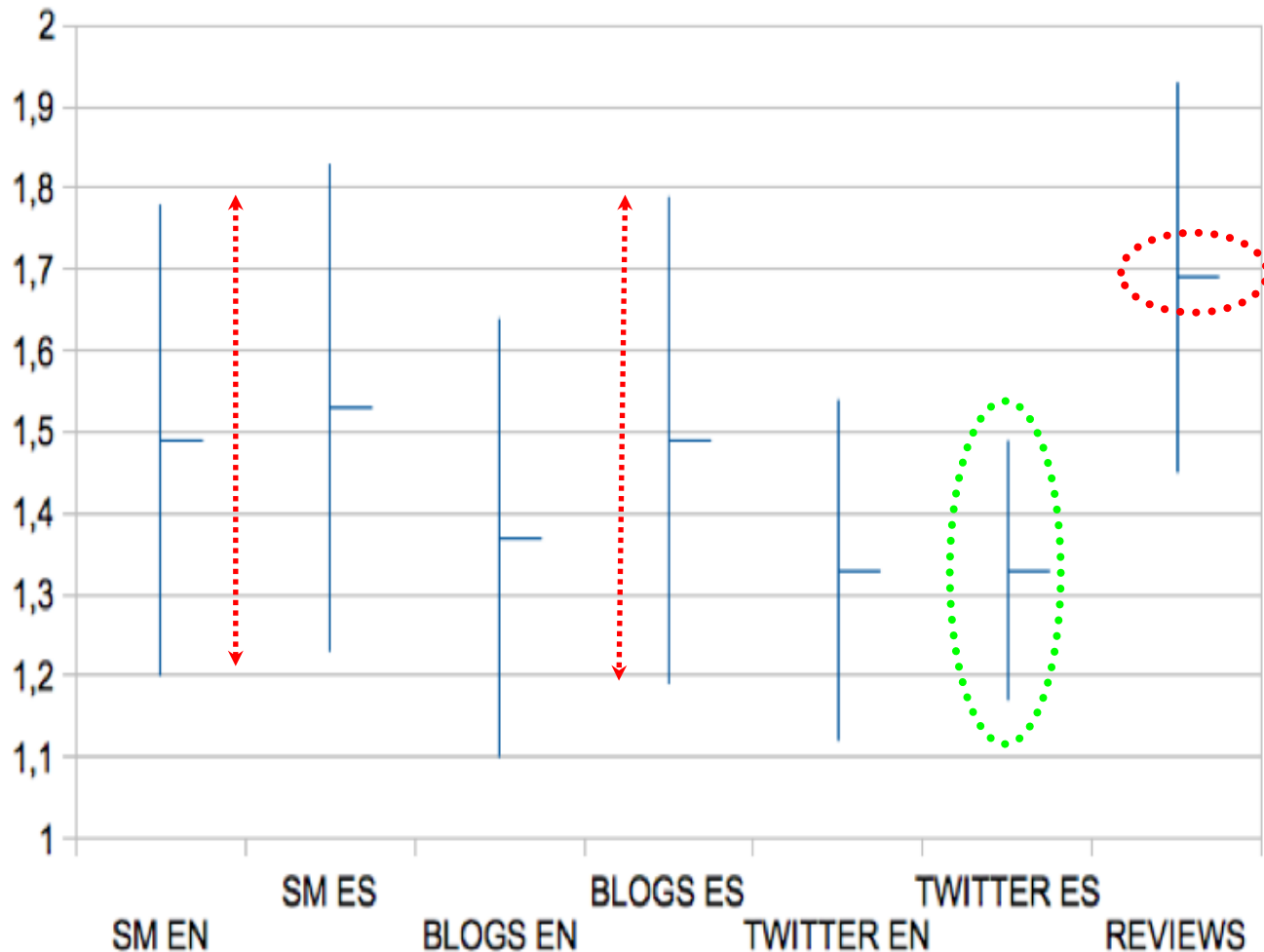


The problem of
deceptive opinions

Distances in misclassified age



Distances in misclassified age



Twitter: more spontaneous way to communicate

Approaches: features

- Similar features than in 2013:
content (bag of words, word n-grams) and stylistic
- frequency of words related to different psycholinguistic concepts, extracted from: LIWC and MRC psycholinguistic database

F. Rangel, P. Rosso, I. Chugur, M. Potthast, M. Trenkman, B. Stein, B. Verhoeven, and W. Daelemans. Overview of the 2nd Author Profiling Task at PAN 2014—Notebook for PAN at CLEF 2014. CEUR Workshop Proceedings Vol. 1180, pp. 898-927, 2014.

EmoGraph based discourse analysis

Rangel F., Rosso P. On the impact of emotions on author profiling.
Information, Processing & Management, 52(1): 73-92, 2016

He estado tomando cursos en línea sobre temas valiosos que disfruto estudiando y que podrían ayudarme a hablar en público.

(I) have been taking online courses about valuable subjects that **(I)** enjoy studying and that might help me to speak in public.

VAIP1S0

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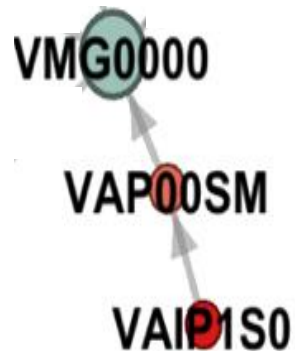
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VAP00SM
VAIP1S0



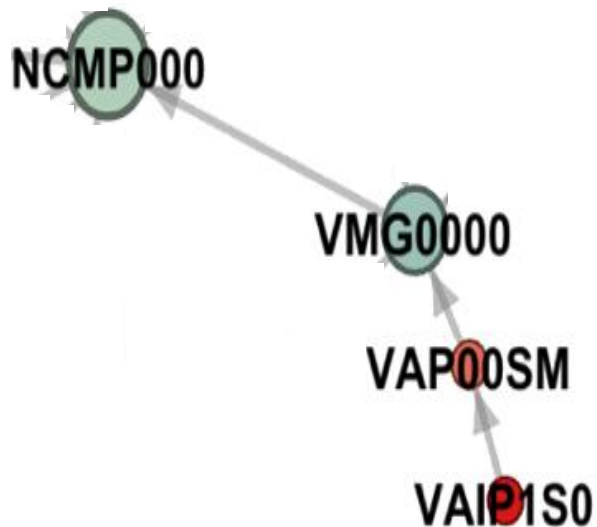
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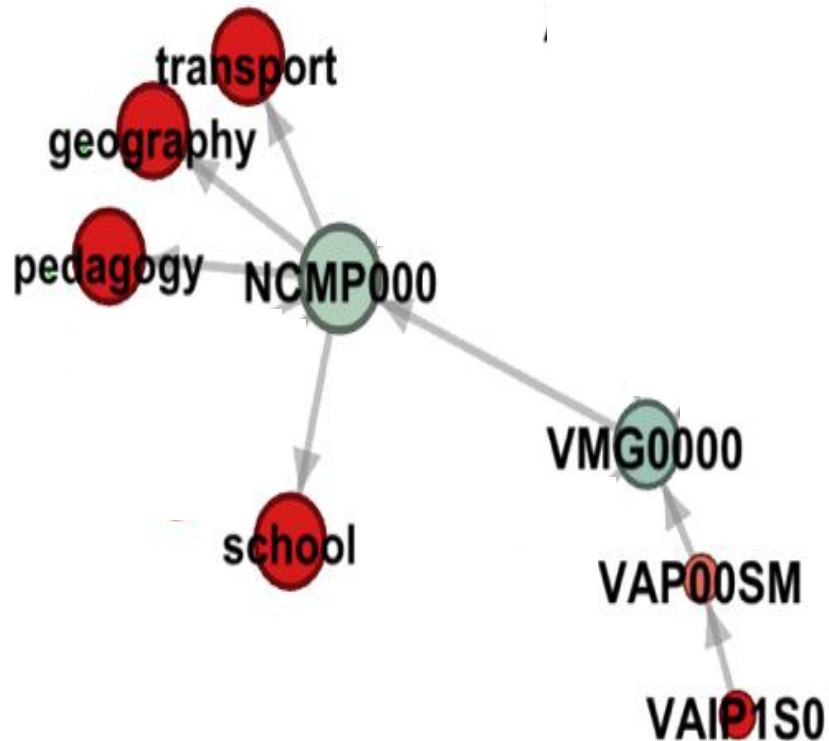
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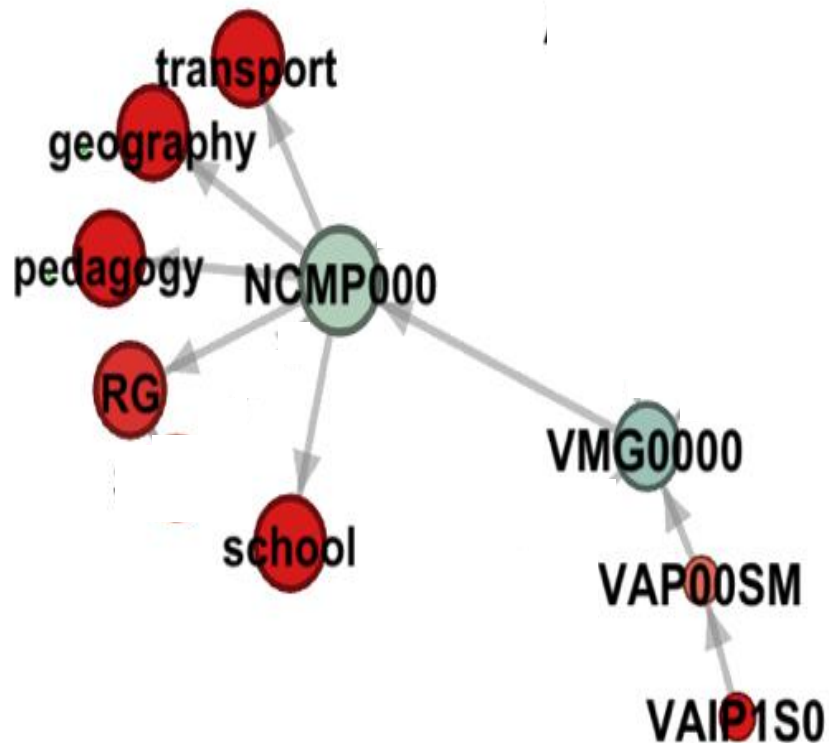
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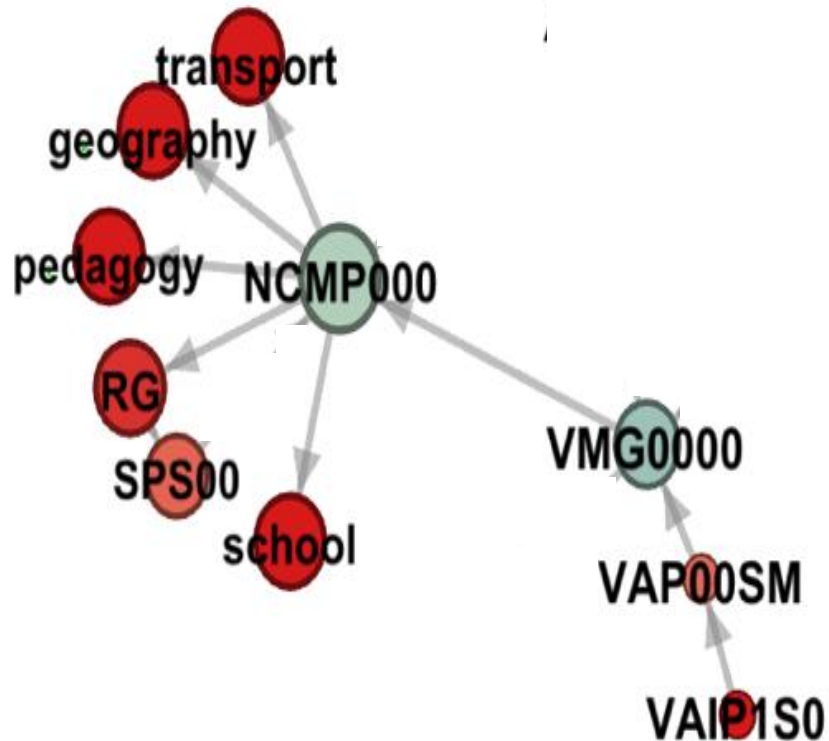
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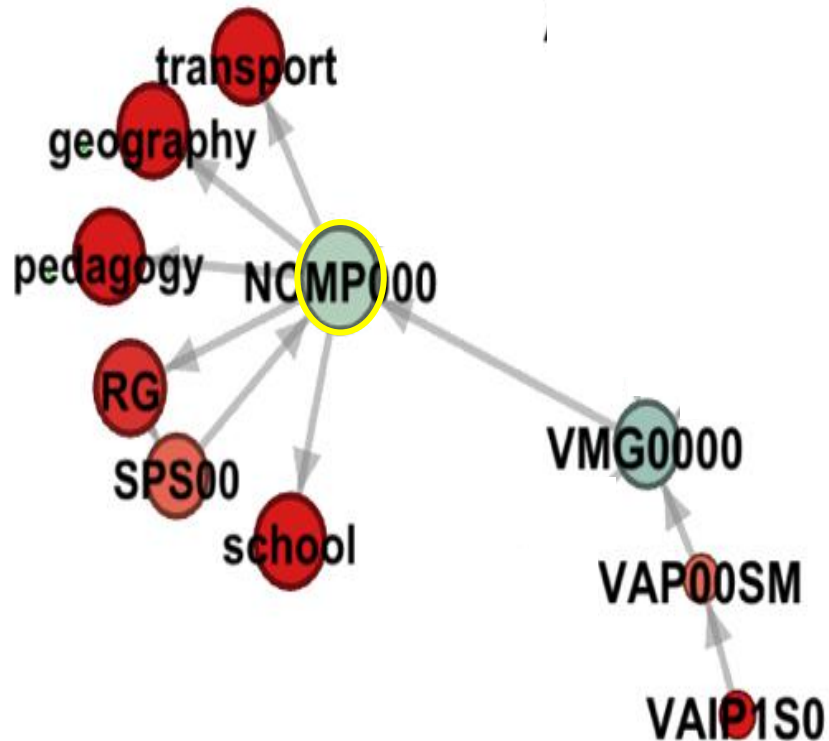
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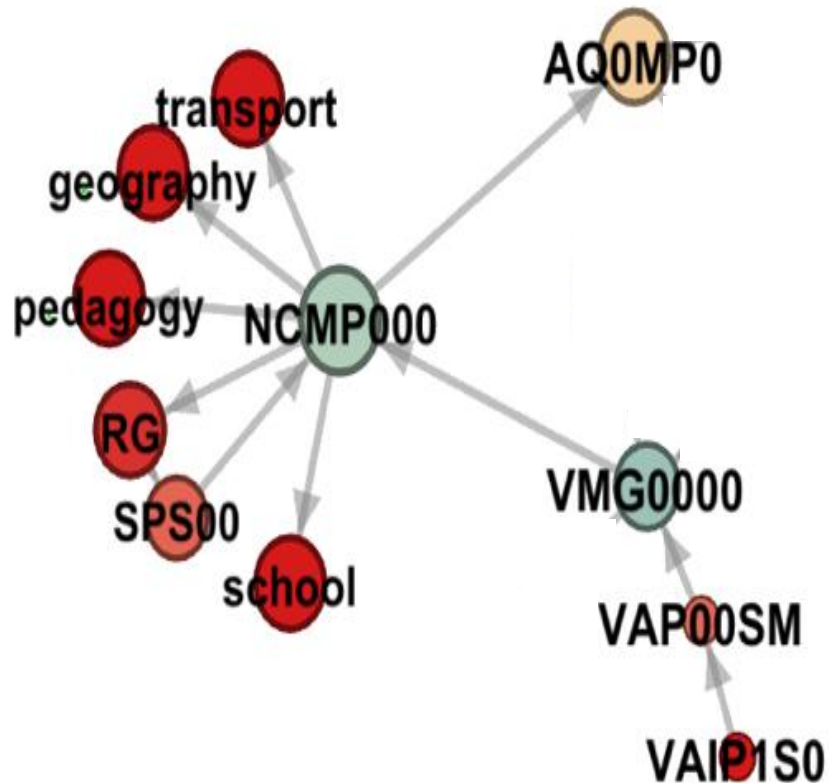
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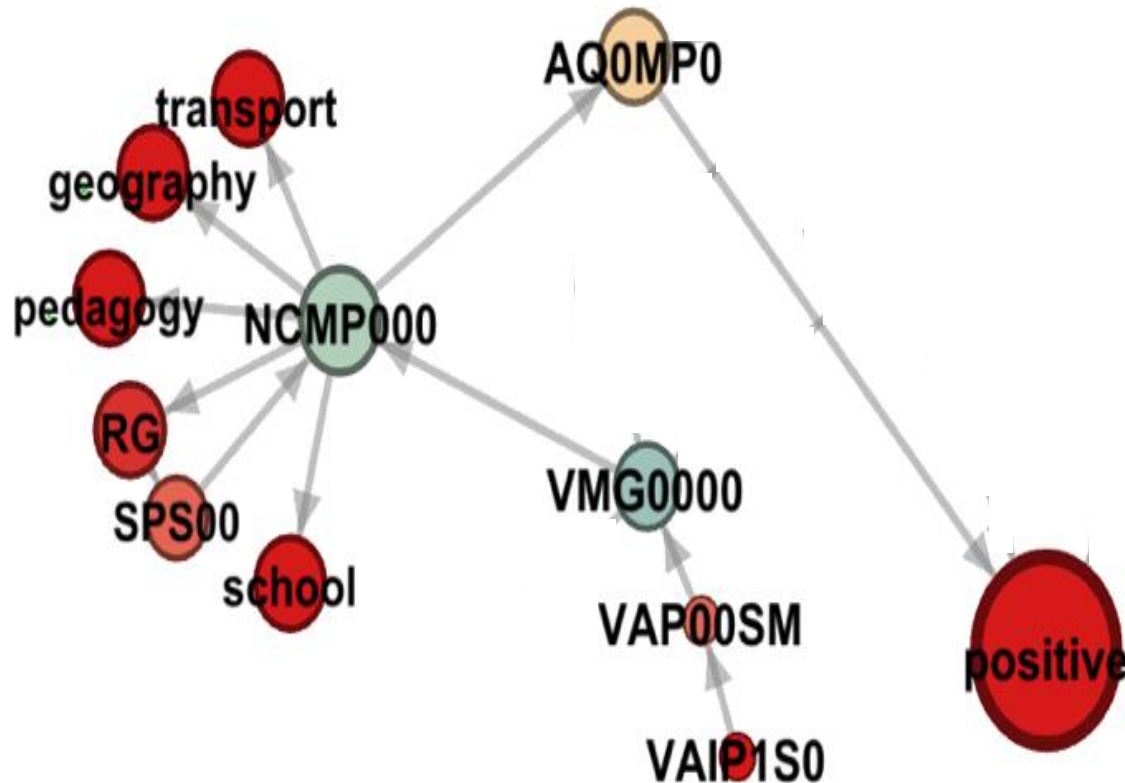
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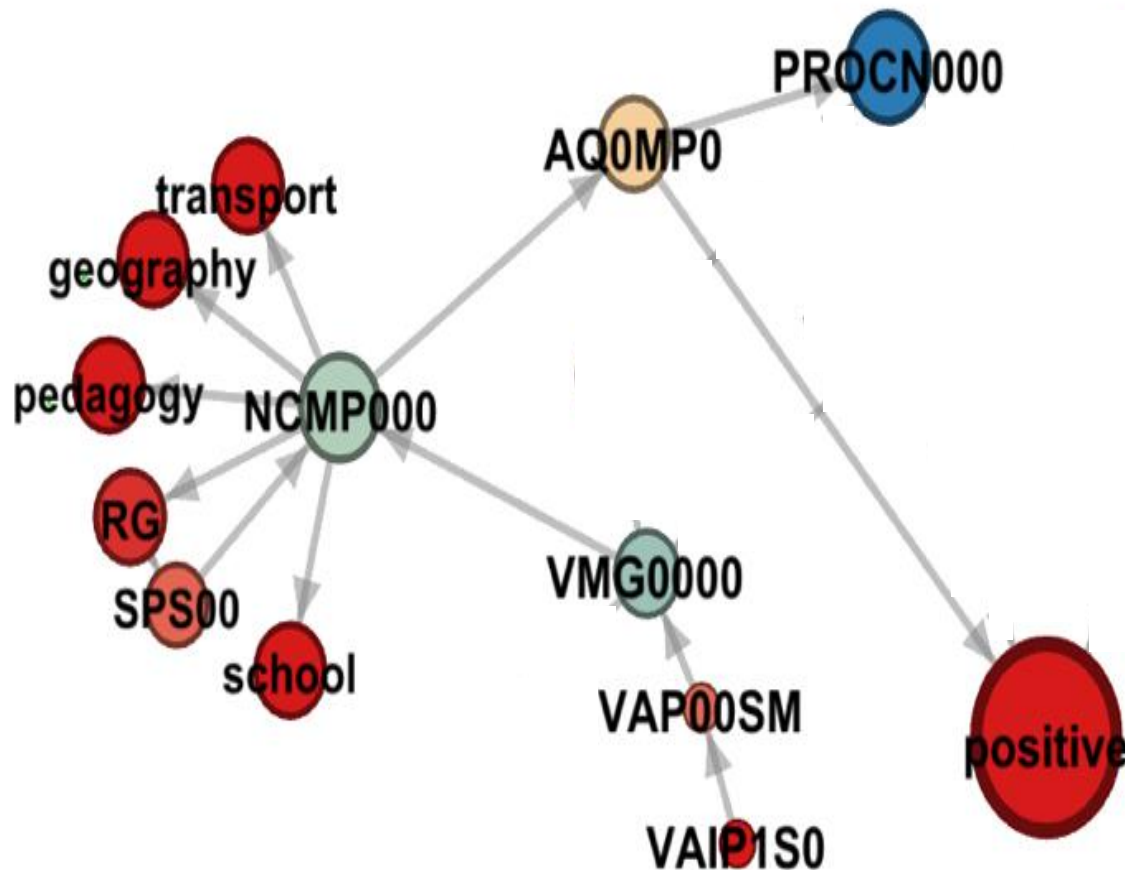
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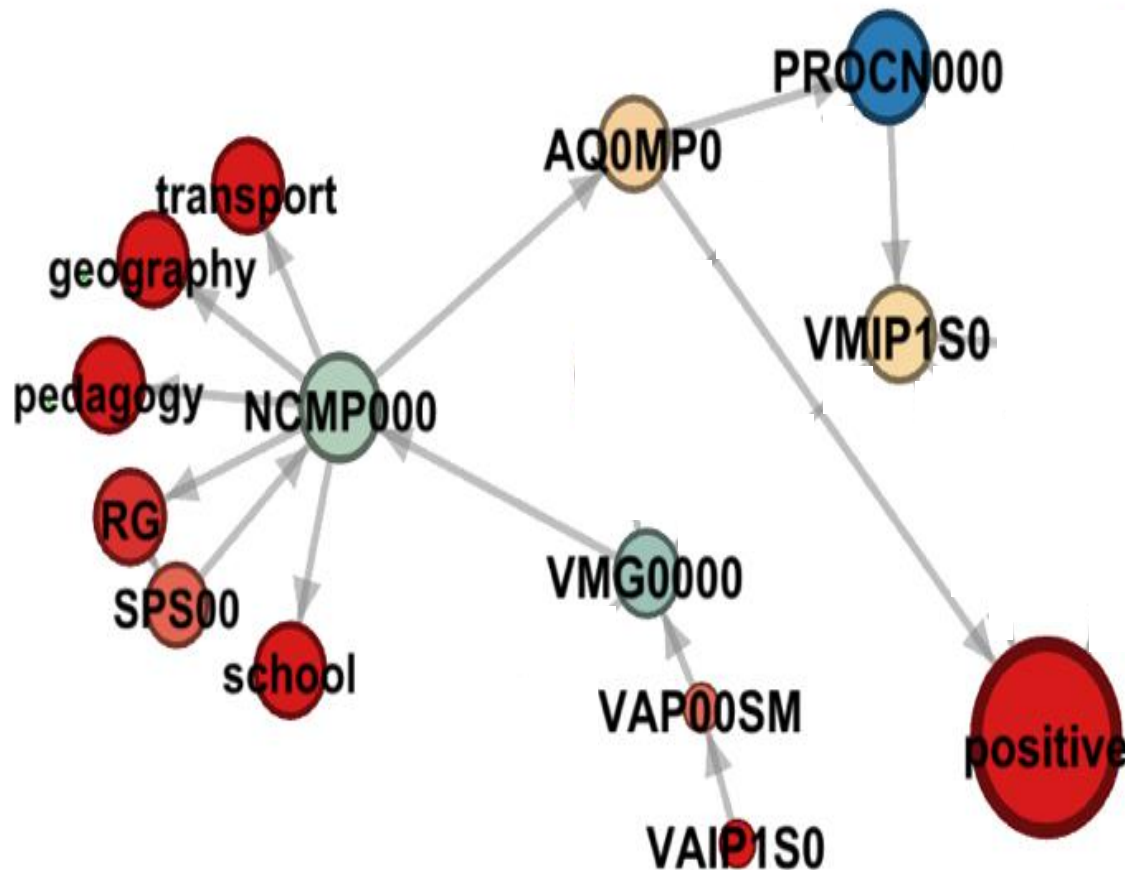
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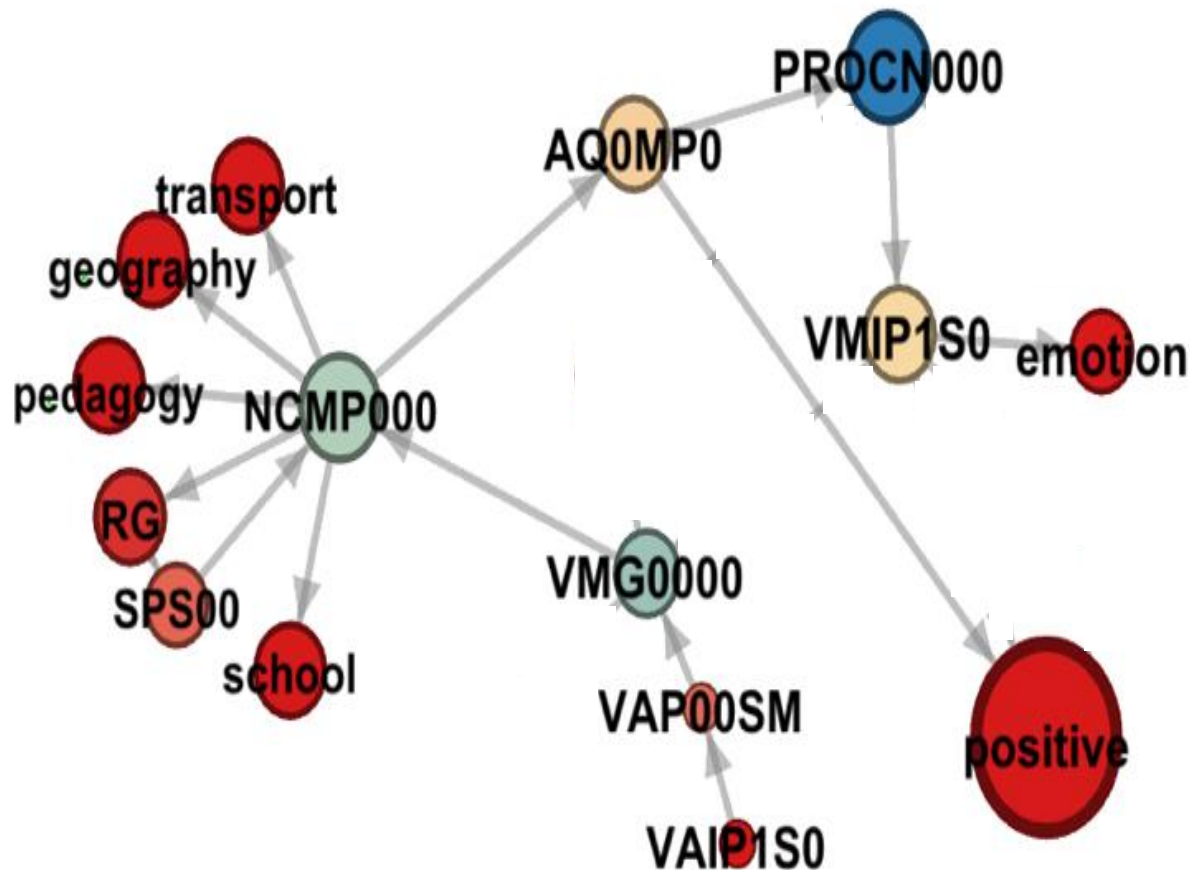
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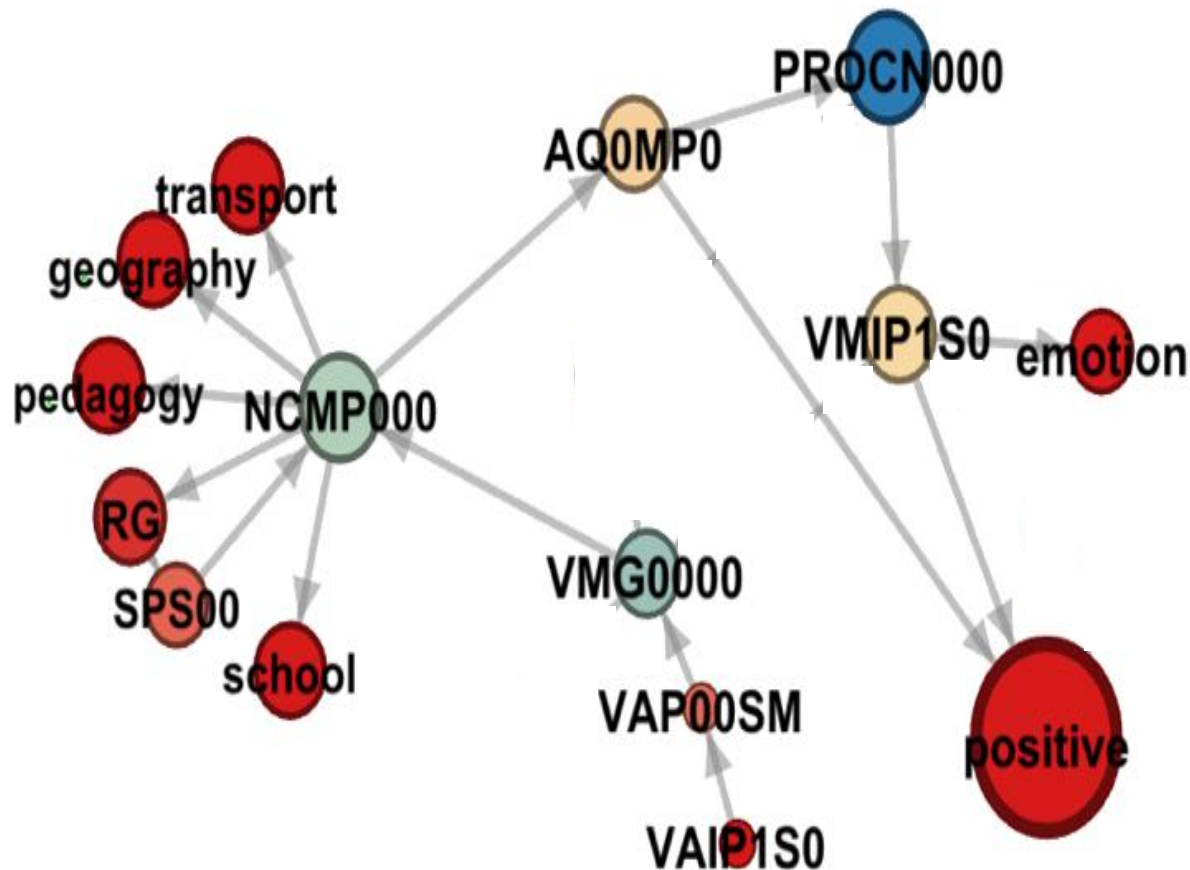
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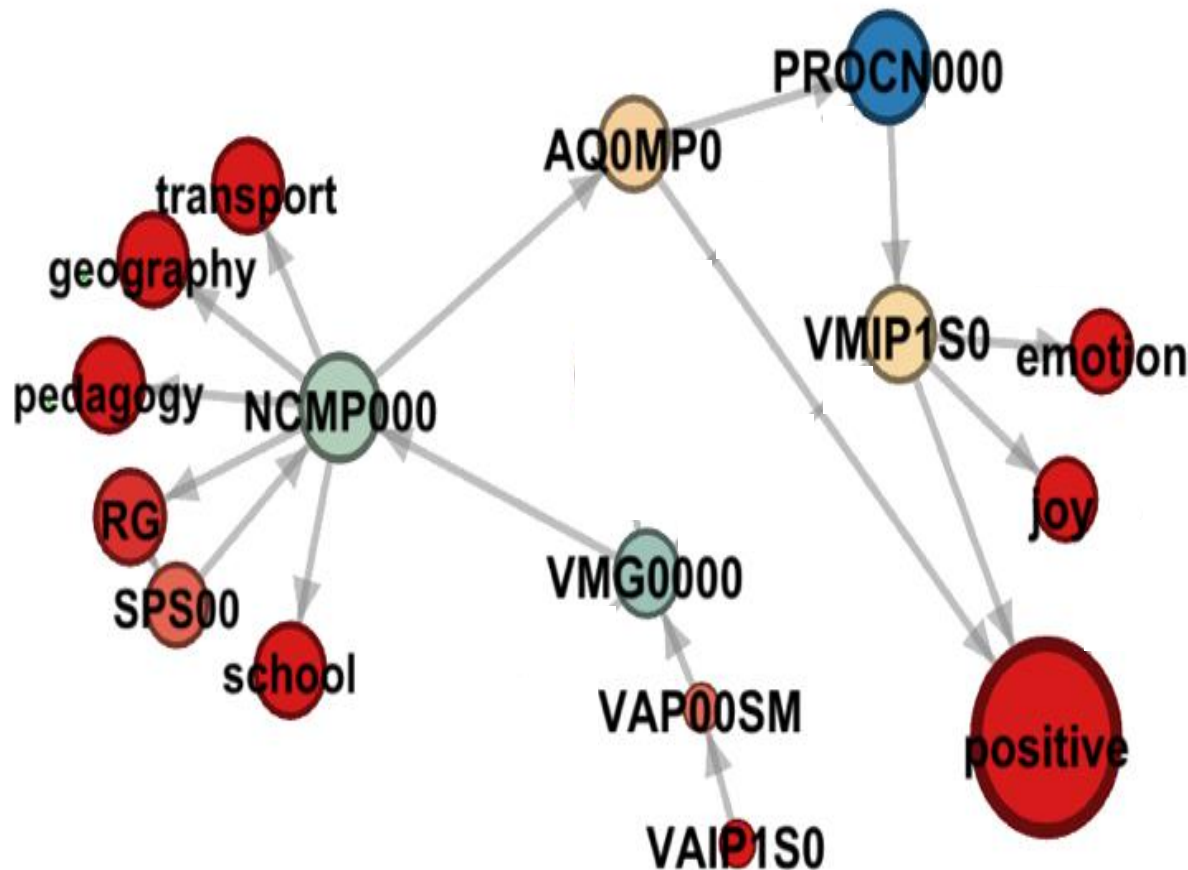
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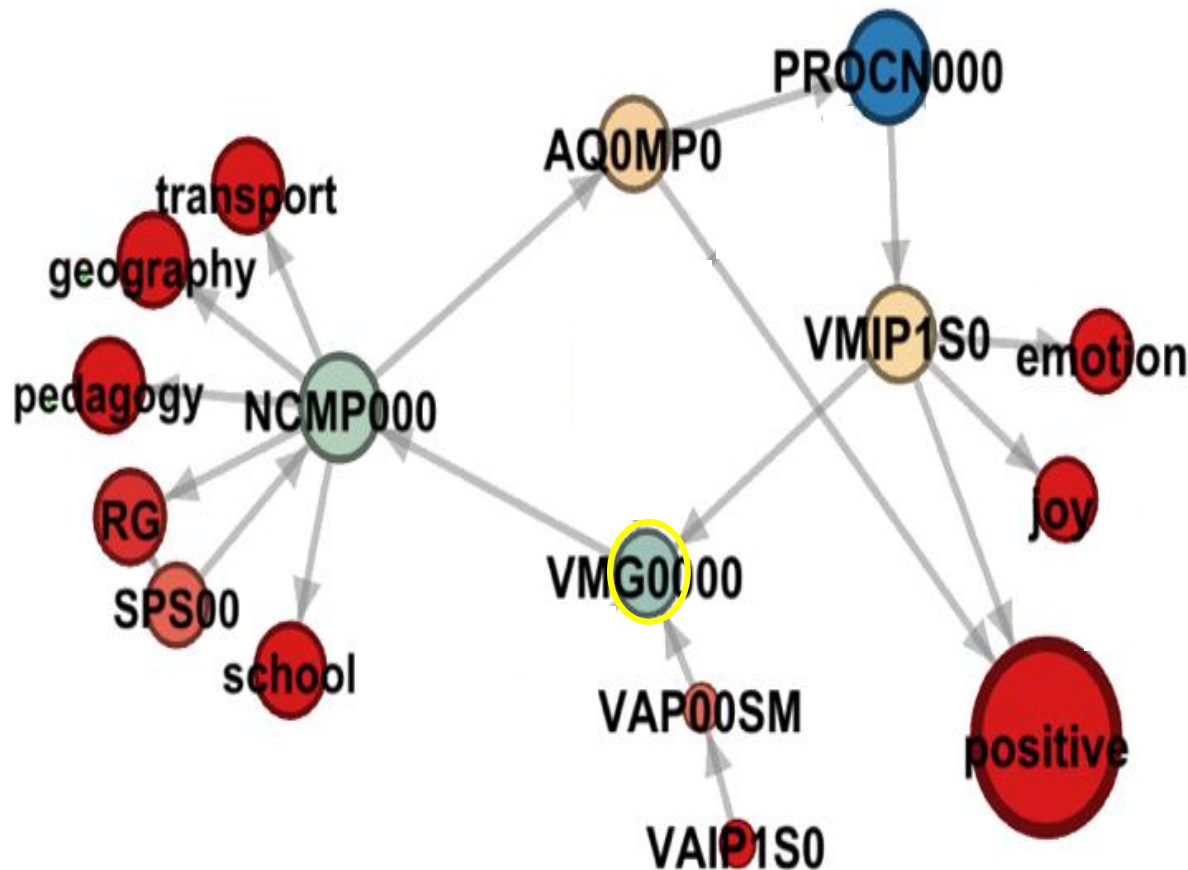
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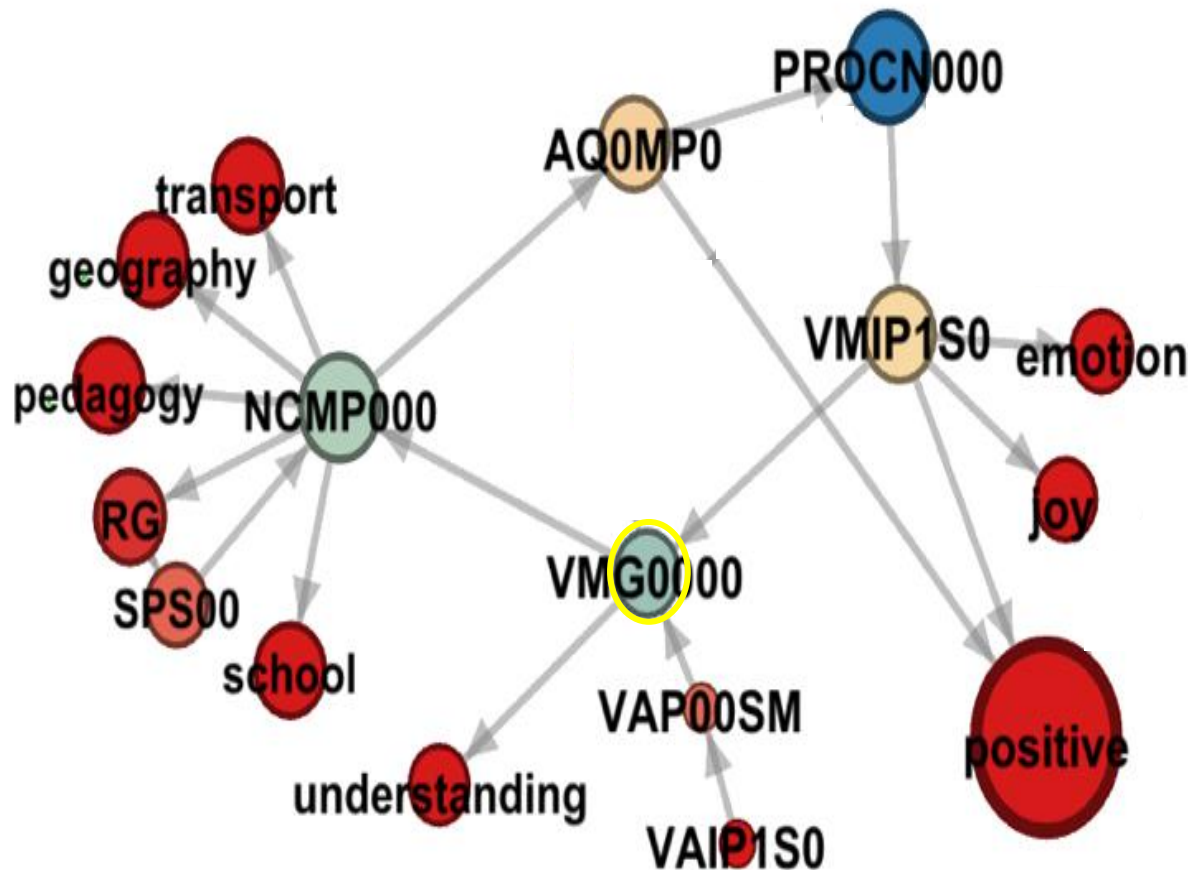
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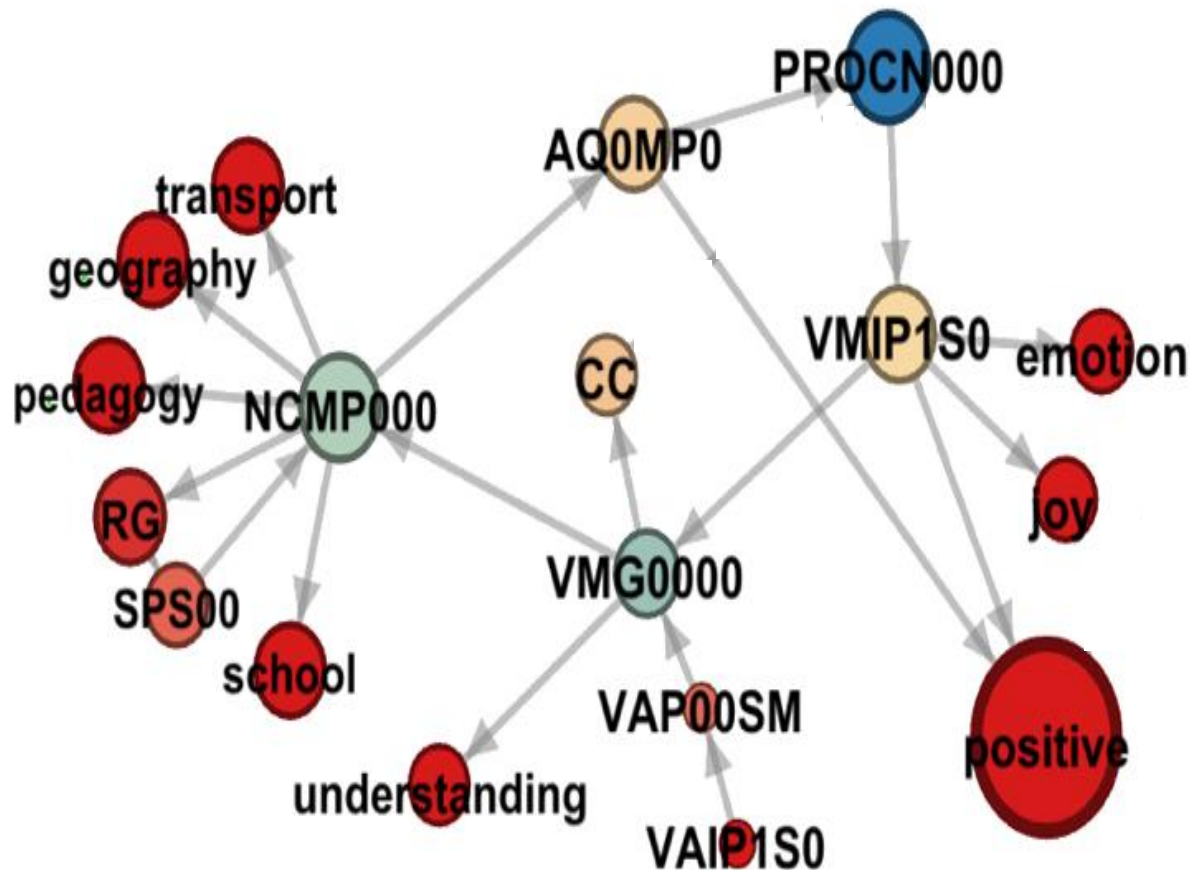
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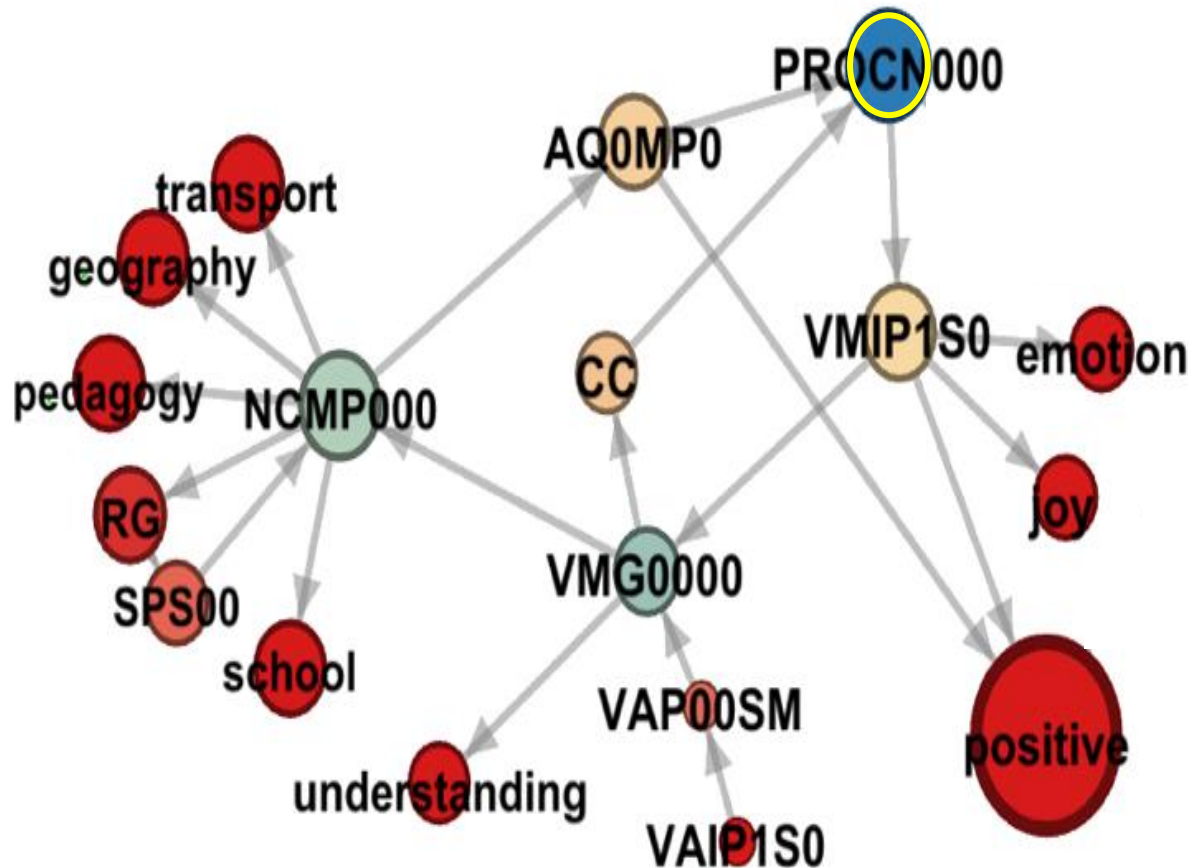
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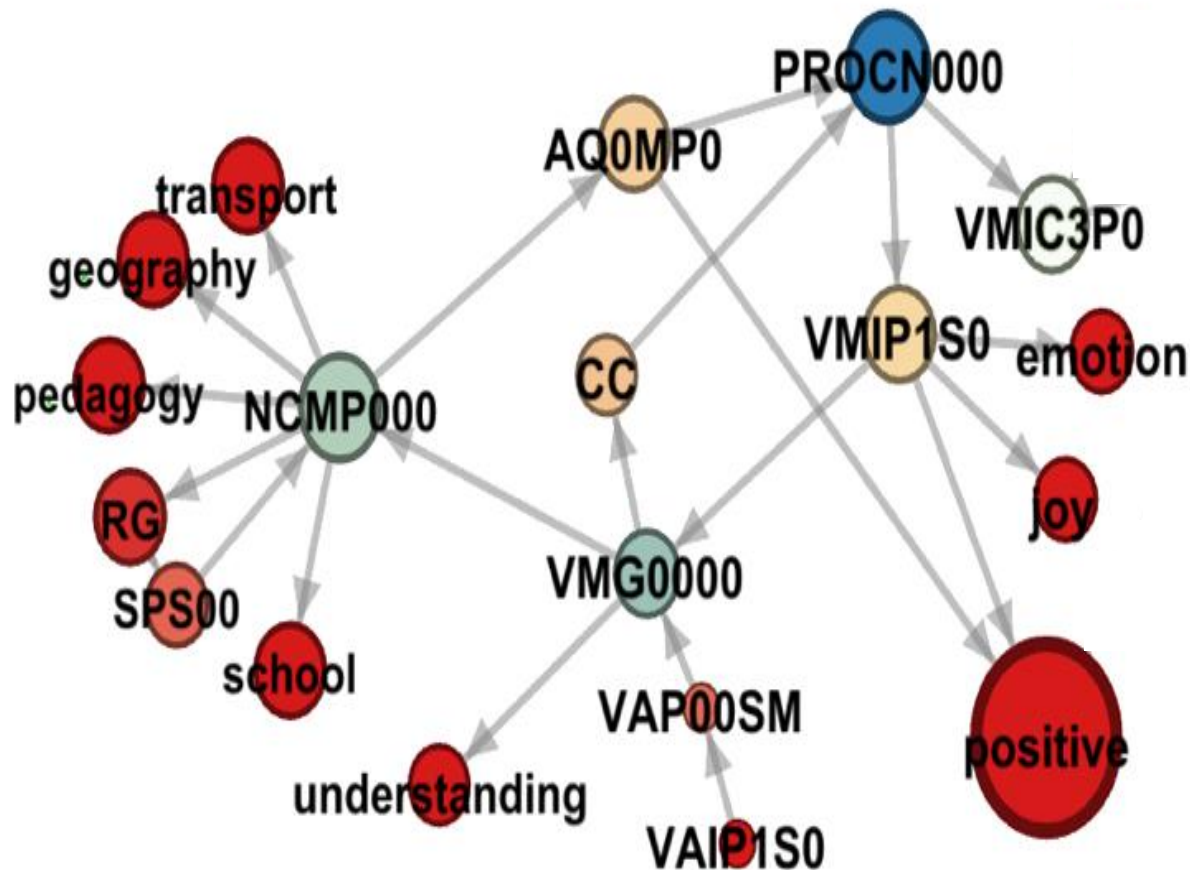
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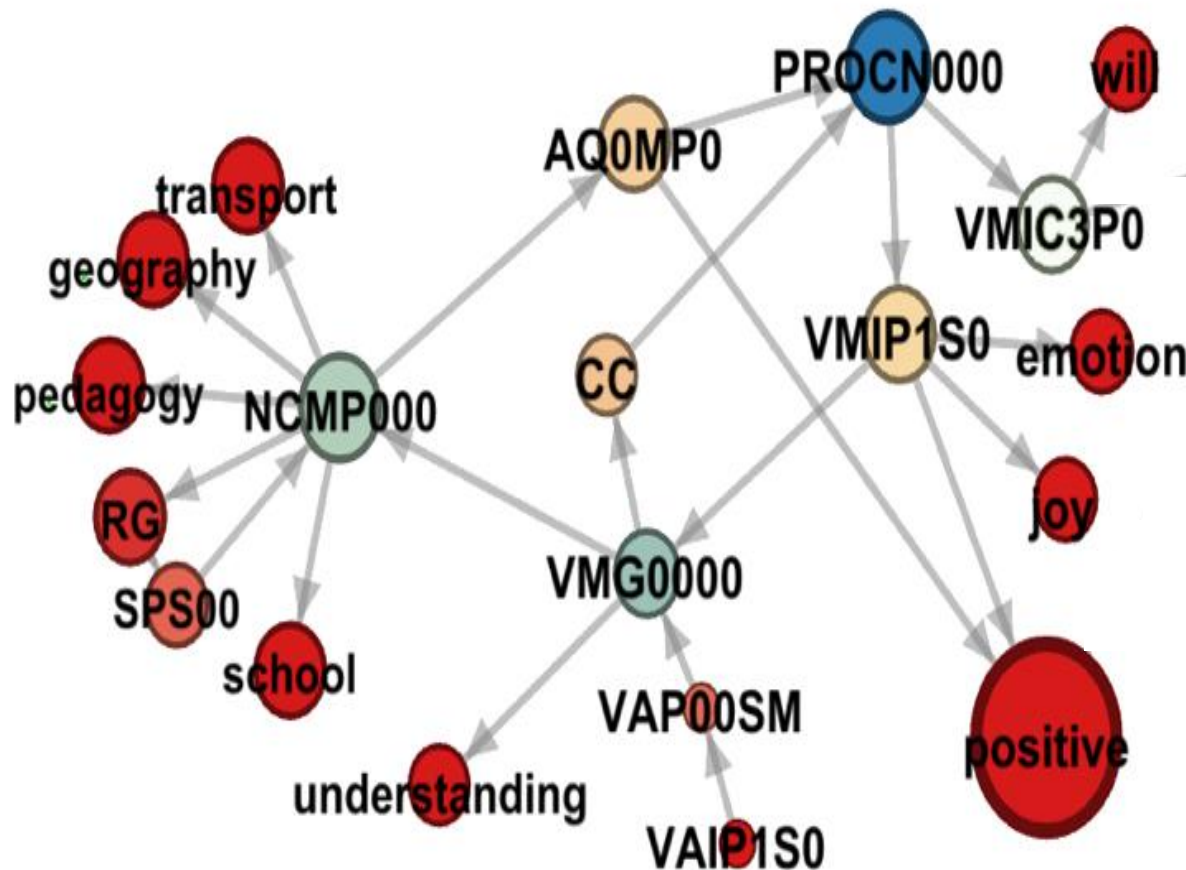
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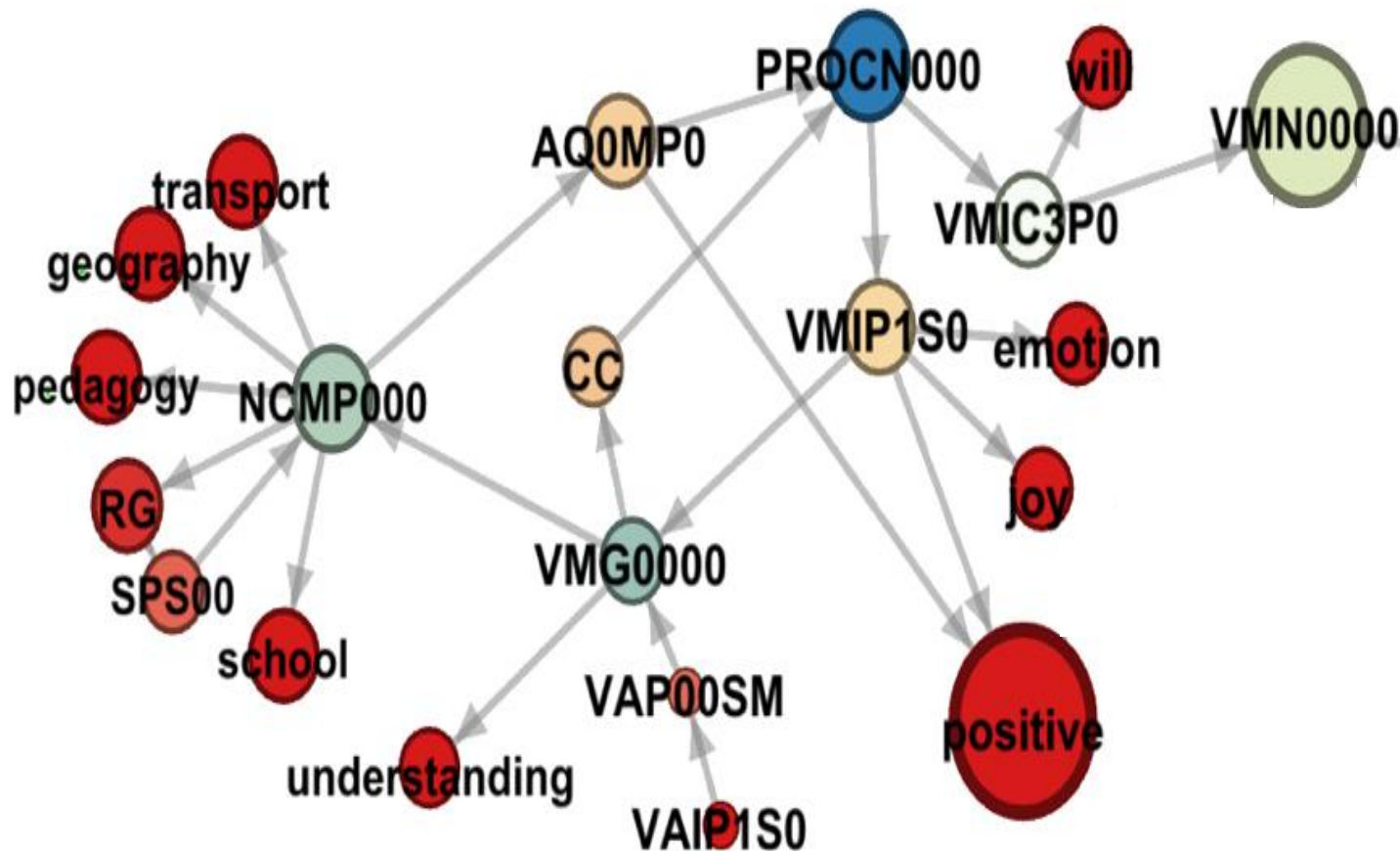
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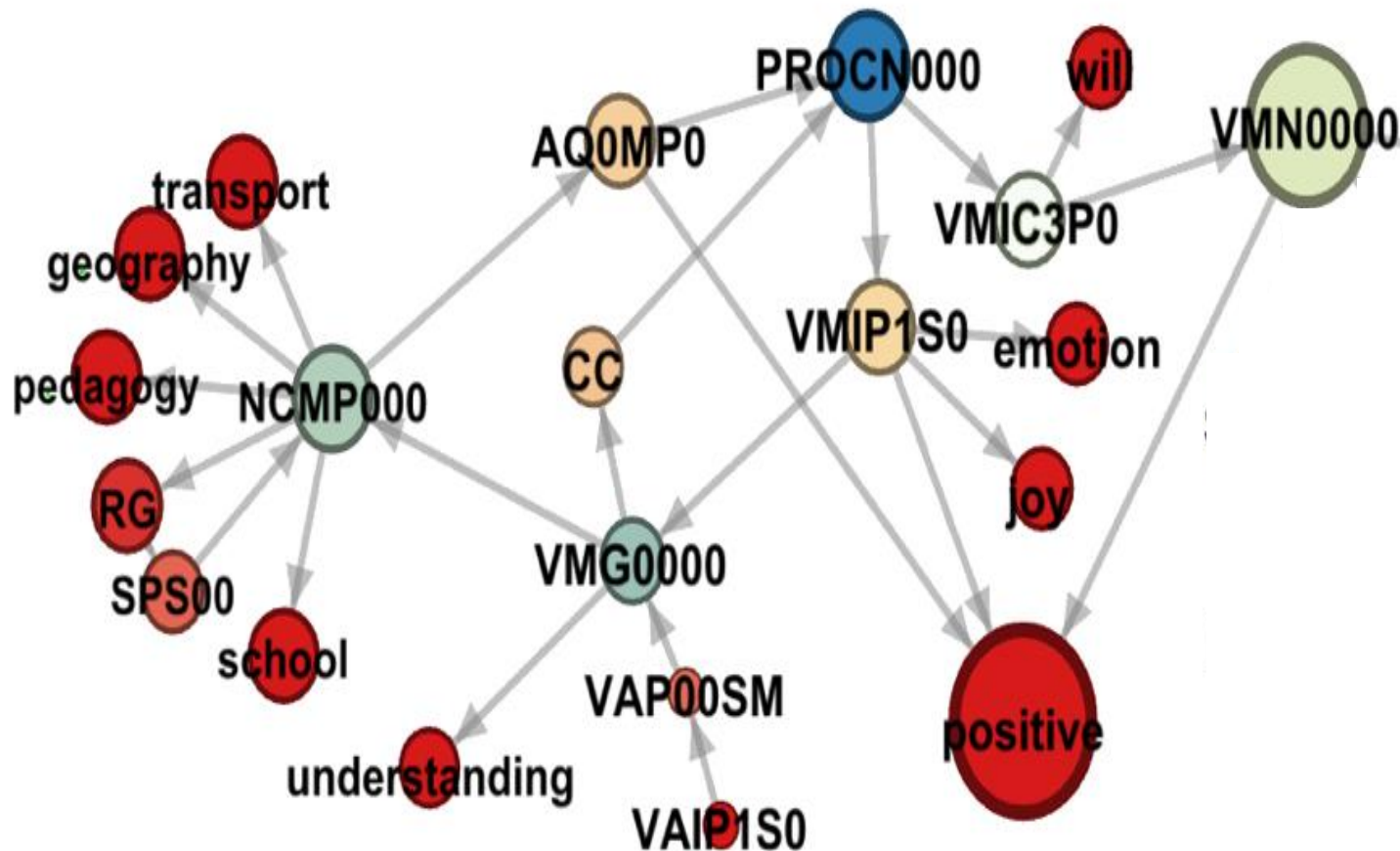
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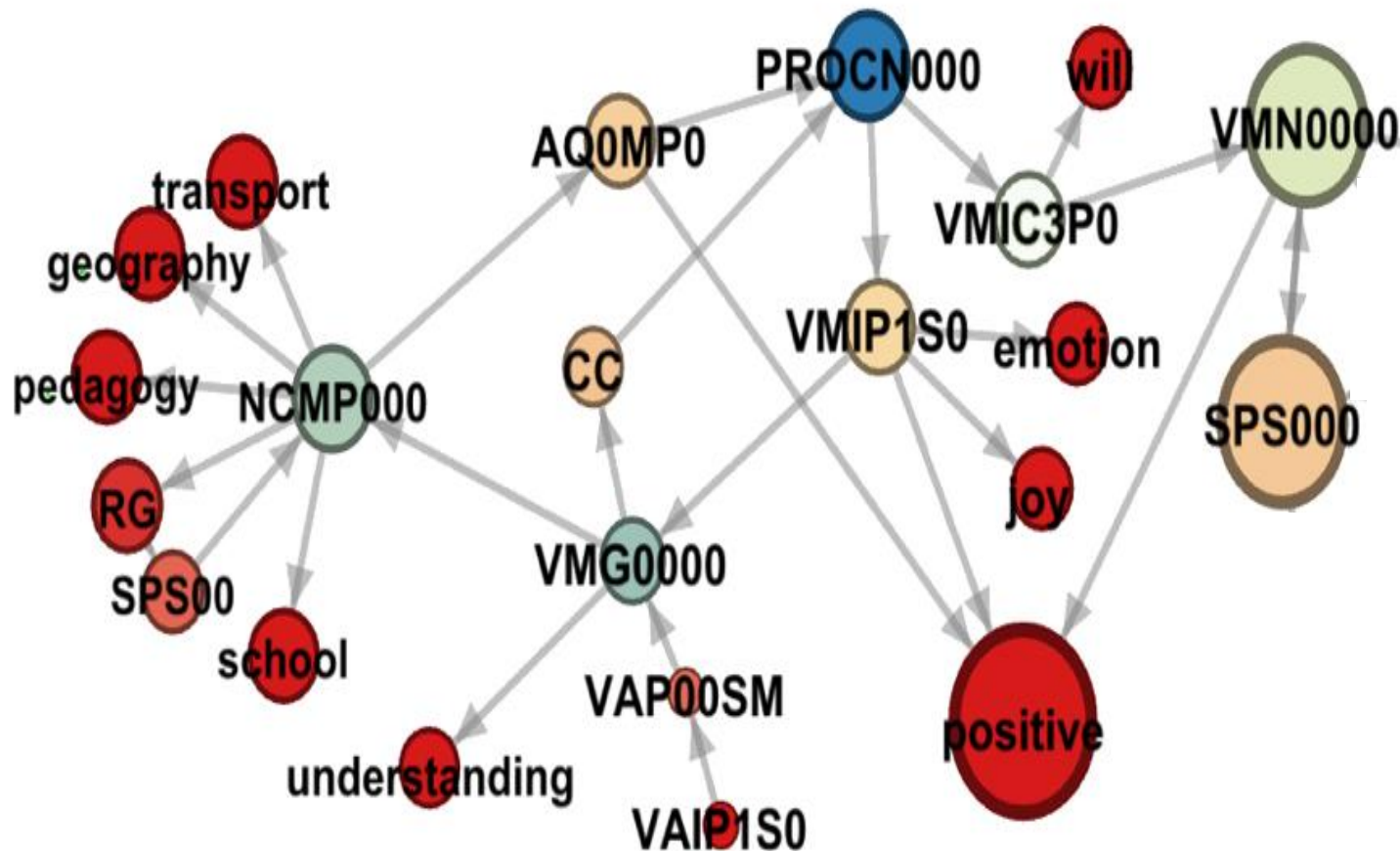
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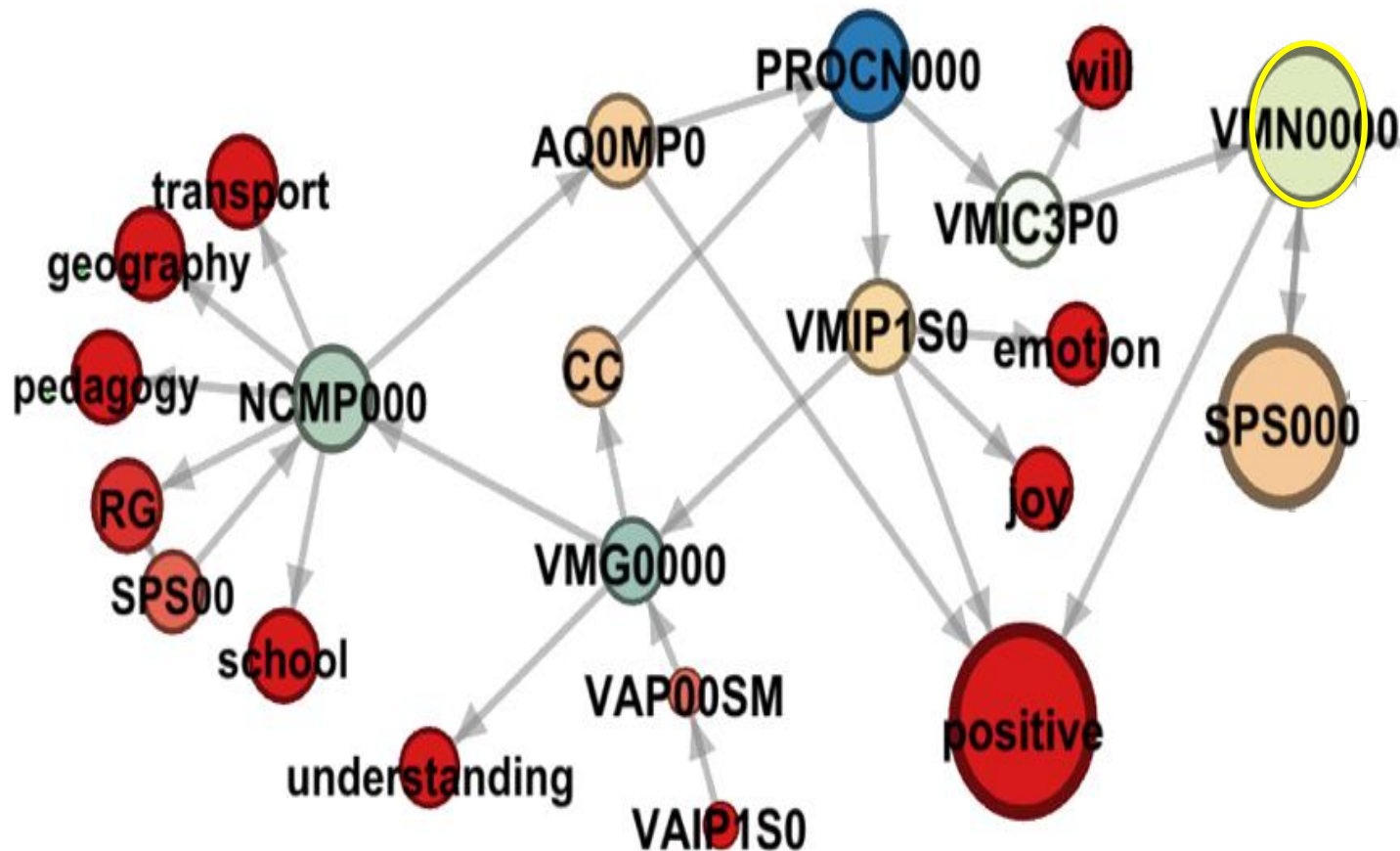
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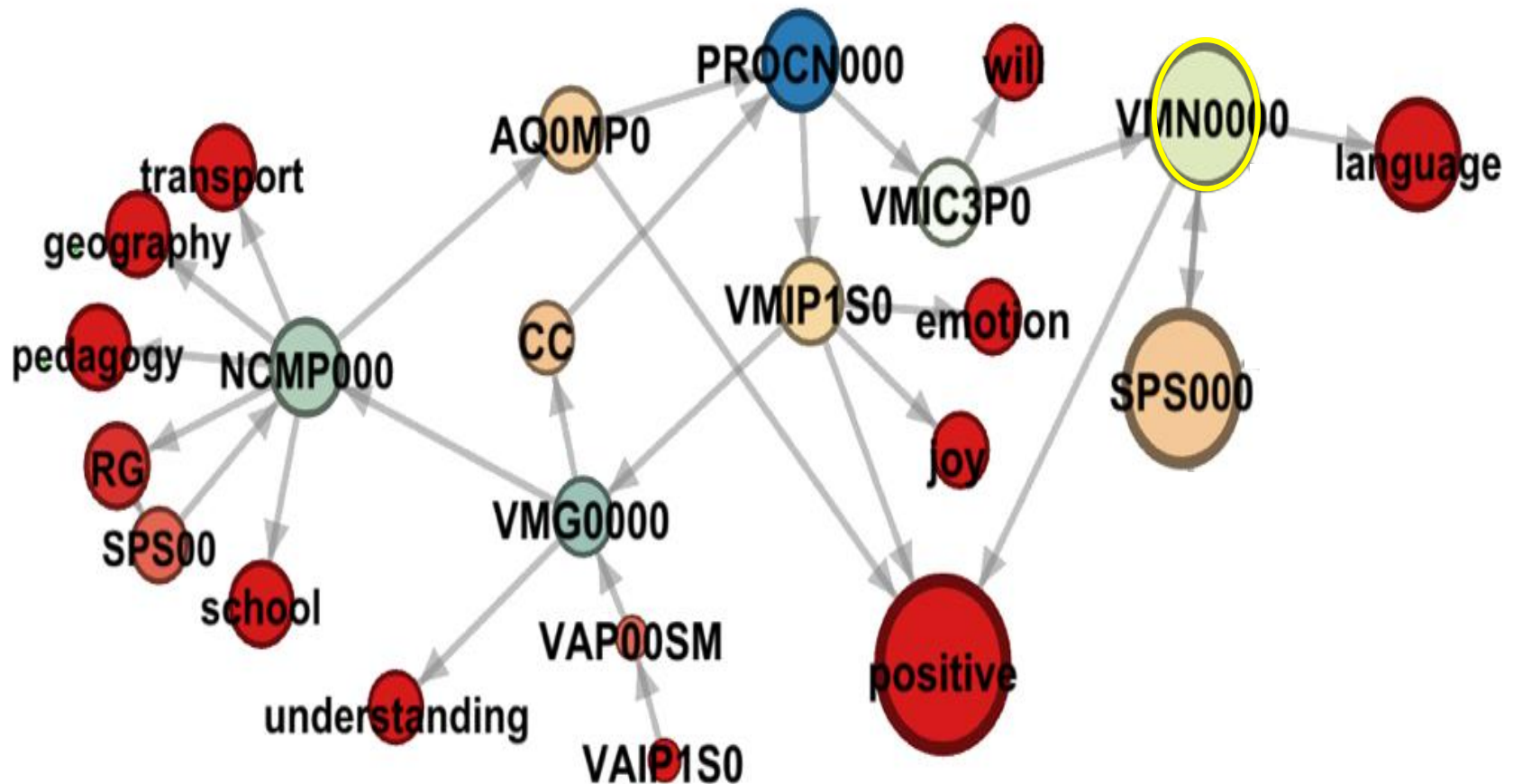


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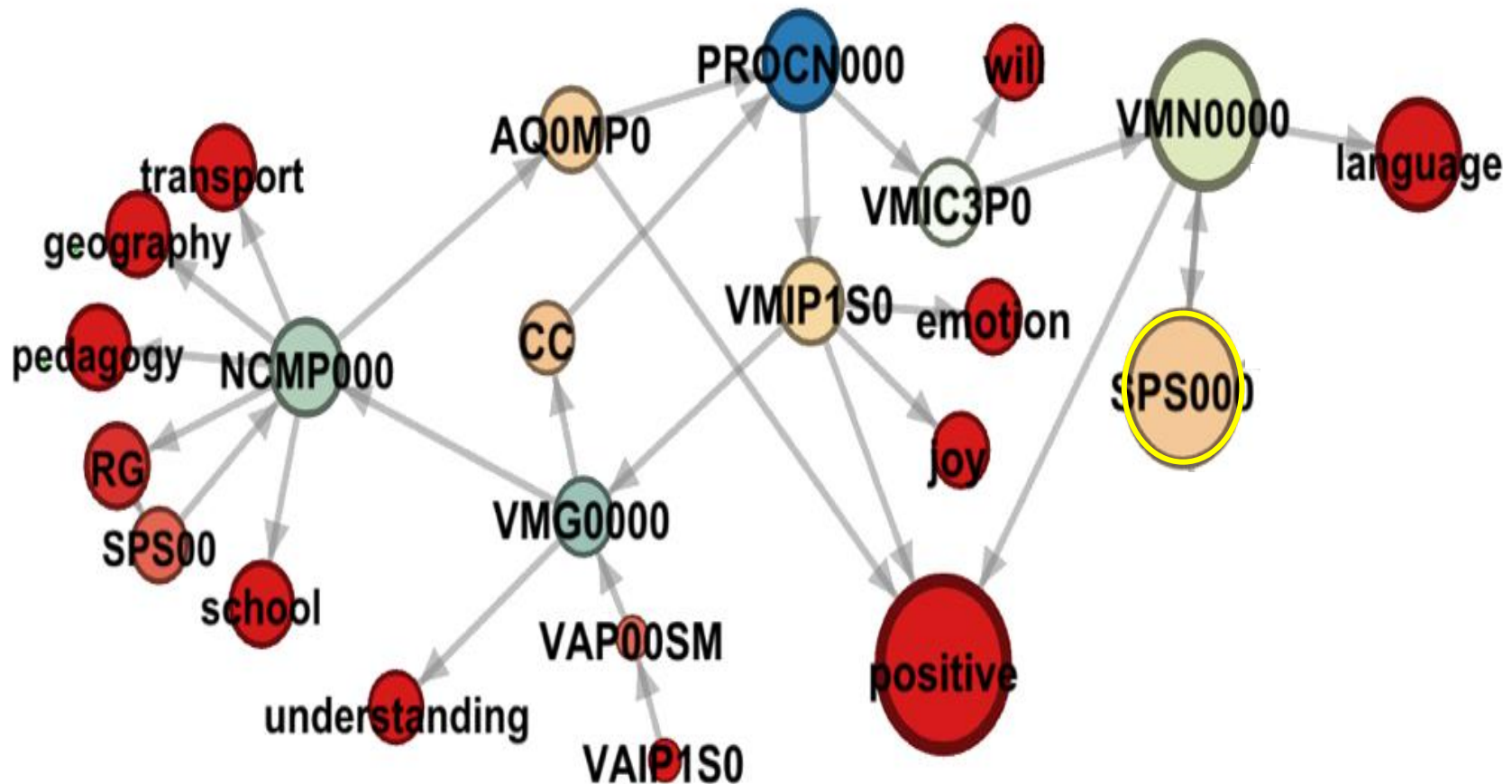


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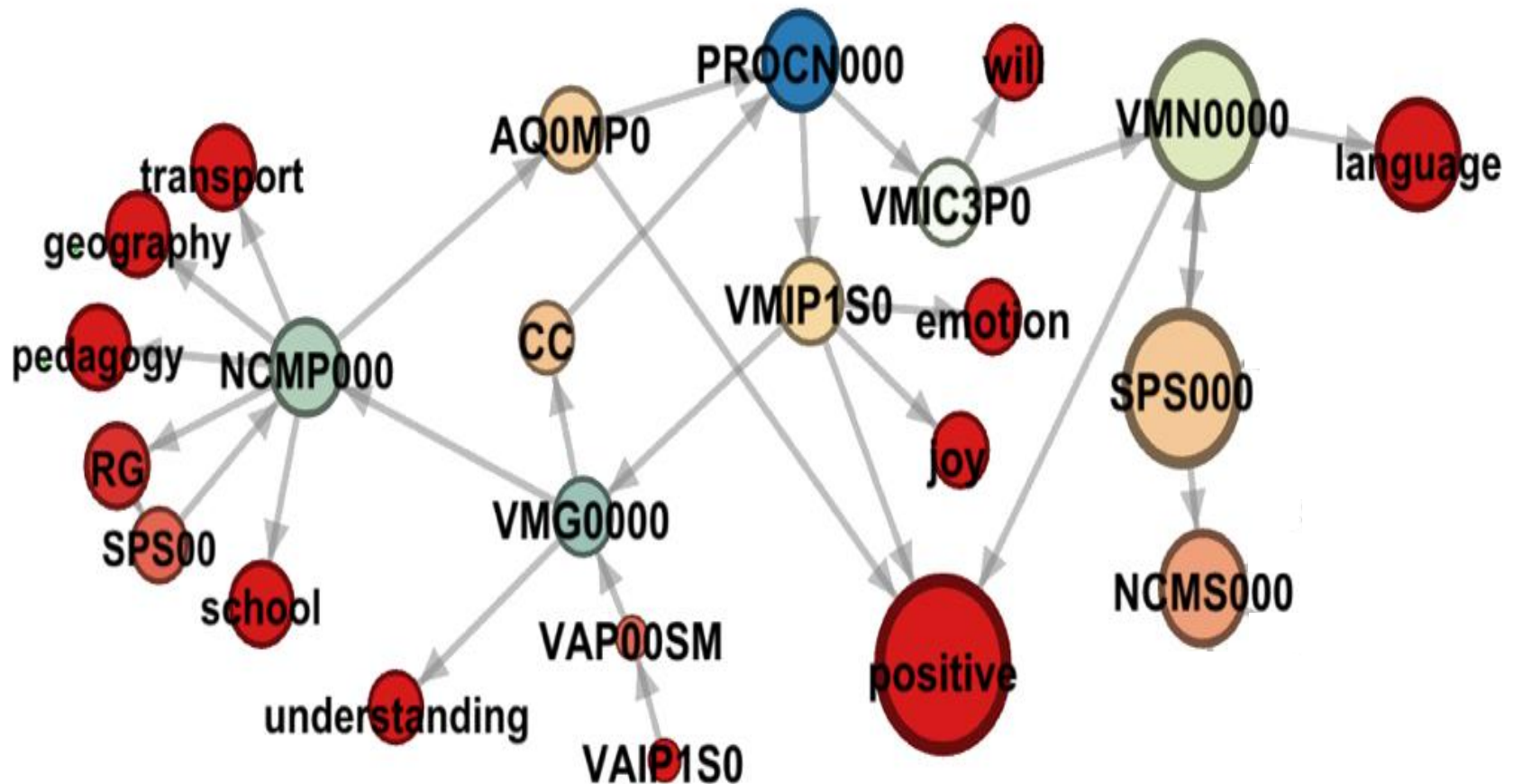
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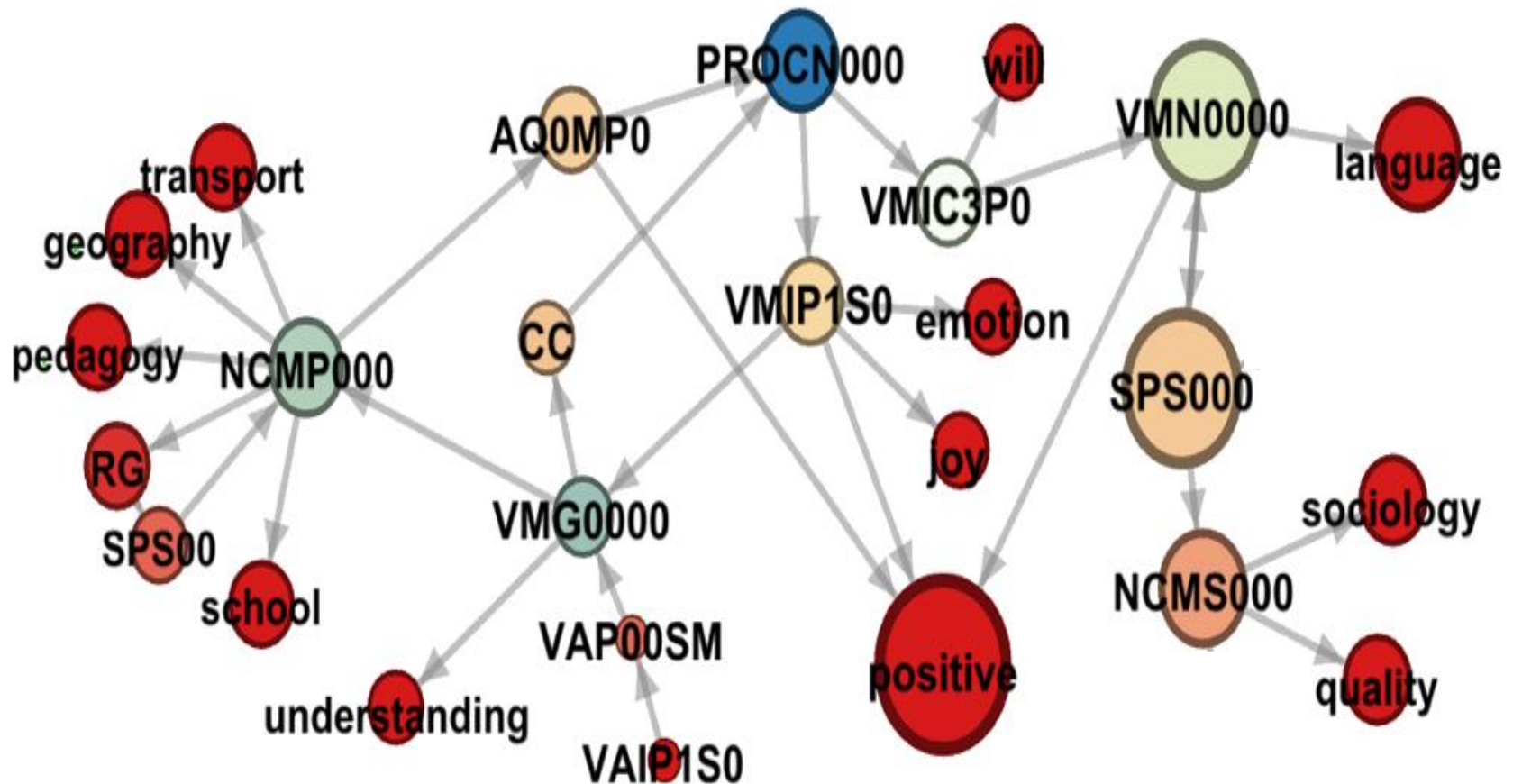
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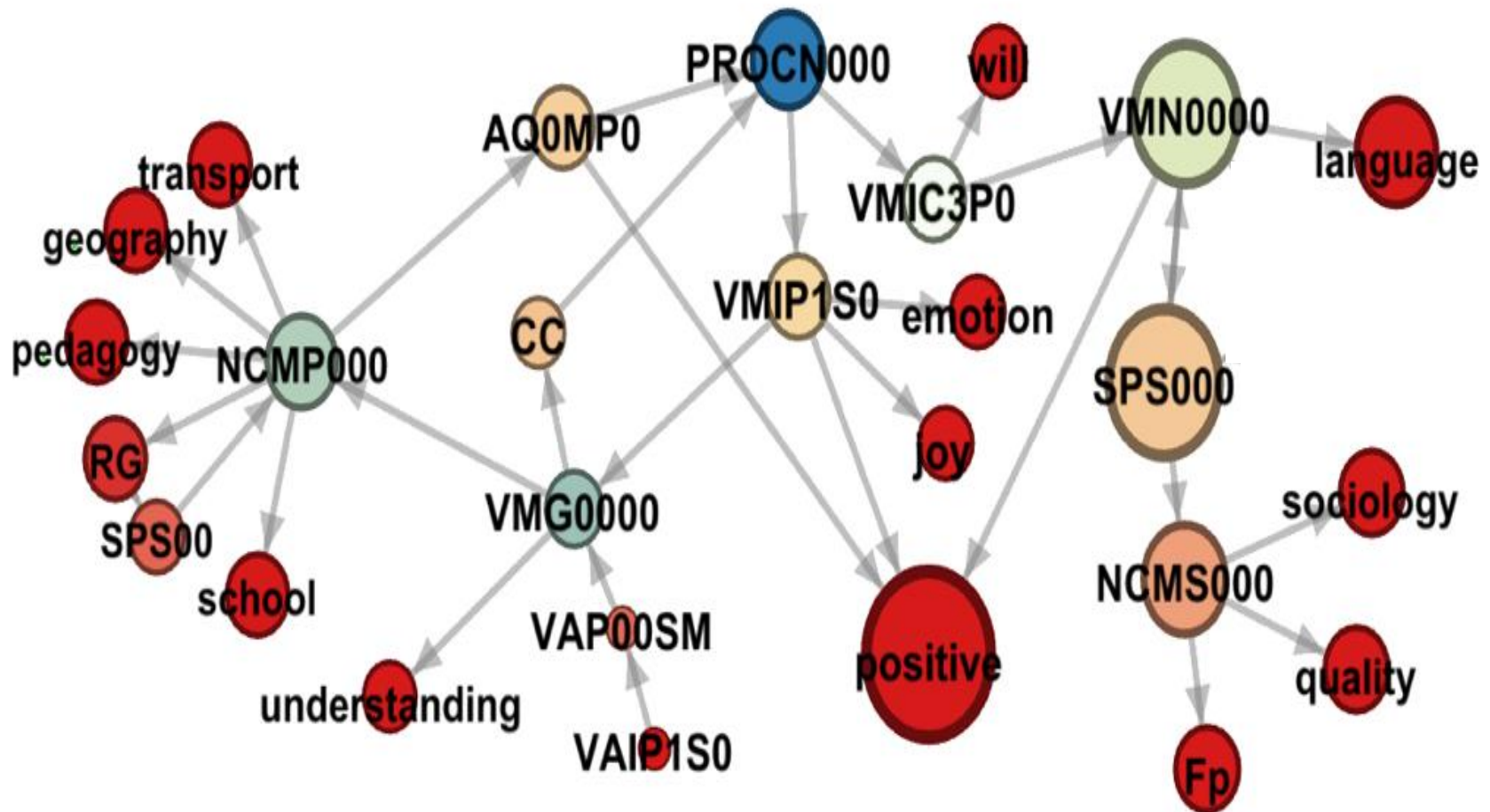
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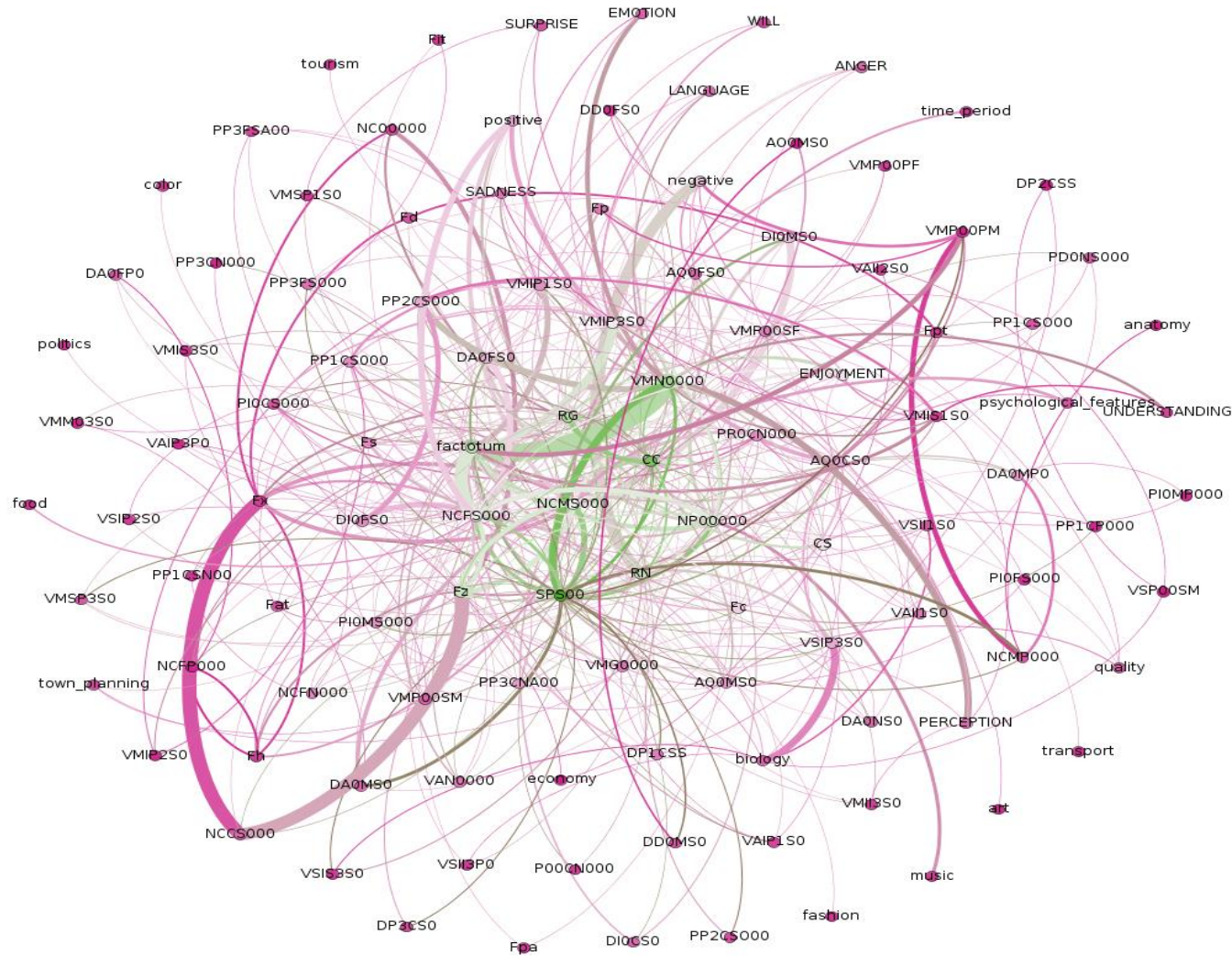


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Representation of texts of a class of authors



Graph-based features

Given a graph $G=\{N,E\}$ where

- N is the set of nodes
- E is the set of edges

We obtain a set of

- **structure-based** features from global measures of the graph
- **node-based** features from node specific measures

We feed a SVM with...

Structure-based features

Nodes-edges ratio	Indicator of how connected the graph is, i.e., how complicated the discourse is
Weighted average degree	Indicator of how much interconnected the graph is, i.e., how much interconnected the grammatical categories are
Diameter	Indicator of the greatest distance between any pair of nodes, i.e., how far a grammatical category is from others, or how far a topic is from an emotion
Density	Indicator of how close the graph is to be complete, i.e., how dense is the text in the sense of how each grammatical category is used in combination with others
Modularity	Indicator of different divisions of the graph into modules (one node has dense connections within the module and sparse with nodes in other modules), i.e., how the discourse is modeled in different structural or stylistic units
Clustering coefficient	Indicator of the transitivity of the graph (if a is directly linked to b and b is directly linked to c, what's the probability that a node is directly linked to c), i.e., how different grammatical categories or semantic information are related to each other
Average path length	Indicator of how far some nodes are from others, i.e., how far some grammatical categories are from others, or some topics are from some emotions

Node-based features

EigenVector	It gives a measure of the influence of each node. In our case, it may give what are the grammatical categories with the most central use in the author's discourse, e.g. which nouns, verbs or adjectives
Betweenness	It gives a measure of the importance of a each node depending on the number of shortest paths of which it is part of. In our case, if one node has a high betweenness centrality means that it is a common element used for link among parts-of-speech, e.g. prepositions, conjunctions or even verbs and nouns. Hence, this measure may give us an indicator of what the most common connectors in the linguistic structures used by authors

Most discriminative features

Ranking	Gender	Age	Ranking	Gender	Age
1	punctuation-semicolon	words-length	11	BTW-NC00000	EIGEN-SPS00
2	EIGEN-VMP00SM	Pron	12	BTW-Z	BTW-NC00000
3	EIGEN-Z	BTW-SPS00	13	EIGEN-DA0MS0	punctuation-exclamation
4	EIGEN-NCCP000	BTW-NCMS000	14	BTW-Fz	emoticon-happy
5	Pron	Intj	15	BTW-NCCP000	BTW-Fh
6	words-length	EIGEN-Fh	16	EIGEN-AQ0MS0	punctuation-colon
7	EIGEN-NC00000	BTW-PP1CS000	17	SEL-disgust	punctuation
8	EIGEN-administration	EIGEN-Fpt	18	EIGEN-DP3CP0	BTW-Fpt
9	Intj	EIGEN-NC00000	19	EIGEN-DP3CS0	EIGEN-DA0FS0
10	SEL-sadness	EIGEN-NCMS000	20	SEL-anger	Verb

EmoGraph vs stylistic + emotion features

- **Word frequency (F)**: words with character flooding; words starting with capital letter; words in capital letters...
- **Punctuation marks (P)**: frequency of use of dots, commas, colon, semicolon, exclamations and question marks
- **Part-Of-Speech**: frequency of use of each grammatical category
- **Emoticons (E)**: number of different types of emoticons representing emotions
- **Spanish Emotion Lexicon (SEL)**: words co-occurring with each emotion: **happiness, anger, fear, sadness, disgust, surprise** (for English: WordNet Affect)

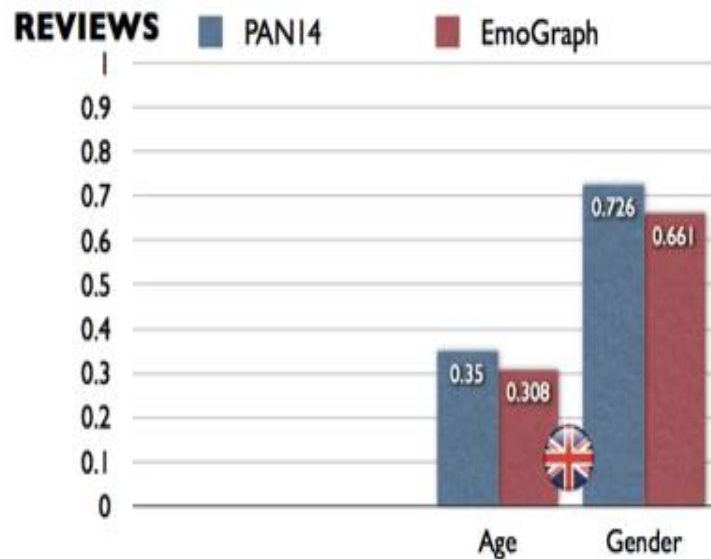
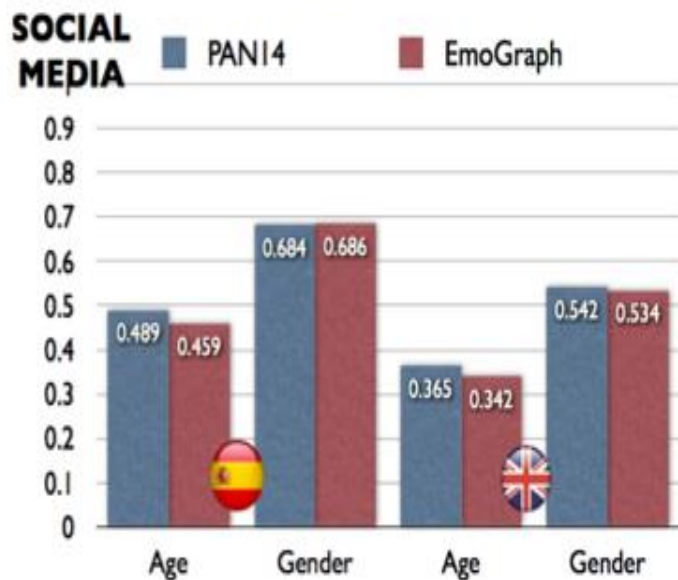
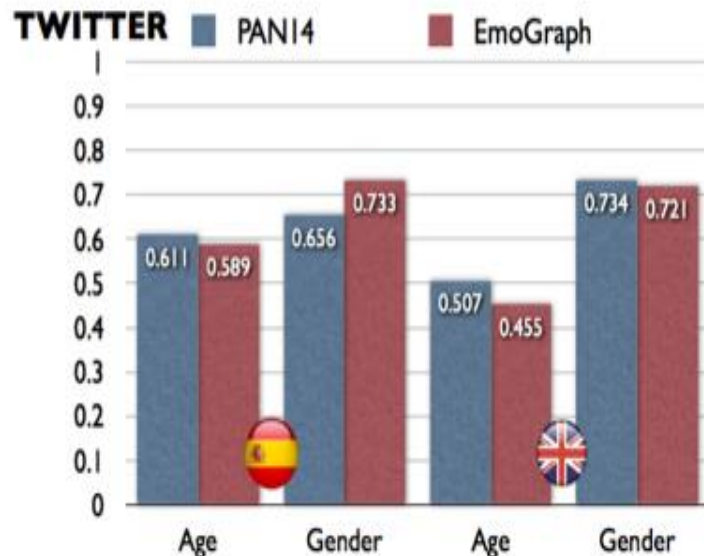
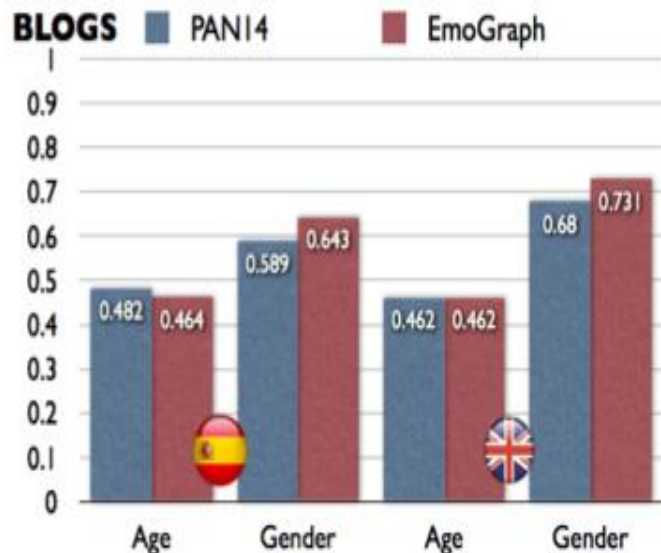
PAN-13

Ranking	Team	Accuracy
1	Rangel-EG	0.6624
2	Pastor	0.6558
3	Santosh	0.6430
4	Rangel-S	0.6350
5	Haro	0.6219
6	Flekova	0.5966
...	...	
21	Baseline	0.3333
...	...	
23	Mechti	0.0512

Ranking	Team	Accuracy
1	Santosh	0.6473
2	Rangel-EG	0.6365
3	Pastor	0.6299
4	Haro	0.6165
5	Ladra	0.6138
...	...	
8	Rangel-S	0.5713
...	...	
18	Baseline	0.5000
...	...	
23	Gillam	0.4784

- EG: EmoGraph features based approach
- S: Stylistic features based approach (Ekman's six basic emotions but not in discourse analysis)

PAN-14: EmoGraph vs. best approach



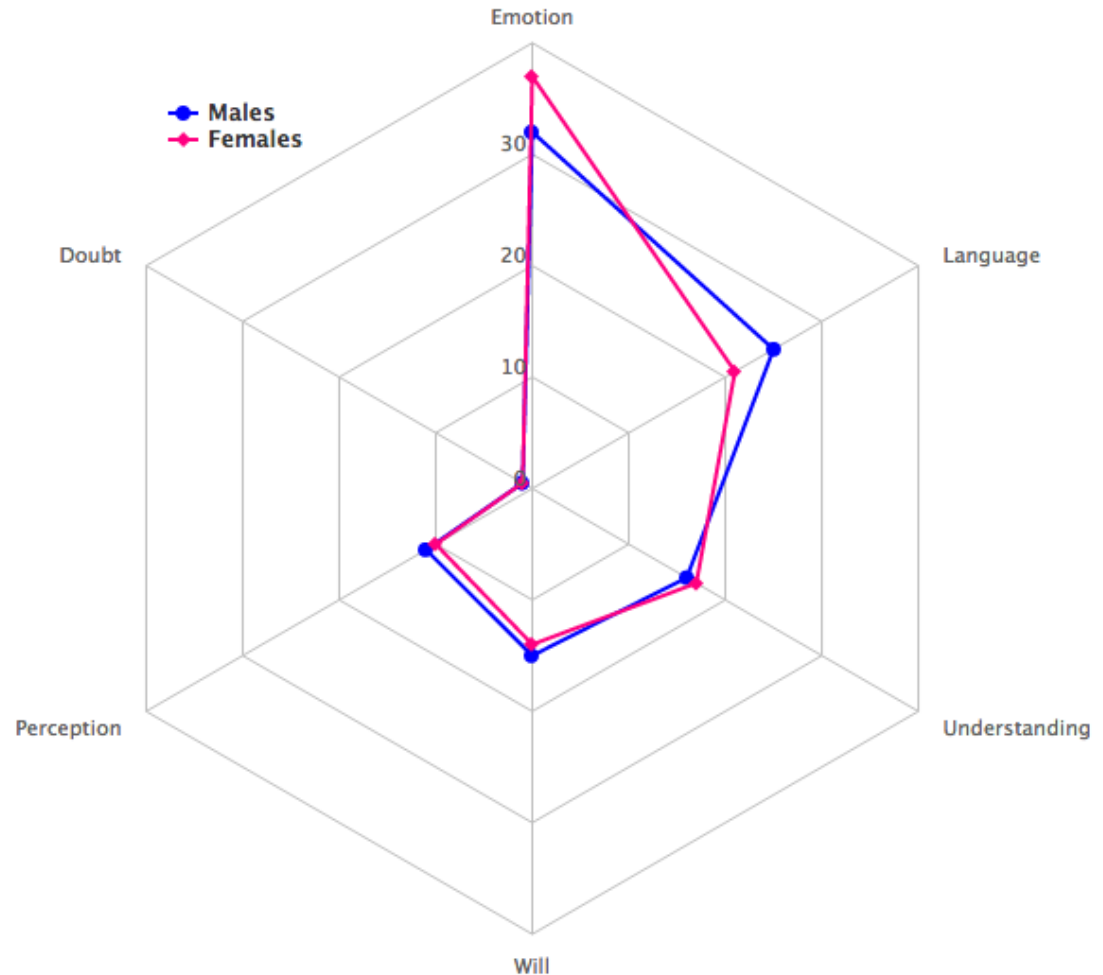
Topics per gender & age (ES)



- Teenagers talk about their studies: e.g. chemistry & linguistics (females) vs. physics & law (males)
- Females talk more about their sexuality and males more about shopping online (commerce)
- We grow up and we are more interested in religion, animals, and food (gastronomy)

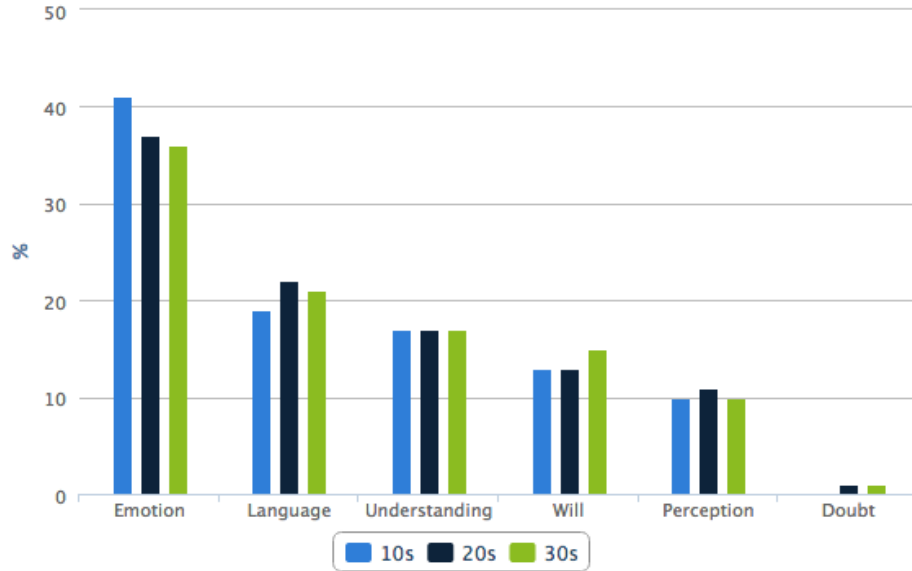
Use of verbs: Gender

- ◆ **Emotion:** feel, love, want...
- ◆ **Language:** say, tell, speak...
- ◆ **Understanding:** know, think, understand...
- ◆ **Perception:** see, listen...
- ◆ **Will:** must, forbid, allow...
- ◆ **Doubt:** doubt, ignore...

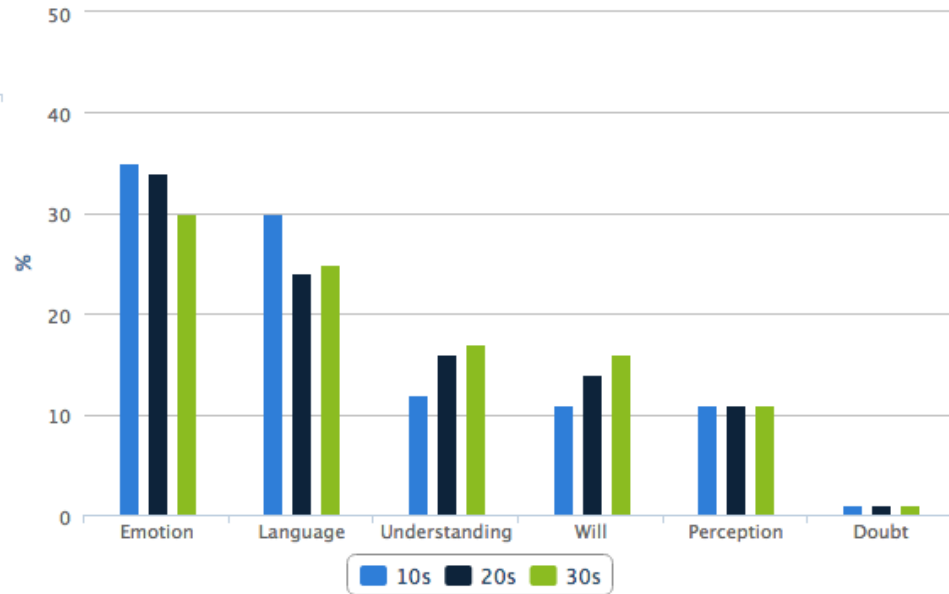


B. Levin. English Verb Classes and Alternations.
University of Chicago Press, Chicago, 1993.

Use of verbs: Gender & age

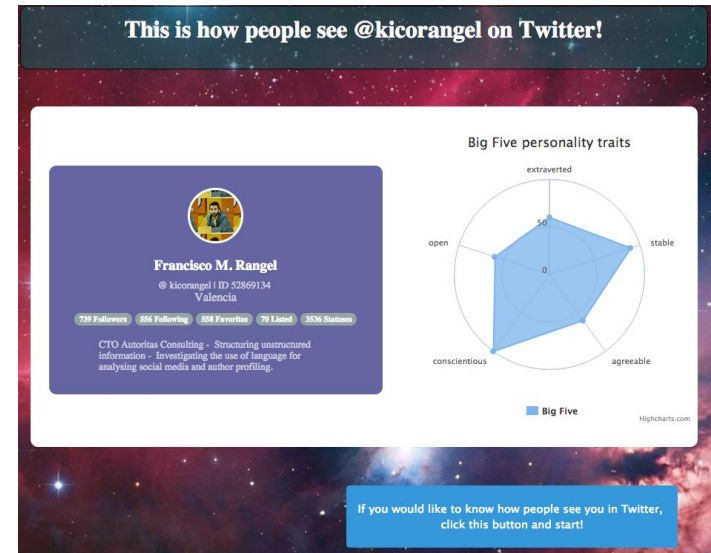
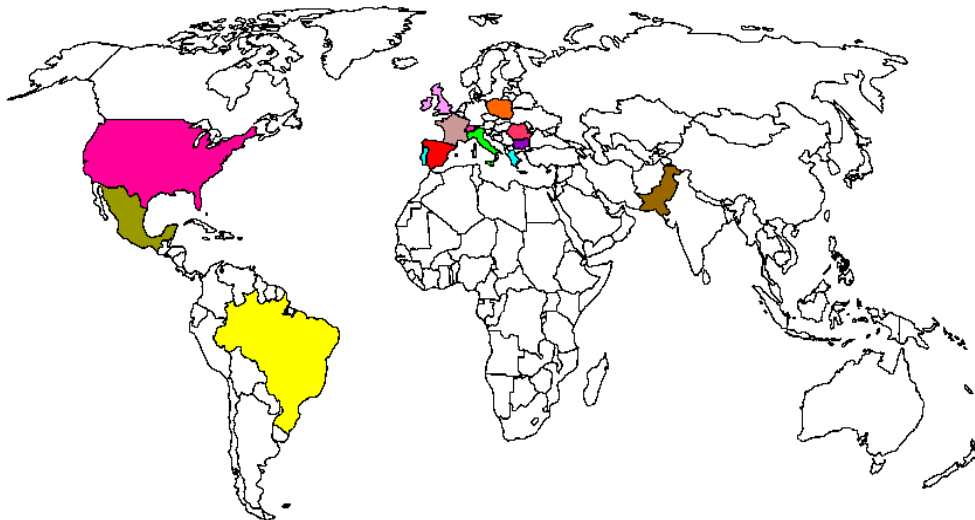


Females vs. Males



2015: Age, gender and personality

- Teams submitting results: 22



- **Personality:** <http://your-personality-test.com/>
- **Age classes:** 18-24, 25-34, 35-49, 50+
- **Languages:** English, Spanish, Dutch, Italian
- **Results:** Gender EN: 85.9% Age EN: 83.8% **Personality** openness (root-mean-square error 0.097)

Personality questionnaire

1. I am a reserved person
2. I trust other people
3. I tend to be lazy
4. I am generally relaxed, not stressed
5. I have few artistic interests
6. I am sociable
7. I tend to find fault with others
8. I do my job well
9. I get nervous easily
10. I have an active imagination

(answers from 1 to 5: <http://mypersonality.autoritas.net/>)

Big Five personality traits

Big Five personality traits (given a text, determine if the author is):

O**pen** to new experiences

C**onscientious**: tends to be careful and scrupulous

E**xtroverted**: gets energy from being around people

A**greeable**: prefers to agree with others

N**eurotic**: tends to worry about things

PR-*SOCO* : **PAN** @FIRE 2016

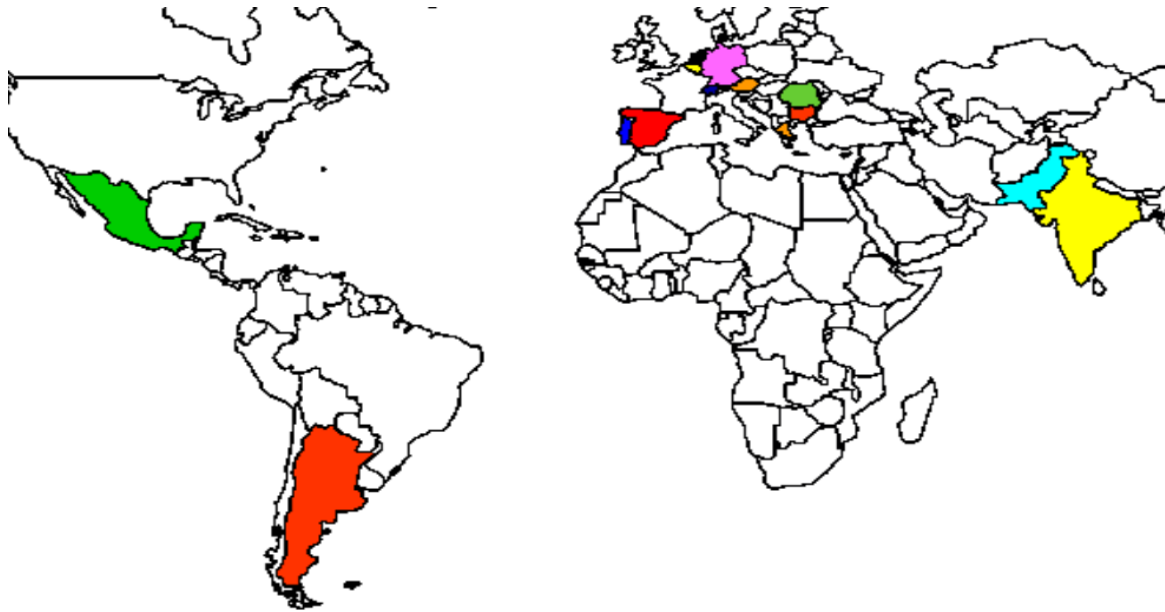
Personality Recognition in SOurce Code

- Big five personality traits from Java source codes
- 11 teams submitted 49 runs
- <http://www.autoritas.es/prsoco/>
- Best results for the **openness** trait
in line with the results obtained on Twitter data
at  **PAN** @ CLEF 2015

Rangel F., González F., Restrepo-Calle F., Montes M., Rosso P. PAN at FIRE: Overview of the PR-SOCO Track on Personality Recognition in SOurce COde. Notebook for PAN at FIRE 2016. CEUR Workshop Proceedings. CEUR-WS.org, vol. 1737, pp. 1-5. 2016

2016: Cross-genre age and gender

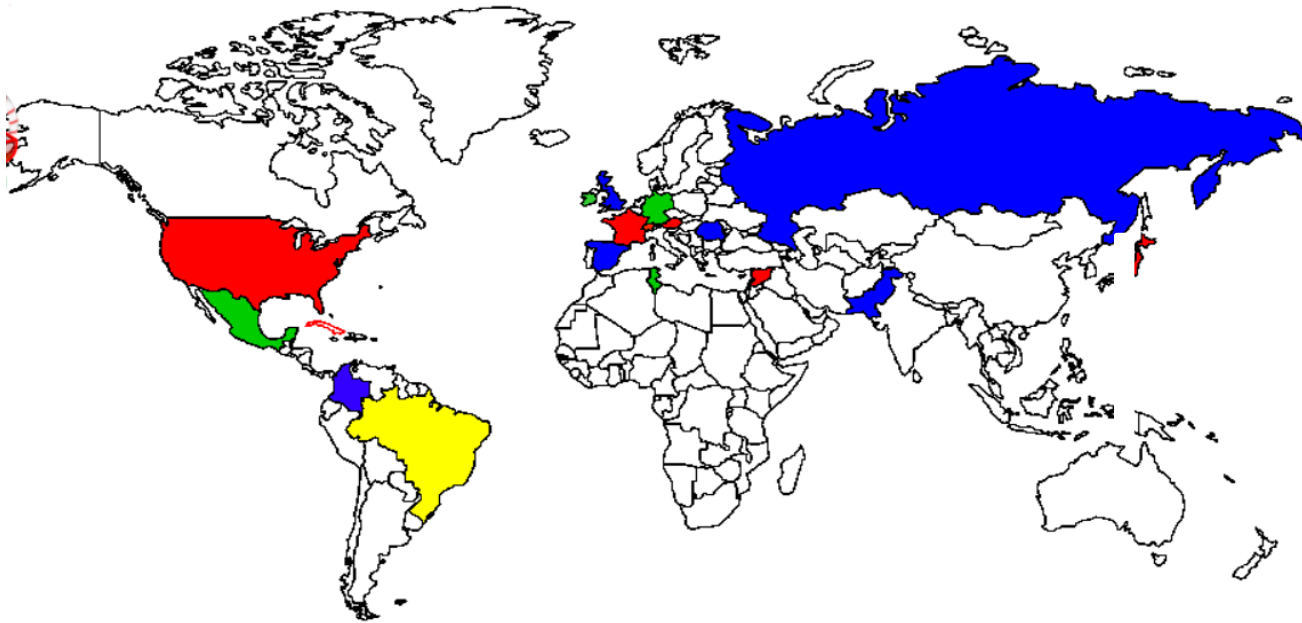
- Teams submitting results: 22
- **Training**  Test: social media and blogs



- **Age classes:** 18-24, 25-34, 35-49, 35-49, 50-64, 65+
- **Languages:** English, Spanish and Dutch
- **Results (blogs): Gender EN: 75.6% Age EN: 58.9%**

2017: Gender and language variety

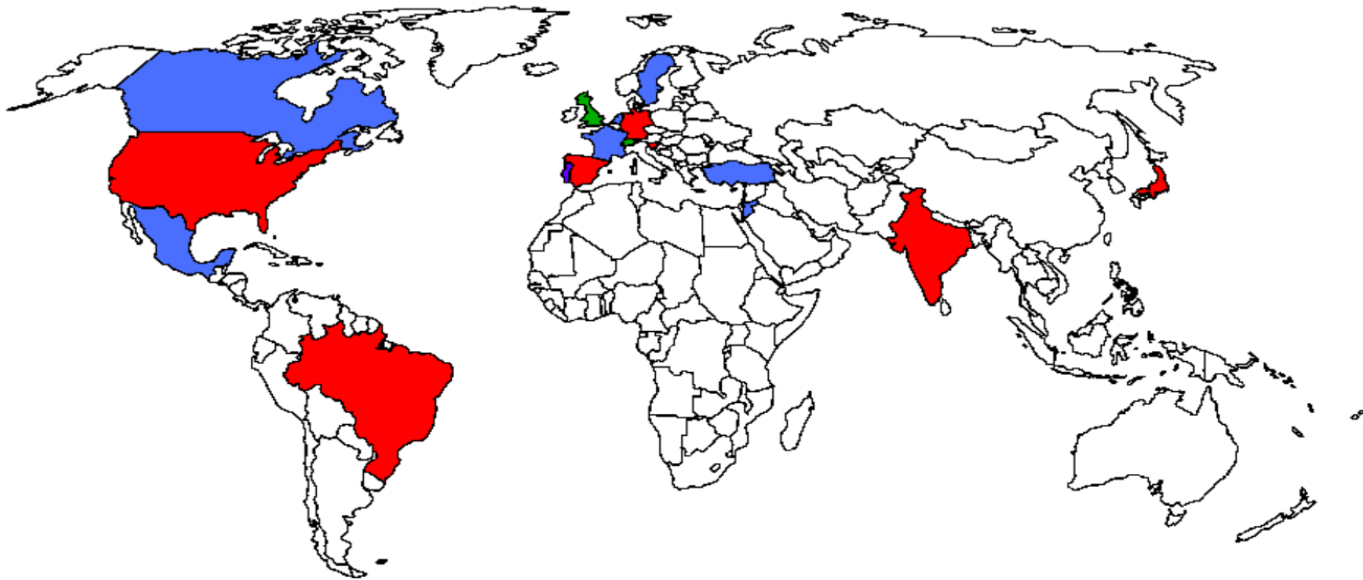
- Teams submitting results: 22



- Languages:** English, Spanish, Portuguese, Arabic
- Results:** **Gender** PT: 87% EN: 82% ES: 83% AR: 80%
Language variety (lowest) EN: 90% AR: 83%

2018: Multimodal gender identification

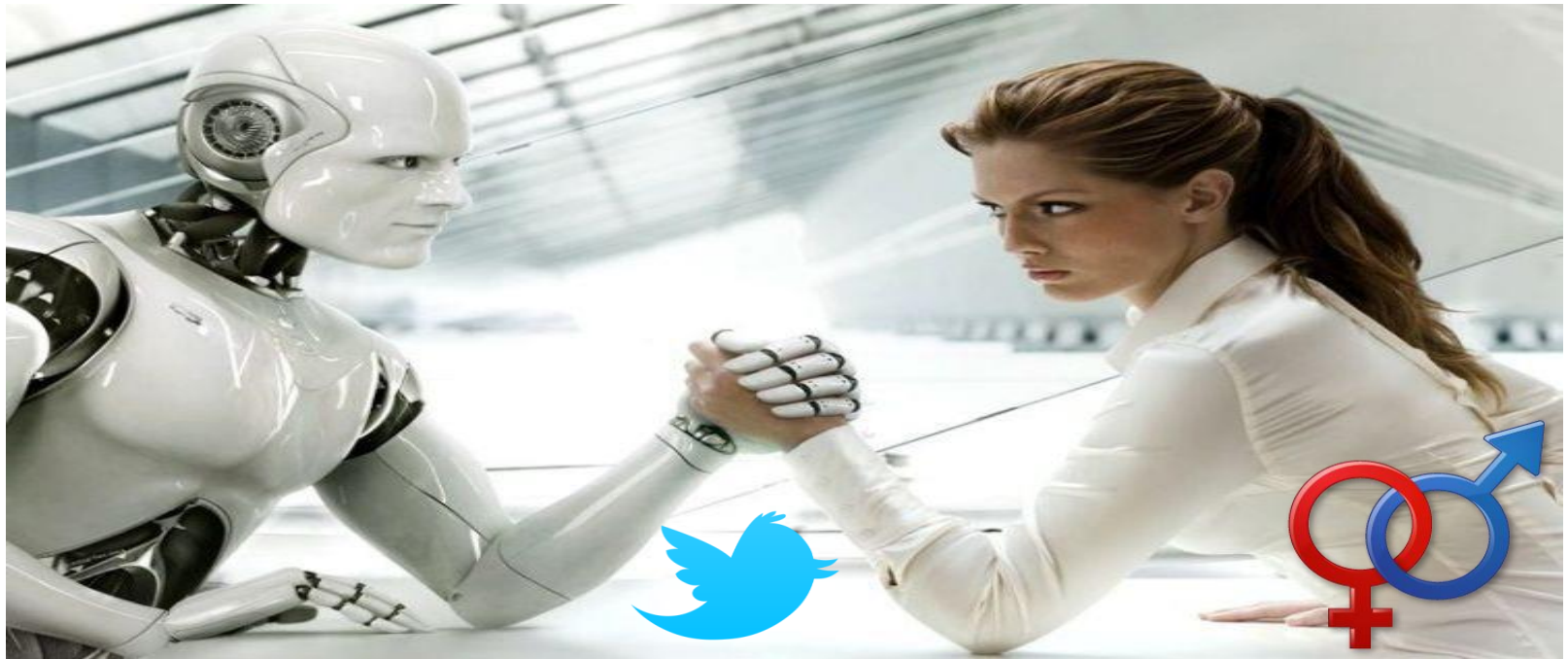
- Teams submitting results: 23



- **Languages:** English, Spanish, Arabic
- **Results:** EN, ES: 82% AR: 81.7%

Author profiling:  **PAN**@CLEF 2019

Bots and gender profiling



Rangel F., Rosso P. Overview of the 7th Author Profiling Task at PAN 2019: Bots and Gender Profiling in Twitter. In: L. Cappellato, N. Ferro, D. E. Losada and H. Müller (eds.) CLEF 2019 Labs and Workshops, Notebook Papers. CEUR Workshop Proceedings.CEUR-WS.org, vol. 2380, 2019

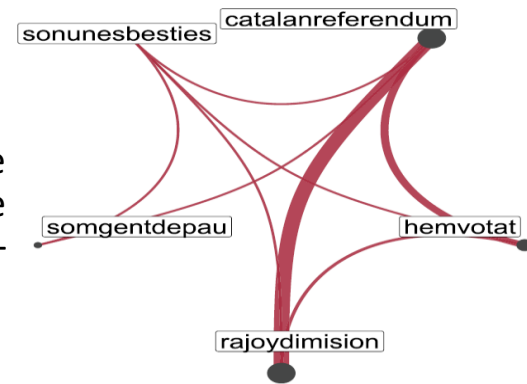
Bots: propaganda, fake news, inflammatory content

- Bots may **influence** users with **comercial, political or ideological** purposes...
- **Polarization** and spread **disinformation and fake news**
- US 2016 Presidencial election, Brexit, 1 Oct 2017 referendum for the Catalan independence:

Bots: propaganda, fake news, inflammatory content

- Bots may **influence** users with **comercial, political or ideological** purposes...
- **Polarization** and spread **disinformation and fake news**
- US 2016 Presidencial election, Brexit, 1 Oct 2017 referendum for the Catalan independence: **23.5%** of 3.6 million tweets generated **by bots** **19%** of the interactions were **from bots to humans**

Massimo Stella, Emilio Ferrara, and Manlio De Domenico. Bots increase exposure to negative and inflammatory content in online social systems. Proc. of the National Academy of Sciences of the United States of America, 115(49):12435–12440, 2018.

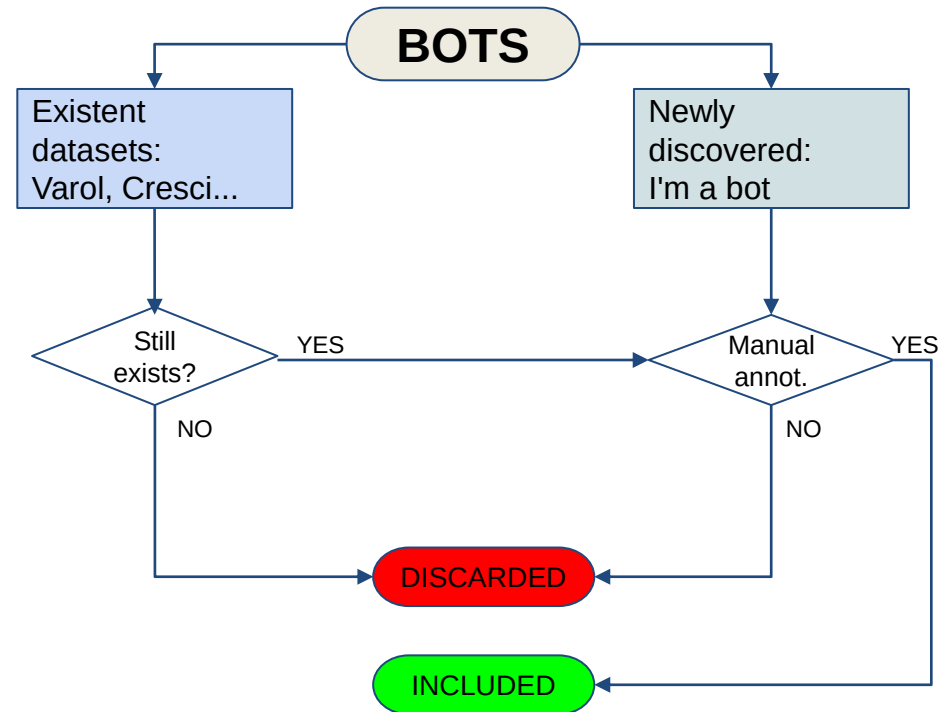


Bots and gender profiling

- How difficult / easy is to discriminate **bots from humans** on the basis only on **textual features**?
- What are the **most difficult types of bots**?



Bots and humans accounts



Humans selected from PAN-AP'17 author profiling + manual annotation

Dataset

- Twitter accounts identified as bots in existing datasets + new ones
- Each **author (bot or human) feed** is composed by exactly **100 tweets**

		(EN) English				(ES) Spanish			
		Bots	Humans		Total	Bots	Humans		Total
			F	M			F	M	
Training	Training	1,440	720	720	2,880	1,040	520	520	2,080
	Development	620	310	310	1,240	460	230	230	920
	Total	2,060	1,030	1,030	4,120	1,500	750	750	3,000
Test		1,320	660	660	2,640	900	450	450	1,800
Total		3,380	1,690	1,690	6,760	2,400	1,200	1,200	4,800

Types of bots

TEMPLATE	The Twitter feed responds to a predefined structure or template , such as for example a Twitter account giving the state of the earthquakes in a region or job offers in a sector
FEED	The Twitter feed retweets or shares news about a predefined topic , such as for example regarding Trump's policies
QUOTE	The Twitter feed reproduces quotes from famous books or songs, from celebrities or people, or jokes
ADVANCED	Twitter feeds whose language is generated on the basis of more elaborated technologies such as Markov chains, metaphors , or in some cases, randomly choosing and merging texts from big corpora

Metaphormagnet

For example, the bot
@metaphormagnet
was developed by
Tony Veale and **Goufu Li**
to automatically generate metaphorical
language



MetaphorIsMyBusiness

@MetaphorMagnet

A Metaphor Machine casts a baleful eye on a dull world. Check out my bro-bots for more metaphors: [@MetaphorMirror](#), [@BotOnBotAction](#) & [@BestOfBotWorlds](#) [#botALLY](#)

📍 UCD, Dublin, Ireland

🔗 [RobotComix.com](#)

📅 Se unió en abril de 2014

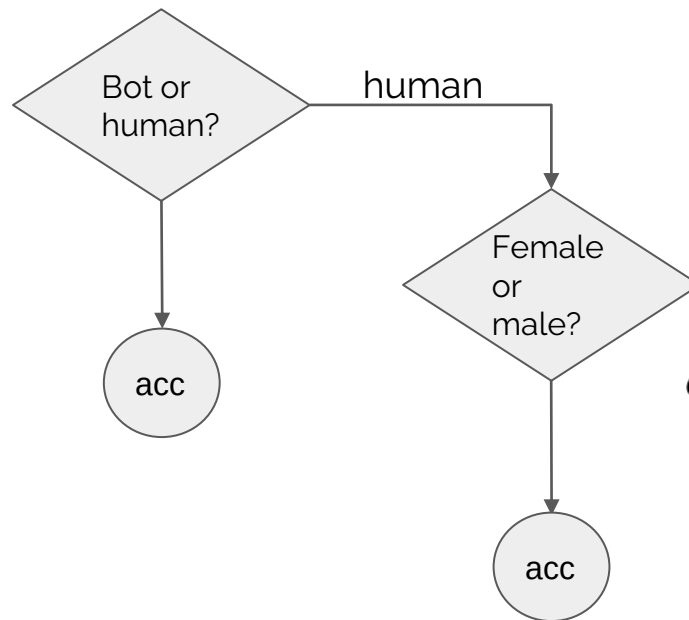


MetaphorIsMyBusiness @MetaphorMagnet · 18 oct. 2016

[#Irony](#): When some playwrights use "inspired" metaphors the way programmers use uninspired hacks. [#Playwright](#)=[#Programmer](#)
[#Metaphor](#)=[#Hack](#)

Evaluation measures

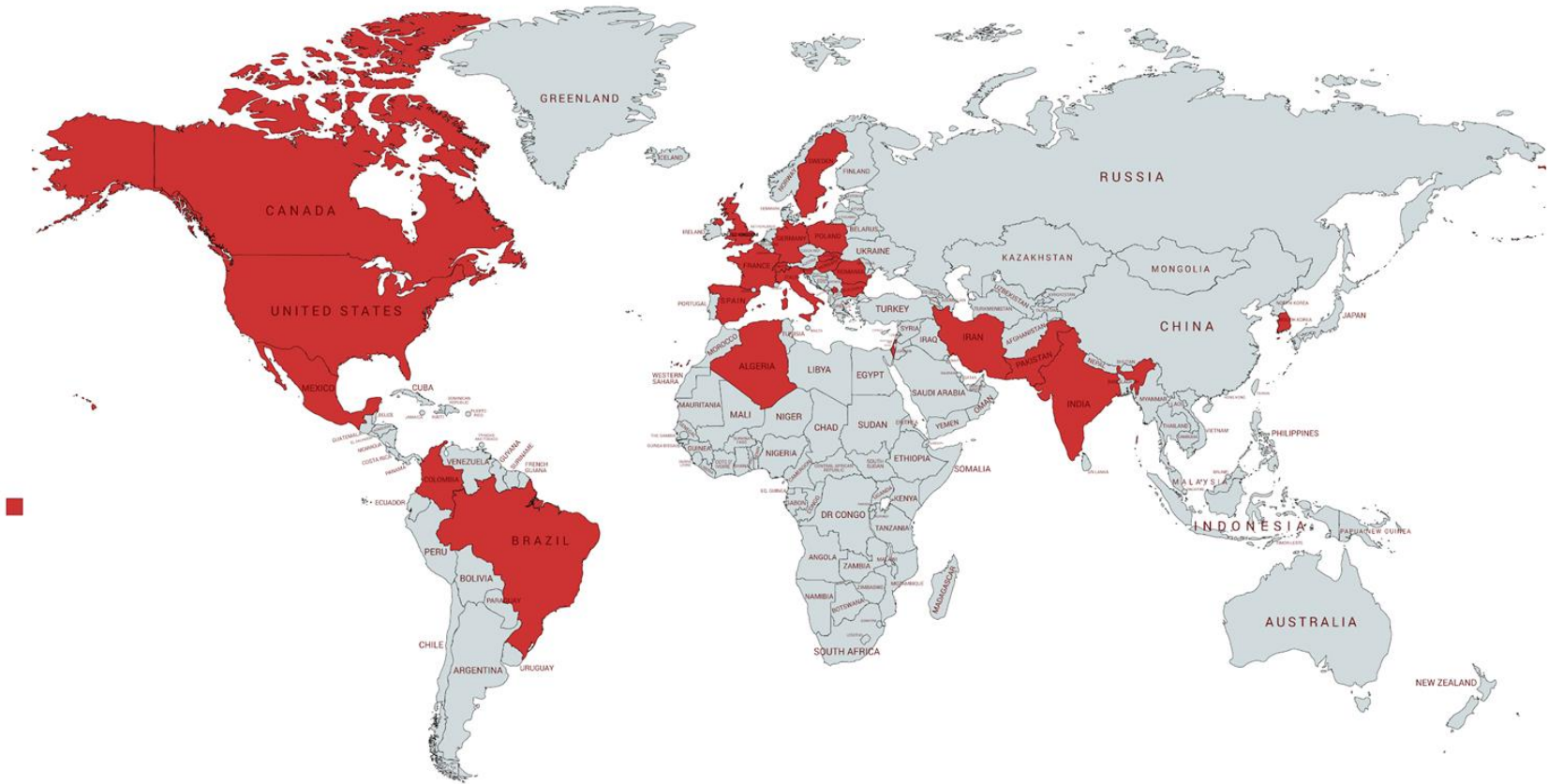
Accuracy is calculated per language and task:



$$acc_{[en|es]} = \frac{acc_{bots} + acc_{gender}}{2}$$

$$ranking = \frac{acc_{en} + acc_{es}}{2}$$

Statistics



55+1 participants
26 countries

Approaches

What kind of ...

Preprocessing

Features

Methods

... did the teams perform?

Approaches: Preprocessing

Twitter elements (URLs, users, hashtags, ...)	Van Halteren; Vogel; Polignano; Giachanou; Gishamer; Puertas; Saeed; Petritk; Valencia; Onose; Babaei; Yacob; Zhechev; Mahmood
Word segmentation	Gishamer; Joo
Tokenisation	Van Halteren; Polignano; Gishamer; Joo; Bacciu; Petritk; Goubin; Zhechev; Mahmood
Stemming / lemmatisation	Ikae; Joo; Saeed; Bacciu; Basile; Petritk; Babaei; Goubin; Zhechev;
Punctuation marks	Vogel; Saeed; Onose; Ribeiro; Goubin; Yacob; Zhechev;
Lowercase	Van Halteren; Vogel; Giachanou; Saeed; Ribeiro
Stopwords	Joo; Saeed; Babaei; Zhechev;
Character flooding	Vogel; Gishamer; Goubin
Latent Semantic Analysis	Rakesh
Short words	Vogel
Infrequent words	Ikae; Gishamer
Contractions and acronyms	Joo; Saeed

Approaches: Features

Stylistic features: - Number of occurrences - Verbs, adjs, pronouns - Number of hashtags, mentions, URLs... - Upper vs. lower case - Punctuation marks - ...	Joo; Goubin; Ashraf; Cimino; Oliveira; Ikae; De la Peña; Johansson; Giachanou; Martinc; Przybyla; Van Halteren; Fernquist
N-gram models	Ispas; Bounaama; Rakesh; Valencia; Mahmood; Fahim; Espinosa; Pizarro; Martinc; Martinc; Dias; Vogel; Giachanou; De la Peña; Babaei; Saeed; Joo; Bacciu; Johansson; Fernquist; HaCohen; Gishamer
Emotional features	Cimino; Giachanou; Oliveira
Lexicon-based features	Gamallo
Compression algorithms	Fernquist
DNA-based approach	Kosmajac
Embeddings	Polignano; Fagni; Halvani; Onose; López-Santillán; Staykovsky; Joo

Approaches: Methods

SVM	Vogel; Cimino; Fagni; Pizarro; Jimenez; HaCohen; Bacciu; Goubin; Srinivasarao; Mahmood; Yacob; Ribeiro; Babaei; Rakesh; Gishamer; Moryossef; Giachanou		
Logistic regression	Gishamer; Moryossef; Valencia; Bolonyai; Przybyła	CatBoost	Fernquist
SpaCy	Moryossef	kNN	Ikae
Random Forest	Johansson; Moryossef	Multilayer Perceptron	Staykovski
Stochastic Gradient Descent	Giachanou; Bounaama	RNN	Dias; Petrik; Bolonyai; Onose
Decision Trees	Saeed	CNN	Dias; Petrik; Polignano; Farber
Multinomial BayesNet	Saeed	BERT	Joo
Naive Bayes	Gamallo	Feedforward NN	Halvani; De la Peña
Adaboost	Bacciu	LSTM	Zhechev

Baselines

MAJORITY	A statistical baseline that always predicts the majority class in the training set. In case of balanced classes, it predicts one of them
RANDOM	A baseline that randomly generates the predictions among the different classes
CHAR N-GRAMS	With values for n from 1 to 10, and selecting the 100, 200, 500, 1,000, 2,000, 5,000 and 10,000 most frequent ones
WORD N-GRAMS	With values for n from 1 to 10, and selecting the 100, 200, 500, 1,000, 2,000, 5,000 and 10,000 most frequent ones
W2V	Texts are represented with two word embedding models: Continuous Bag of Words (CBOW); and Skip-Grams
LDSE	This method represents documents on the basis of the probability distribution of occurrence of their words in the different classes. The key concept of LDSE is a weight, representing the probability of a term to belong to one of the different categories: human / bot, male / female. The distribution of weights for a given document should be closer to the weights of its corresponding category. LDSE takes advantage of the whole vocabulary

Global ranking

Ranking	Team	Bots vs. Human		Gender		Average
		EN	ES	EN	ES	
1	Pizarro	0.9360	0.9333	0.8356	0.8172	0.8805
2	Srinivasarao & Manu	0.9371	0.9061	0.8398	0.7967	0.8699
3	Bacciu et al.	0.9432	0.9078	0.8417	0.7761	0.8672
4	Jimenez-Villar et al.	0.9114	0.9211	0.8212	0.8100	0.8659
5	Fernquist	0.9496	0.9061	0.8273	0.7667	0.8624
6	Mahmood	0.9121	0.9167	0.8163	0.7950	0.8600
7	Ipsas & Popescu	0.9345	0.8950	0.8265	0.7822	0.8596
8	Vogel & Jiang	0.9201	0.9056	0.8167	0.7756	0.8545
9	Johansson & Isbister	0.9595	0.8817	0.8379	0.7278	0.8517
10	Goubin et al.	0.9034	0.8678	0.8333	0.7917	0.8491
11	Polignano & de Pinto	0.9182	0.9156	0.7973	0.7417	0.8432
12	Valencia et al.	0.9061	0.8606	0.8432	0.7539	0.8410
13	Kosmajac & Keselj	0.9216	0.8956	0.7928	0.7494	0.8399
14	Fagni & Tesconi	0.9148	0.9144	0.7670	0.7589	0.8388
	char nGrams	0.9360	0.8972	0.7920	0.7289	0.8385
15	Glocker	0.9091	0.8767	0.8114	0.7467	0.8360
	word nGrams	0.9356	0.8833	0.7989	0.7244	0.8356
16	Martinc et al.	0.8939	0.8744	0.7989	0.7572	0.8311
17	Sanchis & Velez	0.9129	0.8756	0.8061	0.7233	0.8295
18	Halvani & Marquardt	0.9159	0.8239	0.8273	0.7378	0.8262
19	Ashraf et al.	0.9227	0.8839	0.7583	0.7261	0.8228
20	Gishamer	0.9352	0.7922	0.8402	0.7122	0.8200
21	Petrik & Chuda	0.9008	0.8689	0.7758	0.7250	0.8176
22	Oliveira et al.	0.9057	0.8767	0.7686	0.7150	0.8165
	W2V	0.9030	0.8444	0.7879	0.7156	0.8127
23	De La Peña & Prieto	0.9045	0.8578	0.7898	0.6967	0.8122
24	López Santillán et al.	0.8867	0.8544	0.7773	0.7100	0.8071
	LDSE	0.9054	0.8372	0.7800	0.6900	0.8032
25	Bolonyai et al.	0.9136	0.8389	0.7572	0.6956	0.8013

Global ranking

26	Moryossef	0.8909	0.8378	0.7871	0.6894	0.8013
27	Zhechev	0.8652	0.8706	0.7360	0.7178	0.7974
28	Giachanou & Ghanem	0.9057	0.8556	0.7731	0.6478	0.7956
29	Espinosa et al.	0.8413	0.7683	0.8413	0.7178	0.7922
30	Rahgouy et al.	0.8621	0.8378	0.7636	0.7022	0.7914
31	Onose et al.	0.8943	0.8483	0.7485	0.6711	0.7906
32	Przybyla	0.9155	0.8844	0.6898	0.6533	0.7858
33	Puertas et al.	0.8807	0.8061	0.7610	0.6944	0.7856
34	Van Halteren	0.8962	0.8283	0.7420	0.6728	0.7848
35	Gamallo & Almatarneh	0.8148	0.8767	0.7220	0.7056	0.7798
36	Bryan & Philipp	0.8689	0.7883	0.6455	0.6056	0.7271
37	Dias & Paraboni	0.8409	0.8211	0.5807	0.6467	0.7224
38	Oliva & Masanet	0.9114	0.9111	0.4462	0.4589	0.6819
39	Hacohen-Kerner et al.	0.4163	0.4744	0.7489	0.7378	0.5944
40	Kloppenburger	0.5830	0.5389	0.4678	0.4483	0.5095
	MAJORITY	0.5000	0.5000	0.5000	0.5000	0.5000
	RANDOM	0.4905	0.4861	0.3716	0.3700	0.4296
41	Bounaama & Amine	0.5008	0.5050	0.2511	0.2567	0.3784
42	Joo & Hwang	0.9333	-	0.8360	-	0.4423
43	Staykovski	0.9186	-	0.8174	-	0.4340
44	Cimino & Dell'Orletta	0.9083	-	0.7898	-	0.4245
45	Ikae et al.	0.9125	-	0.7371	-	0.4124
46	Jeanneau	0.8924	-	0.7451	-	0.4094
47	Zhang	0.8977	-	0.7197	-	0.4044
48	Fahim et al.	0.8629	-	0.6837	-	0.3867
49	Saborit	-	0.8100	-	0.6567	0.3667
50	Saeed & Shirazi	0.7951	-	0.5655	-	0.3402
51	Radarapu	0.7242	-	0.4951	-	0.3048
52	Bennani-Smires	0.9159	-	-	-	0.2290
53	Gupta	0.5007	-	0.4044	-	0.2263
54	Qurdina	0.9034	-	-	-	0.2259
55	Aroyehun	0.5000	-	-	-	0.1250

Best results

Johansson

- Stylistic features
- Random Forest

Valencia

- n-grams
- Logistic Regression

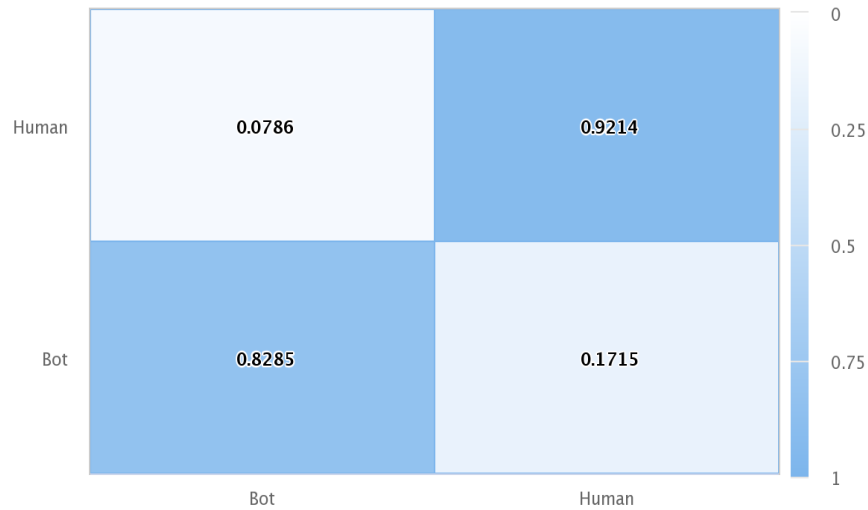
Language	Bots vs. Human	Gender
English	0.9595	0.8417
Spanish	0.9333	0.8172

Pizarro

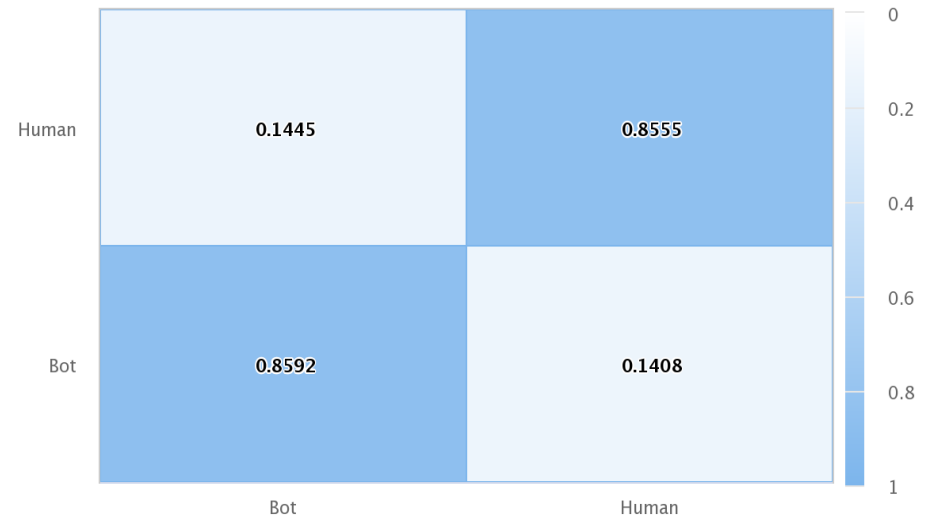
- n-grams
- SVM

Confusion matrices: bots vs. humans

English

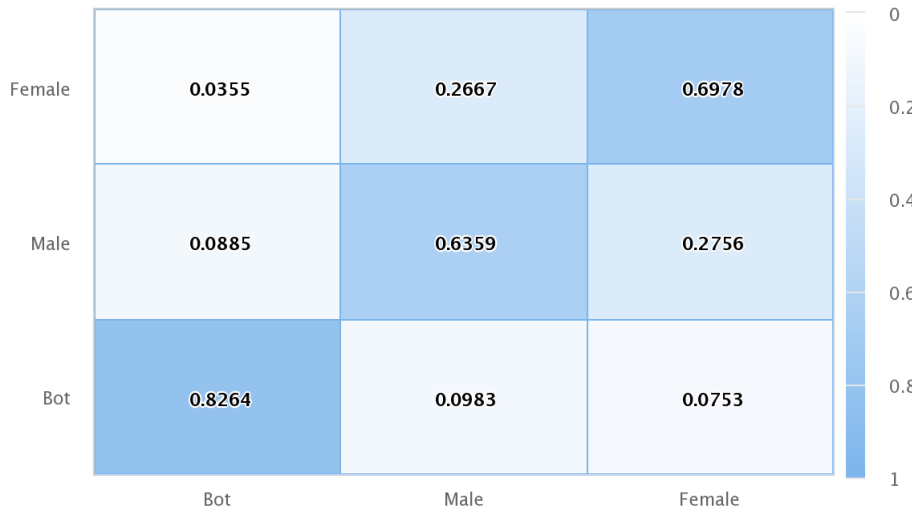


Spanish

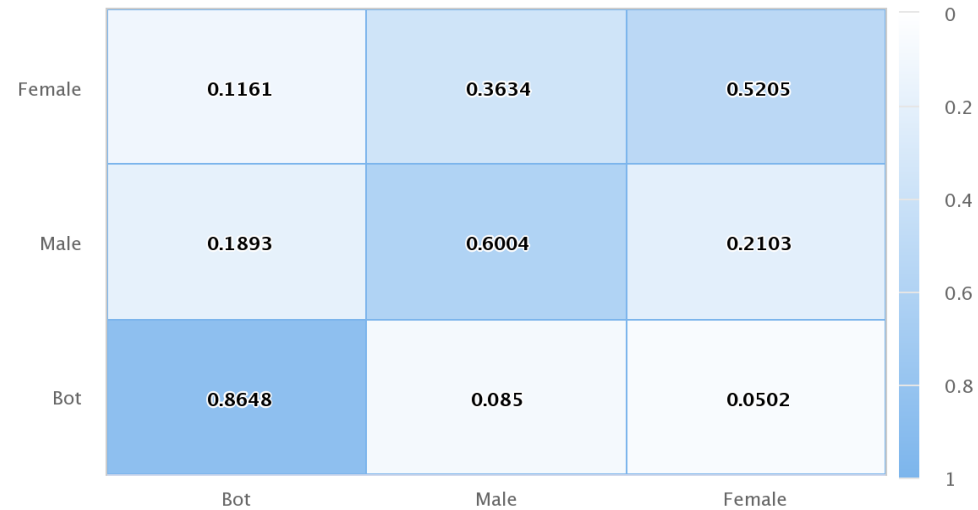


Confusion matrices: gender

English



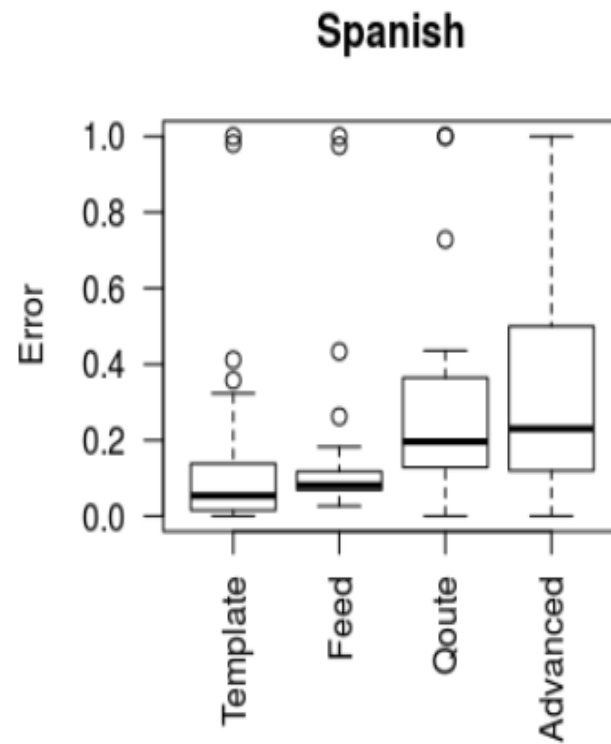
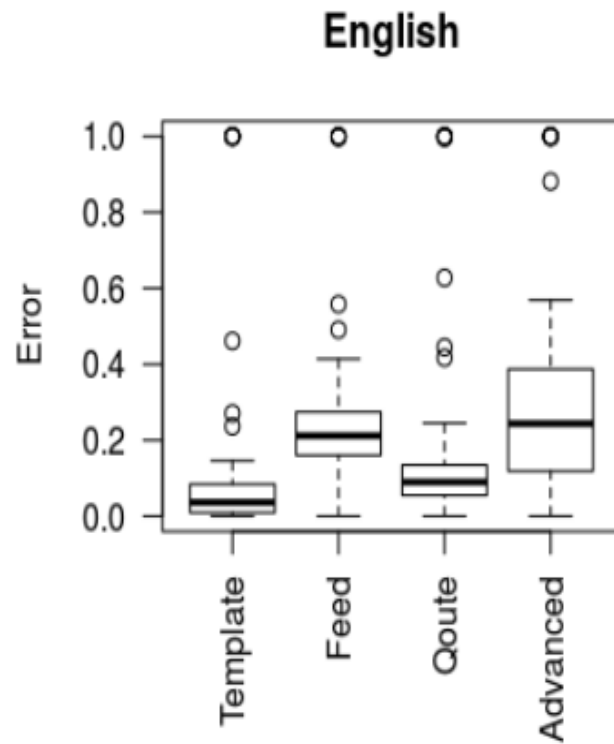
Spanish



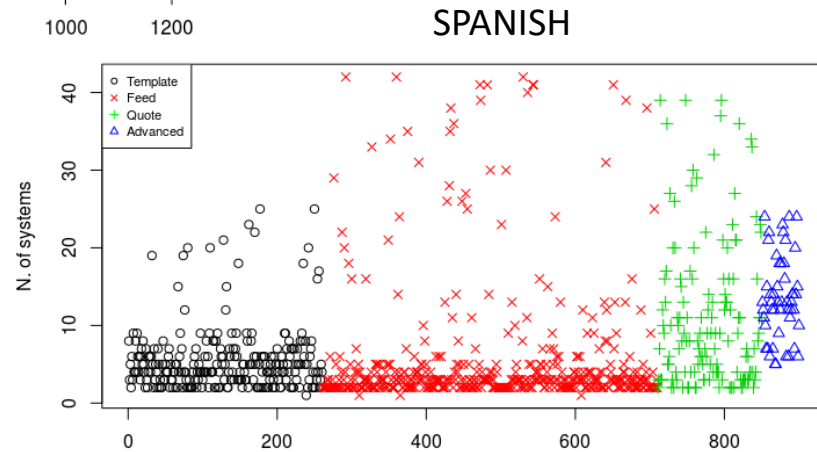
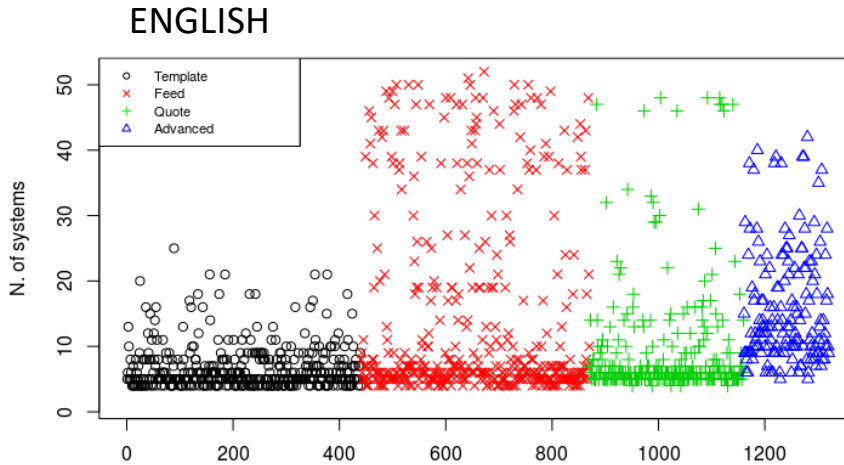
Test dataset: % types of bots

TYPE	ENGLISH	SPANISH
TEMPLATE	33.30%	25.56%
FEED	32.58%	50%
QUOTE	21.97%	15.56%
ADVANCED	12.12%	8.88%

Errors per bot type



Errors per bot type

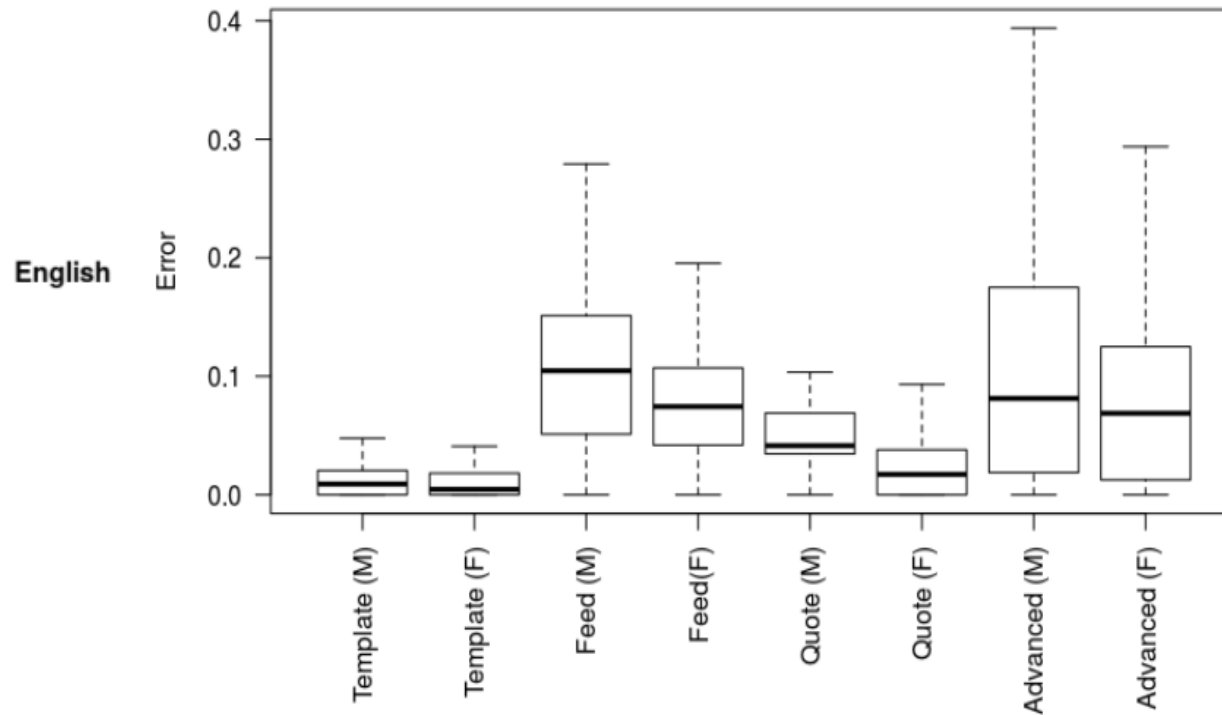


Errors per bot type

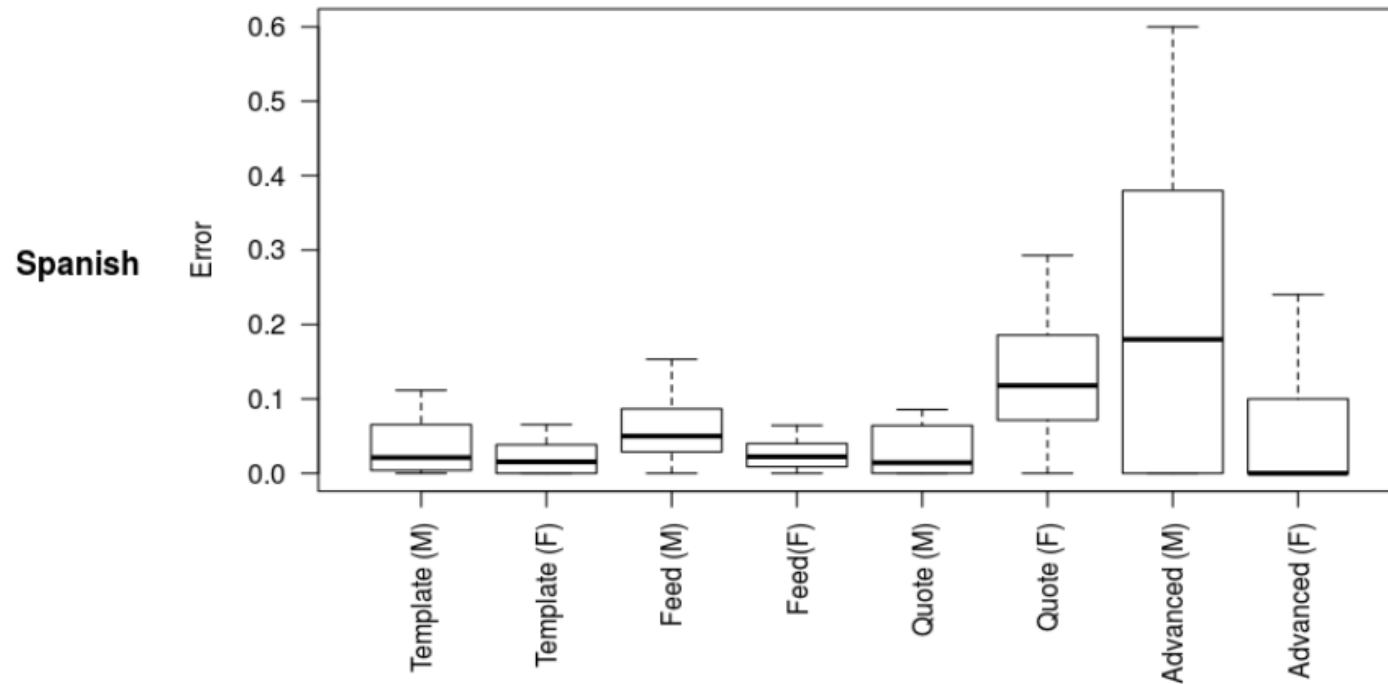
Author Id.	Twitter Account	Type	N. Systems
caf6d82d5dca1598beb5bfac0aea4161	@NasaTimeMachine	template	21 / 53
@wylejw <i>You must be cool, I'll follow you!</i>			
4c27d3c7a10964f574849b6be1df872d	@rarehero	feed	52 / 53
<i>Get a doll, drape fabric and spray the hell out of it with Fabric Quick Stiffening Spray ... https://t.co/C9Ub6xXZWI via @duckduckgo</i>			
8d08e3a0e1fea2f965fd7eb36f3b0b07	@MessiQuote	quote	48 / 53
<i>.@PedroPintoUEFA: "Messi is unstoppable and we should feel privileged to be watching a player who may be the best of all time." https://t.co/TmCR6qCzO2</i>			
6a6766790e1f5f67813afd7c0aa1e60d	@markov_chains	advanced	42 / 53
<i>I have transferred to the local library go you! Just be Crazy John's prepaid sim card.</i>			



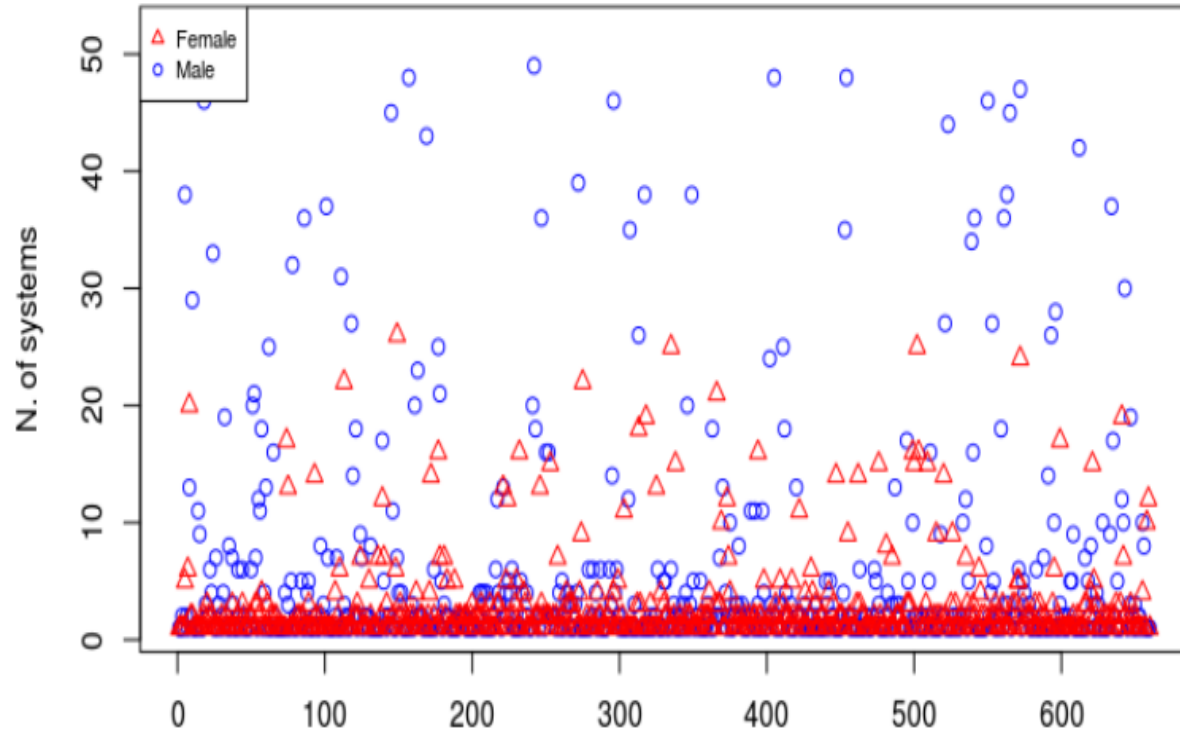
Bot to human per gender errors



Bot to human per gender errors



Human to bot errors



Human to bot errors

Author Id.	Twitter Account	Gender	N. Systems
63e4206bde634213b3a37343cf76e900	@Ask_KFitz <i>#Electric Imp Smart Refrigerator</i> https://t.co/qigh5Womd7	male	49 / 53
b11ffeed0b38eb85e4e288f5c74f704	@iqbalmustansar <i>Trend - What's Dominating Digital Marketing Right Now? -</i> https://t.co/dWp7ovqzCM	male	45 / 53
ba0850ae38408f1db832707f1e0258fd	@CharBar_tweets <i>Hollywood boll #bowling #legs #Sundayfunday</i> https://t.co/cLq9ZlNM38	female	26 / 53
d64be10ecfbbb81d0c6e5b3115c335a5	@RheaRoryJames <i>RT @realDonaldTrump: Employment is up, Taxes are DOWN. Enjoy!</i>	female	25 / 53



<https://botometer.iuni.iu.edu>

Human to bot errors



Francisco M. Rangel
@kicorangel

CTO Autoritas Consulting - Structuring unstructured information - Investigating the use of language for analysing social media and author profiling.

Valencia

kicorangel.com

Se unió en julio 2009

Author Id.	Twitter Account	Gender	N. Systems
a22edd53bb04de0c06a52df897b13dd0	@carlosguadian	male	39 / 42
<i>Tres días para analizar el presente y futuro de la Administración pública: lo que trae el Congreso NovaGob 2018 - NovaGob 2018 https://t.co/Ofc4cDTeym #novagob2018</i>			
cf520c8e810a6a9bae9171d6f23c29be	@kicorangel	male	35 / 42
<i>Google prepara una versión de pago para Youtube http://t.co/UvZdao68wc</i>			
8e4340e95667c8add31f427a09dd3840	@EmaMARredondoM	female	30 / 42
<i>@andrespino007 ¿Se ha preguntado cómo alguien llega a ser científico? Pequeña muestra chilena: https://t.co/fLjJsV0I0J</i>			
6730bdf9686769c4a8a79d2f766a7f67	@Annie_Hgo	female	24 / 42
<i>Wow!! Nuevamente rebasamos expectativas... https://t.co/PJ8bHA1SrG</i>			

Conclusions

- Several approaches to tackle the task:
 - Best approach: n-grams + SVM
- Best results in **bots vs. human**:
 - **Over 84% on average** (EN 86.15%; ES 84.08%)
 - English (95.95%): Johansson - Stylistic features + Random Forest
 - Spanish (93.33%): Pizarro - n-grams + SVM
- Error analysis:
 - **Highest confusion from bots to humans (17.15% vs. 7.86% EN; 14.45% vs. 14.08% ES)**
 - ...mainly towards males (9.83% vs. 7.53% EN; 8.50% vs. 5.02% ES)
 - ...males more confused with bots (8.85% vs. 3.55% EN; 18.93% vs. 11.61% ES)
 - **Error per bot type:**
 - **Advanced bots: 30.11% EN; 32.38% ES**
 - EN: quote (12.64%); template (17.94%); feed (27.89%)
 - ES: quote (26.51%); template (13.20%); feed (14.28%)
 - **Mainly towards males**, except quote bots in ES (6.75% vs. 15.29% towards males)

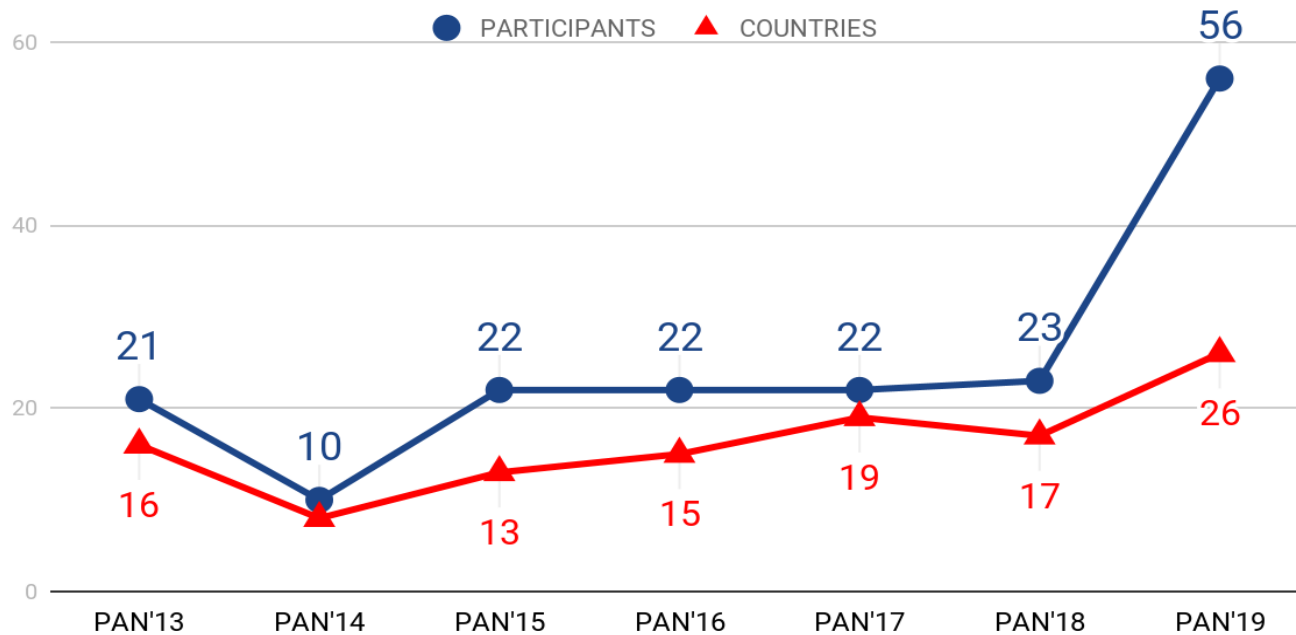
Conclusions

Looking at the results, we can conclude:

- It is feasible to automatically identify bots in Twitter with high precision
 - ...even when **only textual features** are used.
- There are specific cases where the task is difficult due to:
 - ...**the language used by the bots (e.g., advanced bots)**
 - ...**the way the humans use the platform (e.g., to share news)**

In both cases, although the precision is high, a major effort needs to be made to take into account **false positives**.

Participating teams



Industry

Organisation



Sponsors



IBM PartnerWorld



Participant

S



ANCHORMEN



Bitdefender®

Not only industry

Organisation



Sponsors



IBM PartnerWorld



Participant

S



ANCHORMEN



Swedish Defence
Research Agency



2020: Profiling FAKE NEWS spreaders on Twitter



Rangel F., Giachanou A., Ghanem B., Rosso P. (2020) Overview of the 8th Author Profiling Task at PAN 2020: Profiling Fake News Spreaders on Twitter. In: CLEF 2020 Labs and Workshops, Notebook Papers. CEUR Workshop Proceedings.CEUR-WS.org, vol. 2696

2021: Profiling HATE speech spreadeRS on Twitter



Rangel F., De La Peña G., Chulvi B., Fersini E., Rosso P. (2021) Profiling Hate Speech Spreaders on Twitter Task at PAN 2021. In: CLEF 2021 Labs and Workshops, Notebook Papers, CEUR-WS.org, vol. 2936, pp. 1772-1789

2022: Profiling IRONY and STEREOTYPE Spreaders on Twitter

- Author profiling perspective: Profiling **irony and stereotype** spreaders
- Usage of irony and stereotypes for (often implicitly) conveying hurtfulness, although... not always
- Subtask on Profiling **stereotype stance** of ironic authors

How wonderful a **Jew** actually said something bad about Israel. I'm sooo impressed.

If Australia doesn't "DEPORT" 100K **Muslims** a year, what do you propose? Concentration camps?

Didn't you know if they rub against you that you can become **gay**?! Talk about sharing a foxhole!!!

Profiling native language

Koppel M., Schler J., Zigdon Z. Determining an Author's Native Language by Mining a Text for Errors, Proceedings of KDD, Chicago, 2005.

Native language

Given an English text, can we determine the **author's native language?**

Exercise: which is which?

These were written by **Russian**, **French** and **Spanish** speakers, respectively:

In the second part of this author's novel, called Time Passes, time has passed indeed and Mrs Ramsay has died.

There are prejudices of small groups, such as homosexuals, immigrants, AIDS diseaseds, etc. But "political correctness" has have positive and negative consequences.

There is one more kind of films irritating many television viewers - "soap" serials. «Santa Barbara» has even won "Oskar" prize.

Possible clues

Patterns of native language are typically reflected in how other languages are spoken:

- **Word selection**
- **Syntax**
- **Spelling**

Measurable features

- **Frequency of function words**
- **Frequency of letter/char n-grams**
- **Idiosyncrasies (mistakes)**

We will **gather idiosyncrasies data automatically**

Orthographic idiosyncrasies

- Repeated letter (e.g. *remmit* instead of *remit*)
- Double letter appears once (e.g. *comit* instead of *commit*)
- Letter α instead of β (e.g. *firsd* instead of *first*)
- Letter inversion (e.g. *fisrt* instead of *first*)
- Inserted letter (e.g. *friegnd* instead of *friend*)
- Missing letter (e.g. *frend* instead of *friend*)
- Conflated words (e.g. *stucktogether*)

Syntactic idiosyncrasies

- Sentence Fragment
- Run-on Sentence
- Repeated Word
- Missing Word
- Mismatched Singular/Plural
- Mismatched Tense
- *that/which* confusion
- Rare POS pairs

Automatically finding idiosyncrasies

1. Run text through automated spell/grammar checker
2. Compare flagged word to best suggestion
3. Mark error accordingly

e.g. text=*remmit* suggestion=*remit*

mark as “repeated letter”

Features

- 400 function words
- 200 letter sequences
- 185 error types
- 250 rare POS pairs

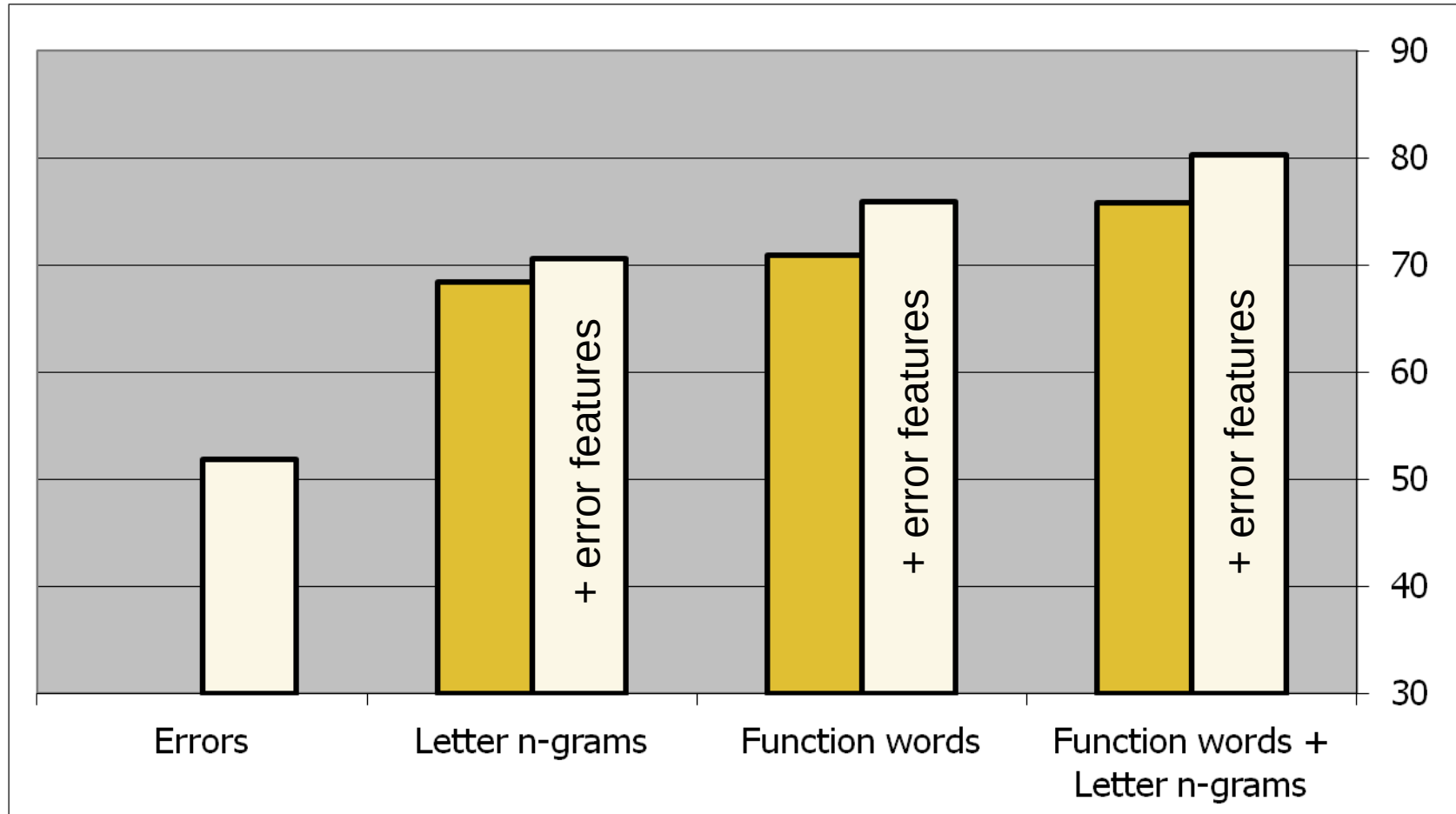
Each document is represented as numerical vector of length 1035

Test corpus

International Corpus of Learner English

- 11 countries
- Subjects same age, proficiency level
- Samples same genre, length
- Actually used in study- 258 docs:
 - French
 - Spanish
 - Bulgarian
 - Czech
 - Russian

Classification accuracy (10-fold CV)



Baseline=20% (5 languages)

Some hints

- Russian: *over, the* (infrequent), **number_relative-adverb**
- French: *indeed, Mr* (no period), misused *o* (e.g. **o**uthor)
- Spanish: *c-q* confusion (e.g. **c**uality), *m-n* confusion (e.g. **con**fortable), undoubled consonant (e.g. **com**it)
- Bulgarian: **most_adverb**, *cannot* (uncontracted)
- Czech: doubled consonant (e.g. **rem**mit)

Exercise: let's try again...

In the second part of this author's novel, called *Time Passes*, time has passed indeed and Mrs Ramsay has died.

There are prejudices of small groups, such as homosexuals, immigrants, AIDS diseaseds, etc. But "political correctness" has have positive and negative consequences.

There is one more kind of films irritating many television viewers - "soap" serials. «*Santa Barbara*» has even won "Oscar" prize.

Exercise: let's try again...

In the second part of this **author's** novel, called Time Passes, time has passed **indeed** and **Mrs** Ramsay has died.

There are **pejudgments** of small groups, such as homosexuals, **inmigrants**, aids diseaseds, etc. But "political correctness" has have positive and negative **consecuences**.

There is **one more** kind of films irritating many television viewers - "soap" serials. «Santa Barbara» has even won **[the]** "**Oskar**" prize.

Translationese

1. Universal effects of the **translation process**
(e.g. **simplification**)
2. **Interference** effects from **source language**

Aims

1. Can we determine **if a text is translated (T) or original (O)?**
2. Can we determine the **source language of a translated text?**
3. Can we train a classifier for T vs. O **using one source language and test using another?**

The Europarl corpus

- Speeches in European Parliament
- Translations to English from 11 languages
- We use five:
Spanish, Italian, French, German, Finnish
- 500,000 words of each
- Also, original English transcripts (2.5M words)
- Translated corpora each chunked to 250 equal chunks
- English corpus chunked to 1250 equal chunks

Function words

- 300+ content-free words
- Work well as features in many style-based application
- Originally used by Mosteller and Wallace
- Avoid cultural conventions

[see van Halteren (2008) on Europarl]

Translationese vs original: Europarl

Aim: To determine if a document is original English or translated into English

Corpus: 1250 translated (5 source langs) + 1250 original English Europarl chunks

Features: function words

Bayesian logistic regression

Experimental setup: 10-fold CV

Results

Learned classifiers distinguish translated English (T) from original English (O) with accuracy of 96.7%.

The differences between T and O

word	freq O	freq T
<i>I</i>	2.552%	2.148%
<i>we</i>	2.713%	2.344%
<i>you</i>	0.479%	0.470%
<i>he</i>	0.286%	0.115%
<i>she</i>	0.081%	0.039%
<i>me</i>	0.148%	0.141%
<i>us</i>	0.415%	0.320%
<i>him</i>	0.066%	0.033%
<i>her</i>	0.091%	0.056%
<i>my</i>	0.462%	0.345%
<i>our</i>	0.696%	0.632%
<i>your</i>	0.119%	0.109%
<i>his</i>	0.218%	0.123%

Pronouns

word	freq O	freq T
<i>therefore</i>	0.153%	0.287%
<i>thus</i>	0.015%	0.041%
<i>consequently</i>	0.006%	0.014%
<i>hence</i>	0.007%	0.013%
<i>accordingly</i>	0.006%	0.011%
<i>however</i>	0.216%	0.241%
<i>nevertheless</i>	0.019%	0.045%
<i>also</i>	0.460%	0.657%
<i>furthermore</i>	0.012%	0.048%
<i>moreover</i>	0.008%	0.036%
<i>indeed</i>	0.098%	0.053%
<i>actually</i>	0.065%	0.042%

Cohesive Adverbs

Discussion

- These differences are consistent with those found on **Italian** and on **Spanish**.
- The phenomena suggest explicitation (e.g. resolving **anaphora** and interpolating chaining).

Universal differences?

- The **International Herald Tribune** (IHT) includes local supplements **translated from Korean, Hebrew and Greek**.
- The **genre** is quite **different than Europarl** (parliamentary speeches vs. newspaper stories) and the languages are from very different families.
- Do the differences we found between T and O hold up in the IHT corpus?

Differences between T and O: IHT

word	freq O	freq T
<i>therefore</i>	0.011%	0.031%
<i>thus</i>	0.011%	0.027%
<i>consequently</i>	0.000%	0.004%
<i>hence</i>	0.003%	0.007%
<i>accordingly</i>	0.003%	0.003%
<i>however</i>	0.078%	0.129%
<i>nevertheless</i>	0.008%	0.018%
<i>also</i>	0.305%	0.453%
<i>furthermore</i>	0.003%	0.011%
<i>moreover</i>	0.009%	0.008%
<i>indeed</i>	0.018%	0.024%
<i>actually</i>	0.032%	0.018%

There is no significant difference between O and T in pronoun usage in IHT.

Cohesive Adverbs

The best feature of all

The frequency of the word *the* (as proportion of function words):

Original English

- Europarl 13.73%
- Newspapers 13.31%

Translations:

- Europarl* 15.32%
- Newspapers** 15.36%

*French, Spanish, Italian, German, Finnish

**Korean, Hebrew, Greek

Discussion

- **Some differences between translations (from many different languages) and original English** are consistent enough to allow accurate classification.
- Such differences must be universal and not language-specific.

Guess the source language

Aim: To determine the source language of a document translated into English

Corpus: 1250 translated Europarl chunks (5 source languages)

Features: function words

Bayesian logistic regression

Experimental setup: 10-fold CV

Guess the source language

Learned classifiers determine source language (out of 5) with accuracy 92.7%

Some salient differences

	over-represented	under-represented
Fr	of, finally	here, also
It	upon, moreover	also, here
Es	with, therefore	too, then
De	here, then	of, moreover
Fi	be, example	me, which

Discussion

- **Translations from different languages are sufficiently distinct** that we can accurately classify a translation according to source language.
- There must be **language-specific interference effects**.

Contrasting the effects

Let's learn a classifier to distinguish T from O
BUT

- train on translations from source language x
- test on translations from source language y

When $x=y$, the effects reinforce each other.
But when $x \neq y$, universal effects help and
interference effects hurt. Who wins?

Contrasting the effects

	Italian	French	Spanish	German	Finnish
Italian	98.3	91.5	86.5	71.3	61.5
French	91	97	86.5	68.5	60.8
Spanish	84.5	88.3	95.8	76.3	59.5
German	82	83.3	78.5	95	80.8
Finnish	56	60.3	56	62.3	97.3

Conclusion: the more closely related x and y, the less noise caused by interference

In fact, this is a nice **measure of the similarity of two languages.**